

BOM VARIANTS

BOM NUMBER	BOM NAME	BOM OPTIONS
639-4206	PCBA, MLB, GOOD, HYNIX, J17	J17_COMMON, J17, CPU:GOOD, GPU:GK187EGX, FB:2G_HYNIX, FBA, FBB, SSD:Y, EEEE:F8YL
639-4418	PCBA, MLB, GOOD, ELPIDA, J17	J17_COMMON, J17, CPU:GOOD, GPU:GK187EGX, FB:2G_ELPIDA, FBA, FBB, SSD:Y, EEEE:FCTC
985-0047	PCBA, MLB, DEV, J17	DEVELOPMENT, J17_DEVEL

CPUs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
33754532	1	H5W, SR14E, PRQ, C0, 3.20, 84W, 4+2, 1.15, 6M, LGA	CPU	CRITICAL	CPU:GOOD
33754505	1	H5W, QE70, QS, C0, 3.26, 84W, 4+2, 1.15, 6M, LGA	CPU	CRITICAL	CPU:GOOD_TDP

CPU SOCKET

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
511S0080	1	SOCKET, MOLEX, LGA1150, CPU-HF	U0500	CRITICAL	

BOM Groups

BOM GROUP	BOM OPTIONS
J17_COMMON	COMMON, ALTERNATE, J17_COMMON1, J17_PROGPARTS
J17_COMMON1	XDP, SPEAKERID, TBTHV:P12V, CPUVCC:4PHASE, EXT_GPU:YES, VDDQ:P1V5
J17_PROGPARTS	SMC:PROG, BOOTROM:PROG, TBTRM:PROG, CIVROM:PROG, CAMROM:PROG, BLCMCU:PROG
J17_DEVEL	XDP_CONN, LPCPLUS, DDRVREF_DAC, DEVEL_SENSORS, DEVEL_AUDIO
DEVEL_SENSORS	TEMPSNSDEV, AP_ISNS:Y
J17_PRODUCTION	AP_ISNS:N

ASICs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337SA561	1	1C, QEW, LPT-D, C2, S05, 23X22MM, FCBG6708	U1100	CRITICAL	
337SA562	1	1C, QEW, LPT-D, C2, S05, 23X22MM, FCBG6708	U1100	CRITICAL	PCH_TDP
338S1113	1	1C, TBT, CR-4C, B1, PRQ, 28B FCBG6, 12X132MM	U2800	CRITICAL	
343S0616	1	1C, BCM57766A1, ENET&SD, 8X8	U3900	CRITICAL	

Programmable Parts

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
341S3858	1	IC, PROGRMD, EFI ROM, V0084, J17	U5210	CRITICAL	BOOTROM:PROG
335S0807	1	IC, 64 MBIT SPI SERIAL FLASH	U5210	CRITICAL	BOOTROM:BLANK
341S3787	1	IC, PROGRMD, SMC, A3, VTBD, J17	U5000	CRITICAL	SMC:PROG
338S1159	1	IC, SMC, LM4FS1BH5BBCIGR, SCPL	U5000	CRITICAL	SMC:BLANK
341S3855	1	IC, EEPROM, CR, V23.8 (B1), J17	U2890	CRITICAL	TBTRM:PROG
335S0865	1	IC, EEPROM, SERIAL, 8KB, M1PB	U2890	CRITICAL	TBTRM:BLANK
341S3735	1	IC, ENET 1MBIT, SPI, ROM, V13.13 J17	U3990	CRITICAL	CIVROM:PROG
335S0862	1	IC, SERIAL FLASH, 2MBIT, 2.7V, REF	U3990	CRITICAL	CIVROM:BLANK
341S3778	1	IC, CAMERA FLASH, V7230, D7/D8	U4202	CRITICAL	CAMROM:PROG
335S0852	1	IC, FLASH, SPI, 1MBIT, V93	U4202	CRITICAL	CAMROM:BLANK
341S3674	1	IC, BLC, MCU, PPROGRAMMED, V924, D8	U8100	CRITICAL	BLCMCU:PROG
337S3978	1	IC, BLC MCU LPC2132FBD64/01, LQFP64	U8100	CRITICAL	BLCMCU:BLANK

Bar Code Labels / EEEE #'s

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
825-7896	1	LABEL, MLB, 2D	EEEE_F8YL	CRITICAL	EEEE:F8YL
825-7896	1	LABEL, MLB, 2D	EEEE_FCTC	CRITICAL	EEEE:FCTC

J17 SCHEMATIC / PCB #'S

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
951-9884	1	SCH, MLB, J17	SCH1	CRITICAL	J17
820-3478	1	PCBF, MLB, J17	PCB1	CRITICAL	J17

J17 ALTERNATES

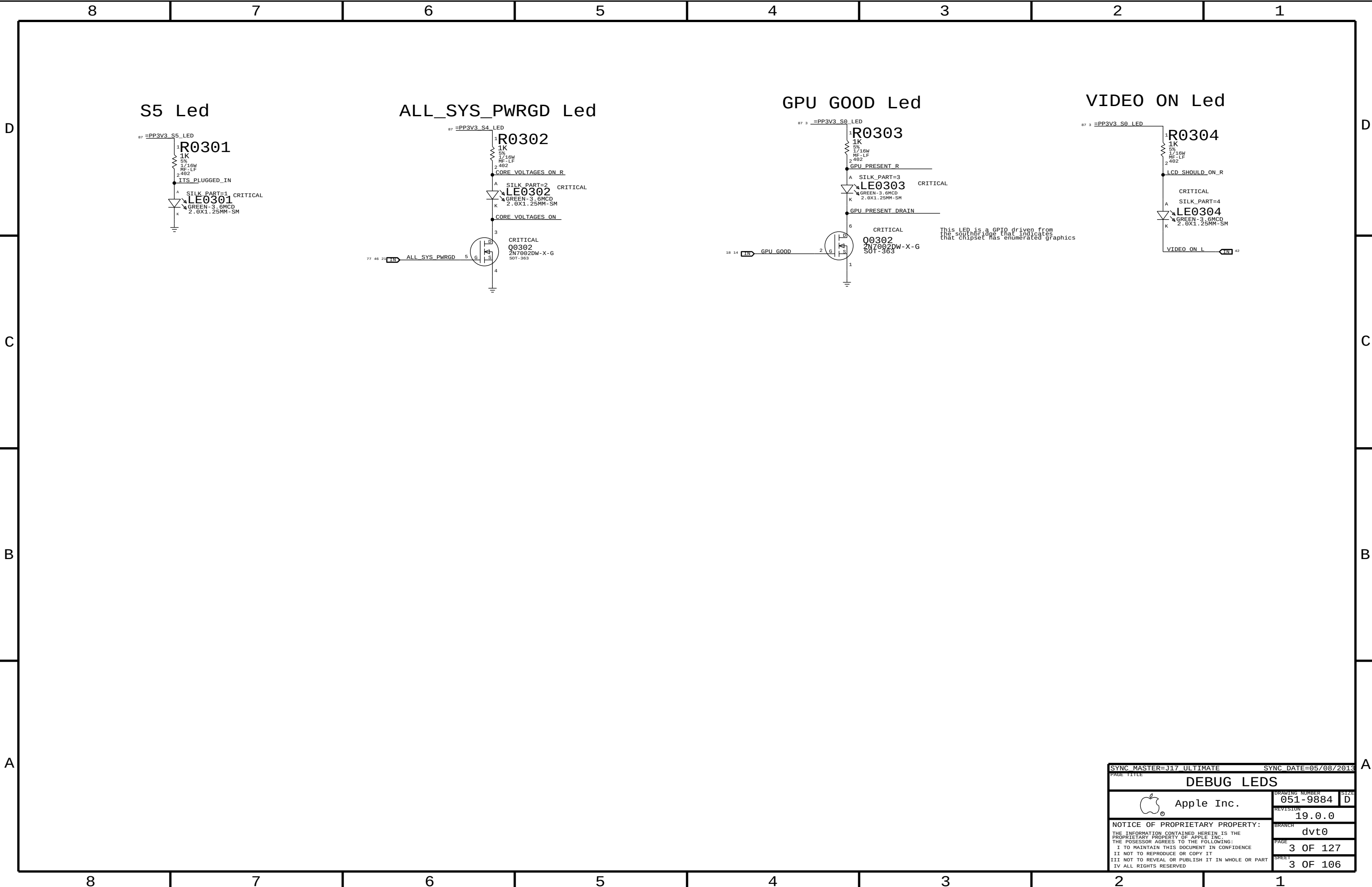
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
377S0147	377S0126		ALL	USB Diodes Array
377S0155	377S0104		ALL	USB Diode
157S0084	157S0058		ALL	Enet Magnetics
376S0975	376S1081		ALL	P/NCH DUAL FET
128S0365	128S0368		ALL	150UF CAPS BLK
138S0803	138S0804		ALL	2.2UF CAPS SOFT
126-0161	126-0160		ALL	56UF CAPS 10X12 THD
197S0481	197S0480		Y1950	25MHz PCH Xtal
197S0479	197S0478		ALL	12MHZ Cam Xtal
372S0186	372S0185		ALL	Alt temp diode
107S0251	107S0249		R5520	Alt 2mOhm sense
107S0254	107S0241		R5450, R5480	Alt 5mOhm sense
107S0255	107S0240		R5450, R5480	Alt 1mOhm sense
341S3747	341S3745		U3990	ENET ROM
138S0681	138S0638		ALL	10uF Caps
376S1106	376S0969		ALL	N-Ch FET
138S0860	138S0775		ALL	Single-source 1uF
138S0859	138S0788		ALL	Single-source 10uF

Y4200 and Y8105

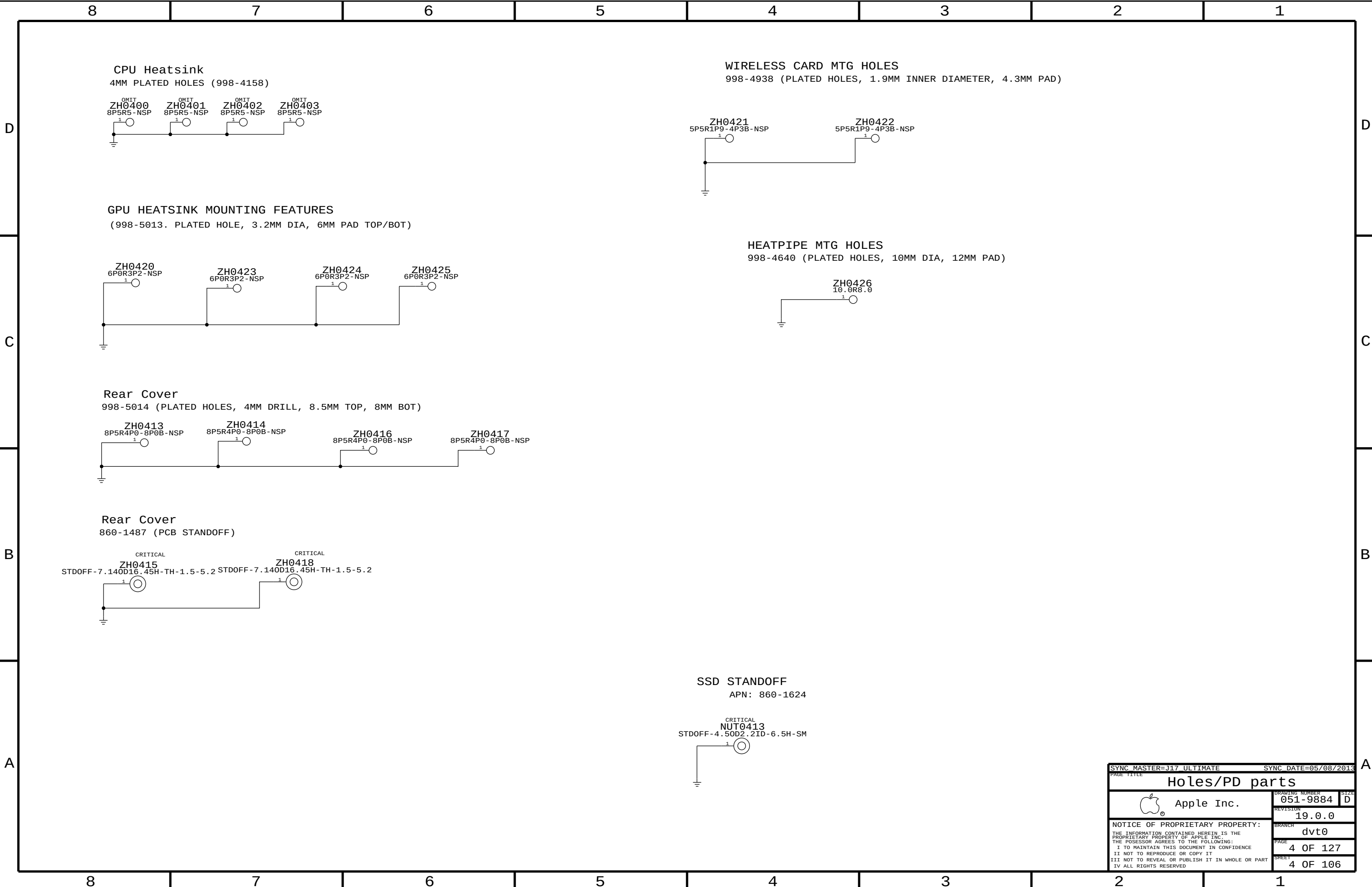
GPU and VRAM

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S4430	1	IC, GPU, NV, GK107-EXX, PS-A2	U8700	CRITICAL	GPU:GK107EGX
333S0630	4	IC, GDDR5, 2GBIT1, 64KX32, GEMMA-DIE, 1708 FBGA	U9200, U9250, U9300, U9350	CRITICAL	FB:2G_HYNIX
333S0695	4	IC, GDDR5, 2GBIT1, 64KX32, B-DIE, 1708 FBGA	U9200, U9250, U9300, U9350	CRITICAL	FB:2G_ELPIDA

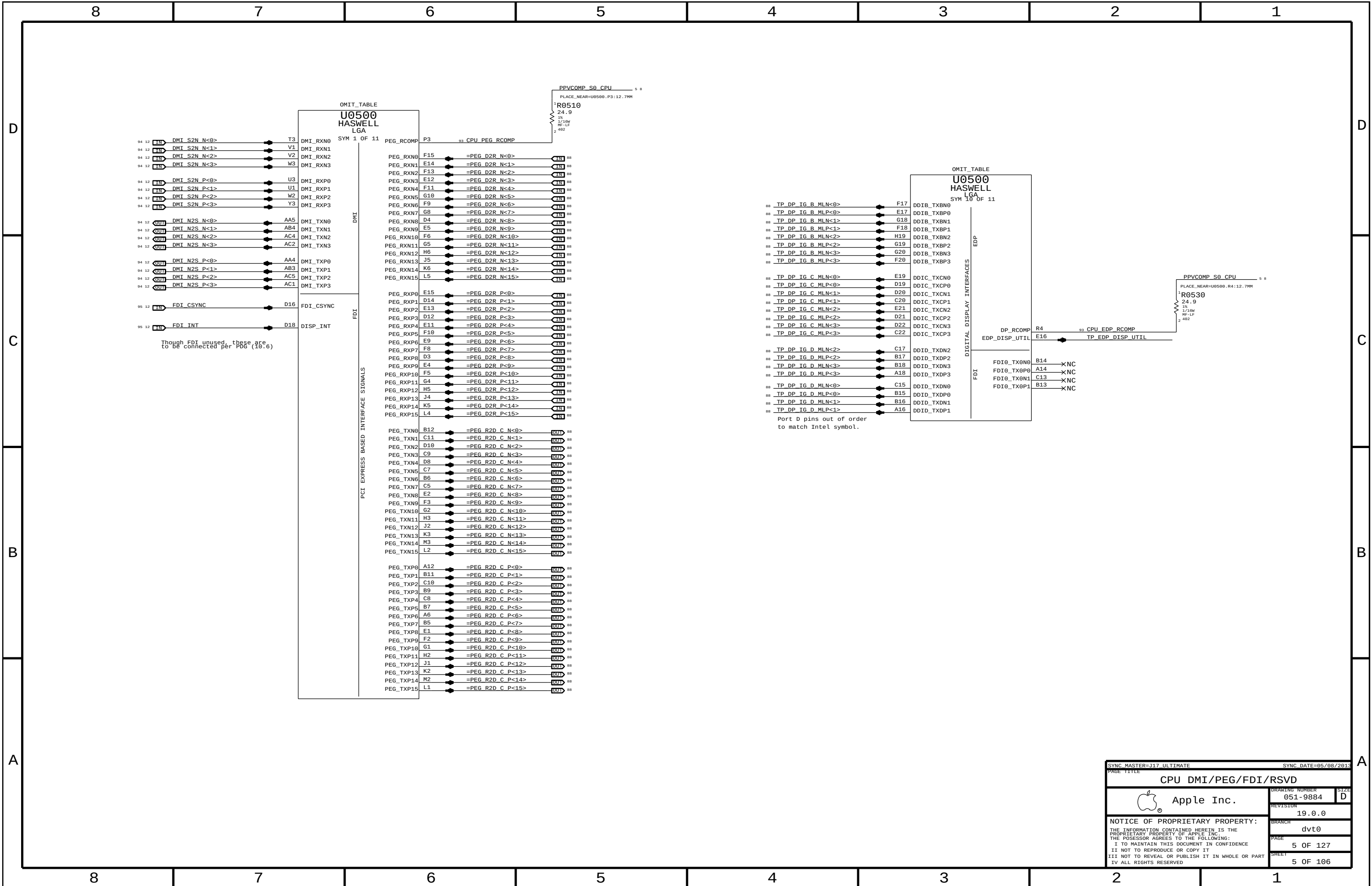
SYNCH MASTER=J17 DINI		SYNCH DATE=02/06/2013	
PAGE TITLE			
BOM Configuration			
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	051-9884	D	
	REVISION	19.0.0	
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		SHEET	2 OF 106



SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/08/2013	
PAGE TITLE			
DEBUG LEDS			
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		PAGE	3 OF 127
		SHEET	3 OF 106



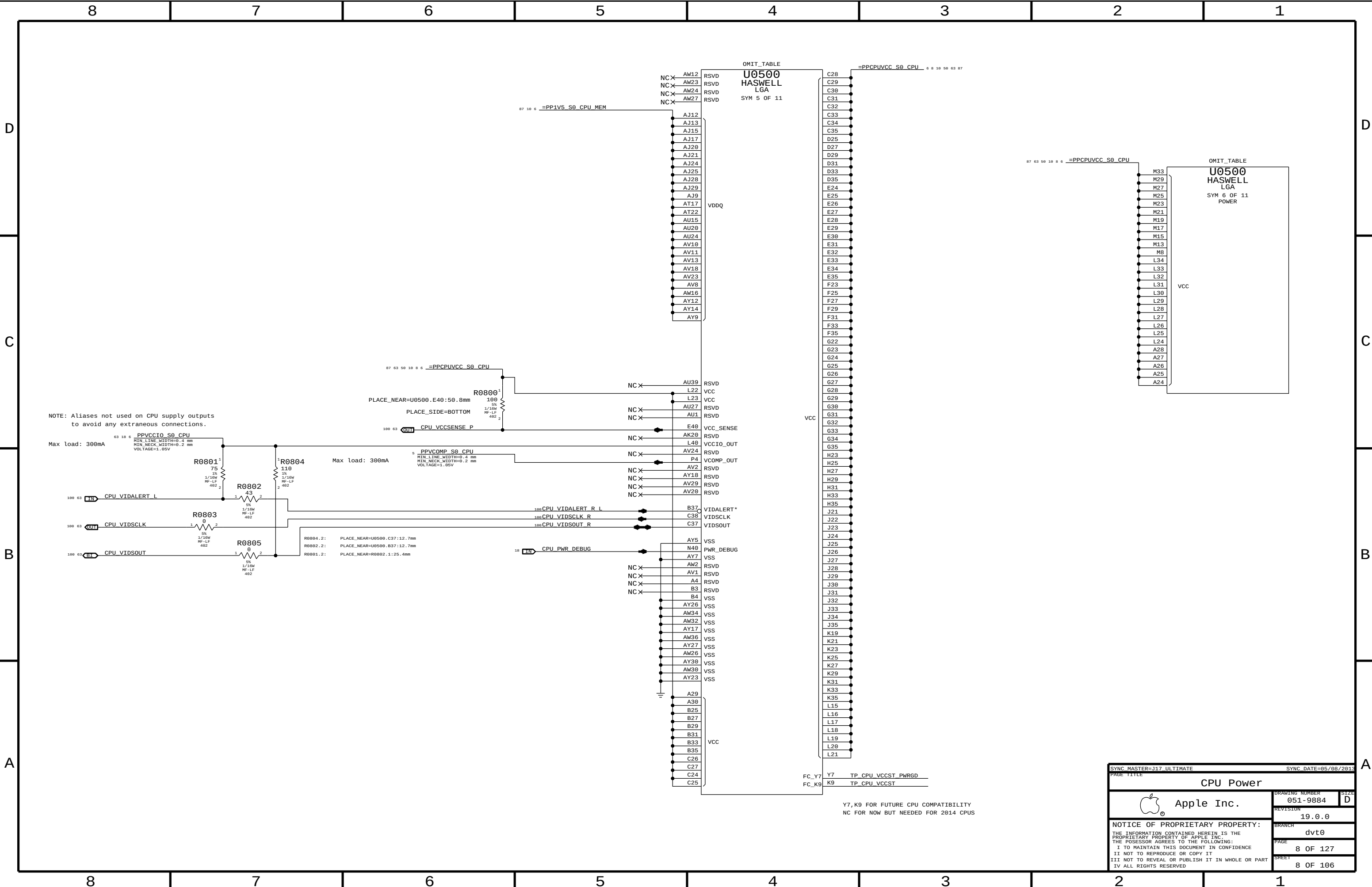
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Holes/PD parts			
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		PAGE	4 OF 127
		SHEET	4 OF 106





INTEL'S HASWELL DT EDS SAYS THAT ALL RSVD AND RSVD_NCTF SHOULD BE NC'D

SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/08/2013	
PAGE TITLE		CPU DDR3 Interfaces	
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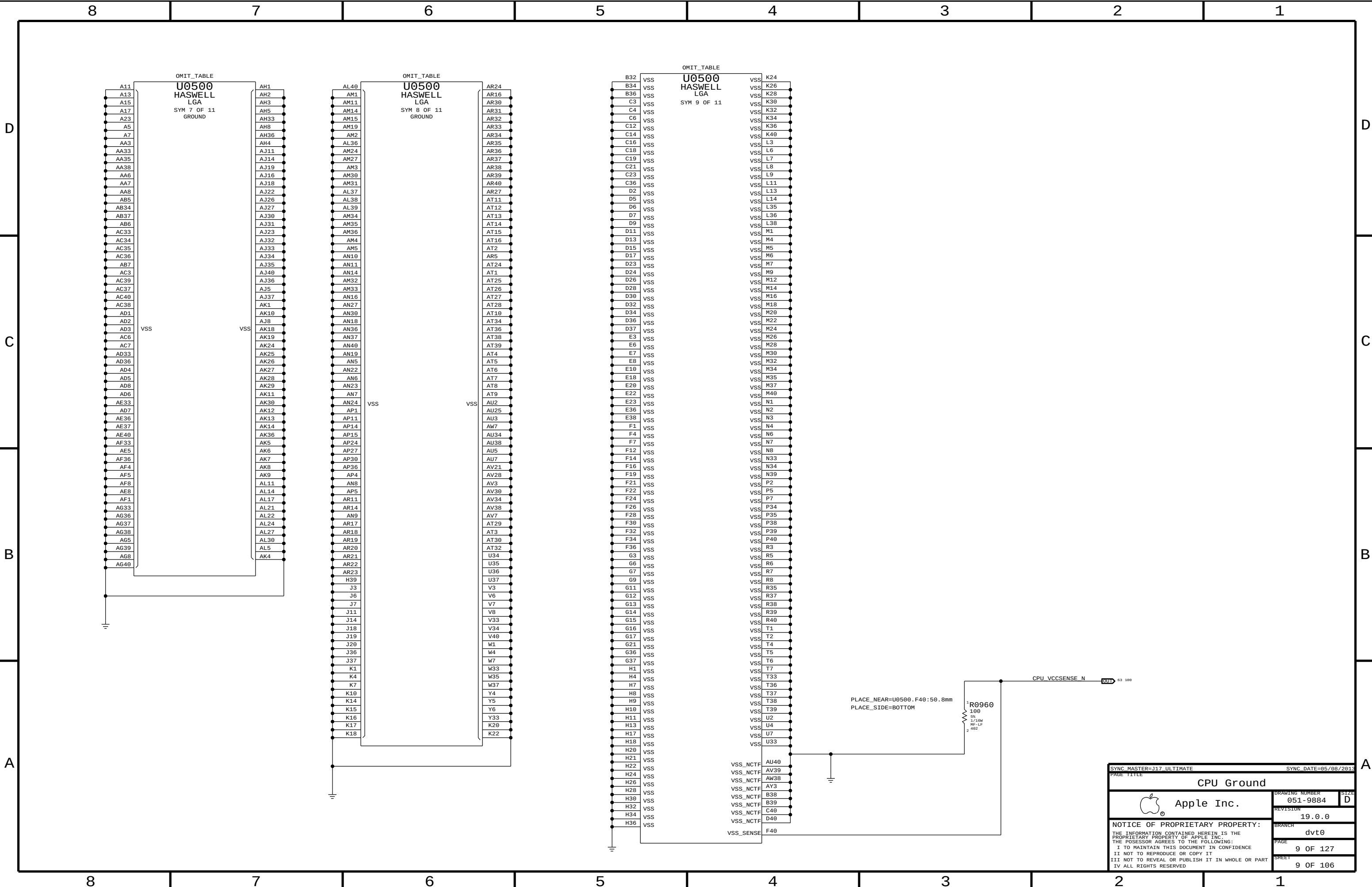
NOTE: Aliases not used on CPU supply outputs to avoid any extraneous connections.

Max load: 300mA

Max load: 300mA

Y7,K9 FOR FUTURE CPU COMPATIBILITY
NC FOR NOW BUT NEEDED FOR 2014 CPUS

SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/08/2013	
PAGE TITLE		CPU Power	
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		PAGE	8 OF 127
		SHEET	8 OF 106



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CPU Ground

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PAGE

9 OF 127

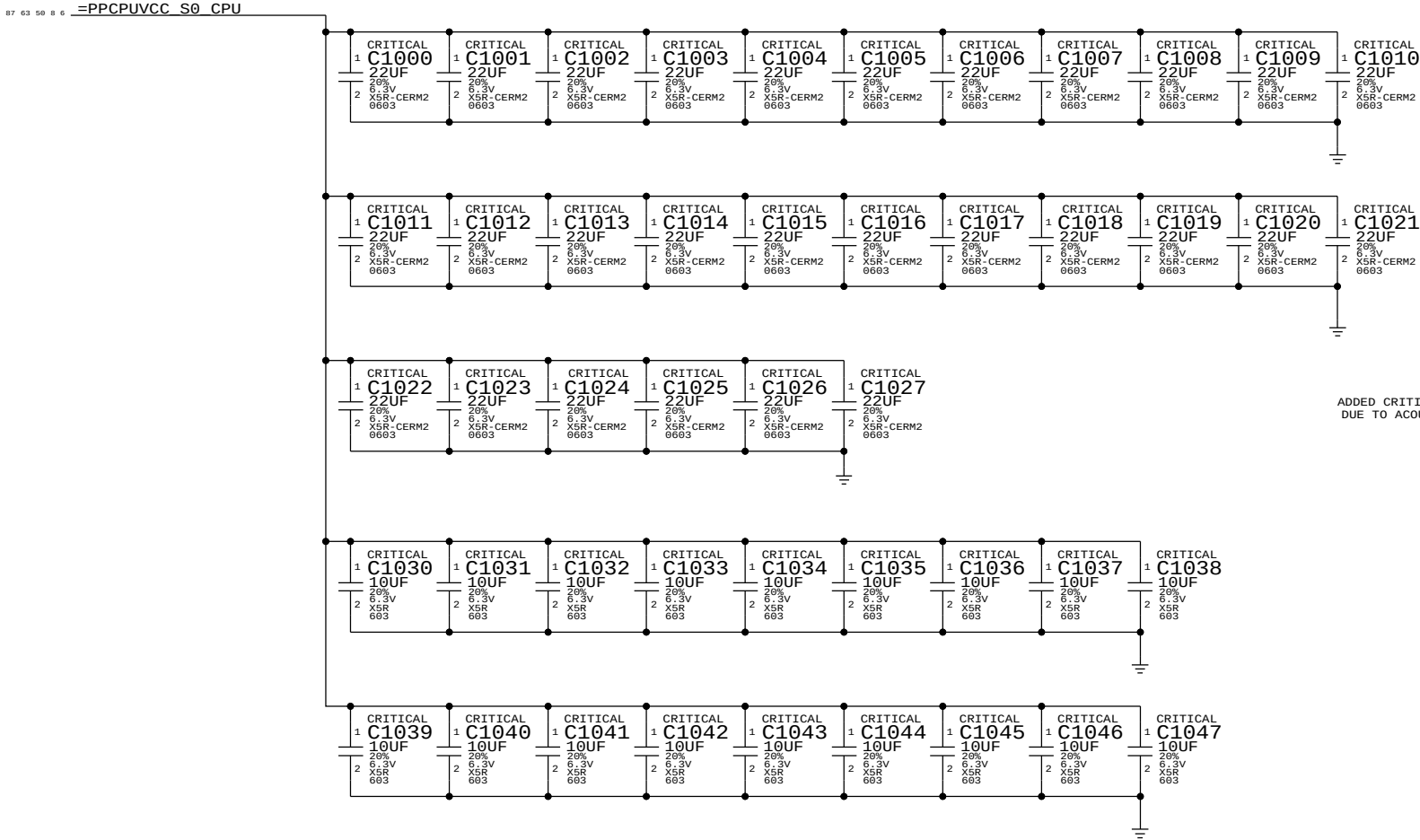
SHEET

9 OF 106

CPU VCORE DECOUPLING

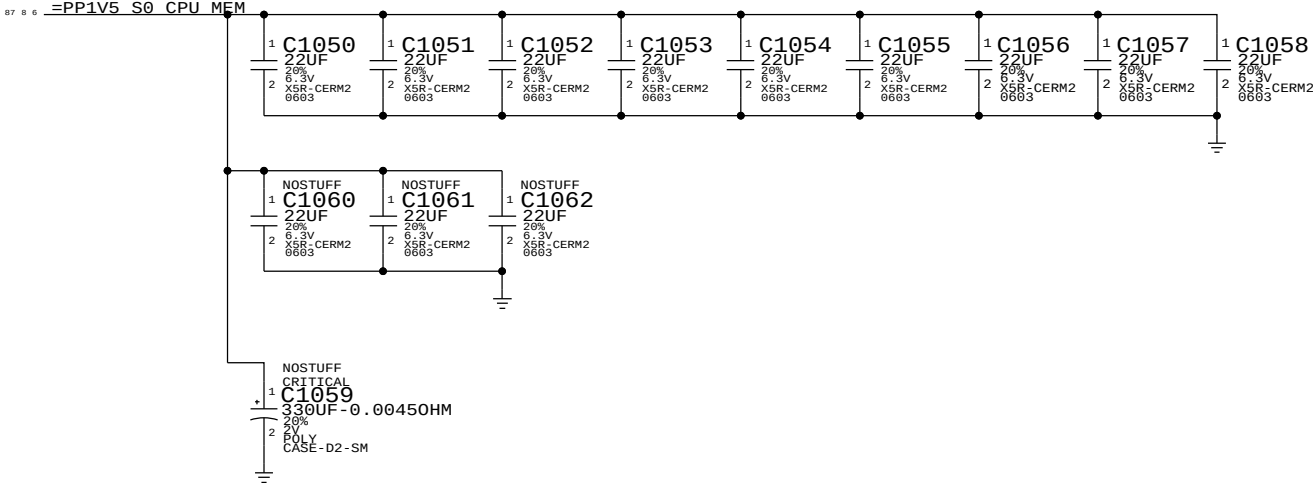
Intel Recommendation:22x 22UF 0805,topside (18 inside cavity, 4 north of processor),8x 470uF bulk caps(5 stuffed,3 no-stuffed)
Apple Implementation:28x 22UF 0603 per Harold
18x 10UF 0603 placed inside socket cavity

Layout Note: These caps should be placed symmetrically on Top and Bottom sides.
BULK CAPS ON CPU VREG PAGE 71



Memory (CPU VCCDDR) DECOUPLING

Intel Recommendation:9x 22UF 0805 near CPU power pins
Apple Implementation:9x 22UF 0603 per Harold
Layout Note: These caps should be placed symmetrically on Top and Bottom sides.



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		BRANCH	dvt0
		PAGE	10 OF 127
		SHEET	10 OF 106



NOTE: SSC control is ganged on PCIe 0-3 and 4-7 clocks.
PEG-attached (CPU) PCIe devices must use one set,
while PCH-attached PCIe devices use the other set.
If 2 or less devices are attached to PEG the
CLKOUT_PEG outputs can be used for those devices.

SATA Port assignments:
PRIMARY HDD

SSD

R1130¹
7.5K
15
1/20W
MF
201 2

Unused clock terminations for FCIM Mode

SYNC MASTER=117 ULTIMATE

SYNC DATE=05/08/2013

PAGE 11111

051-9884

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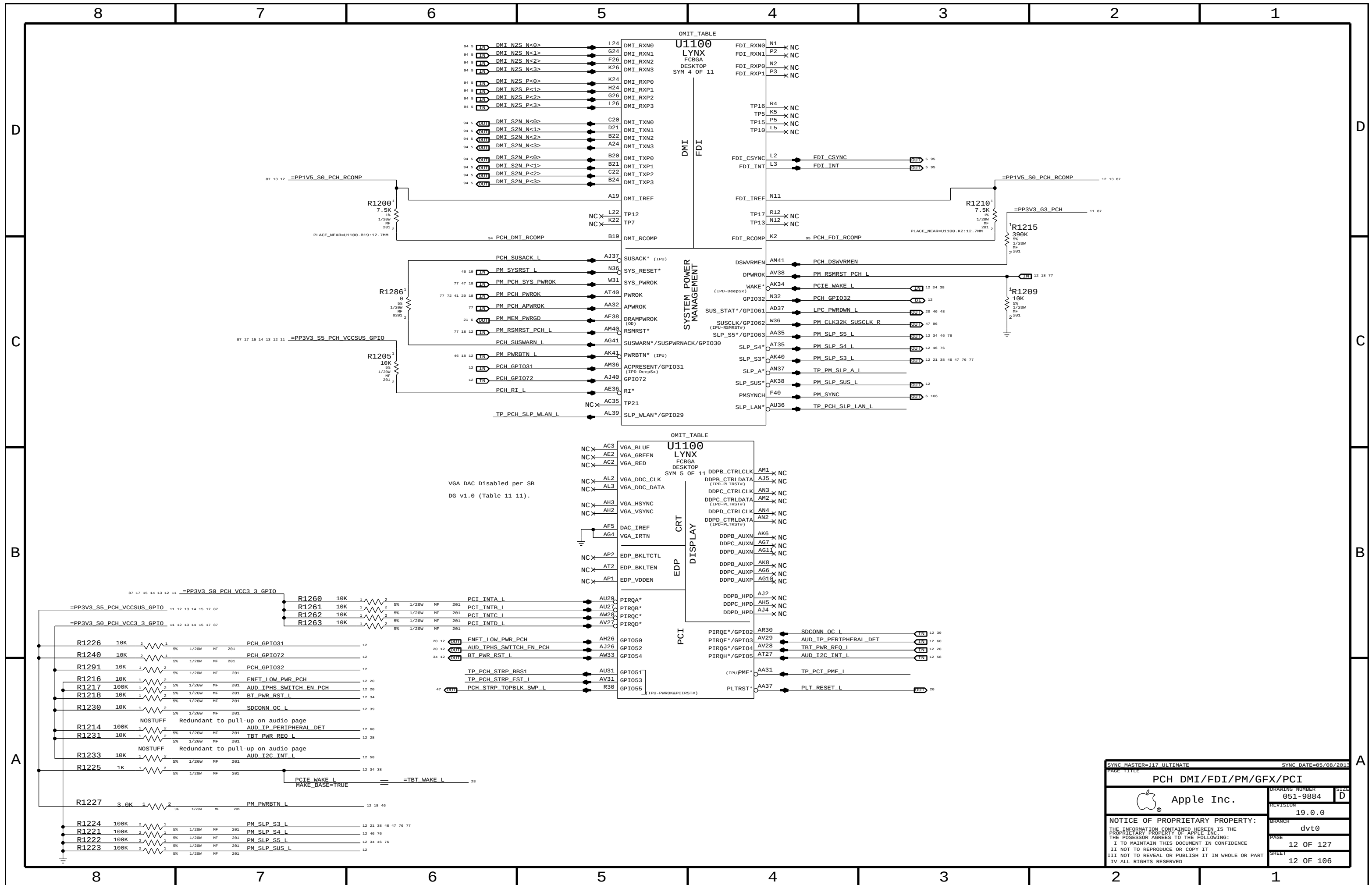
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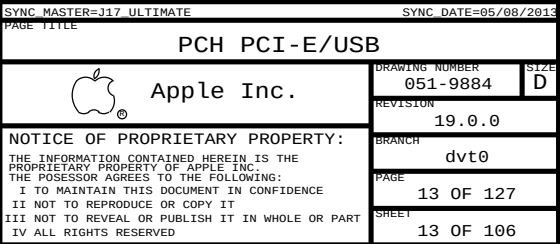
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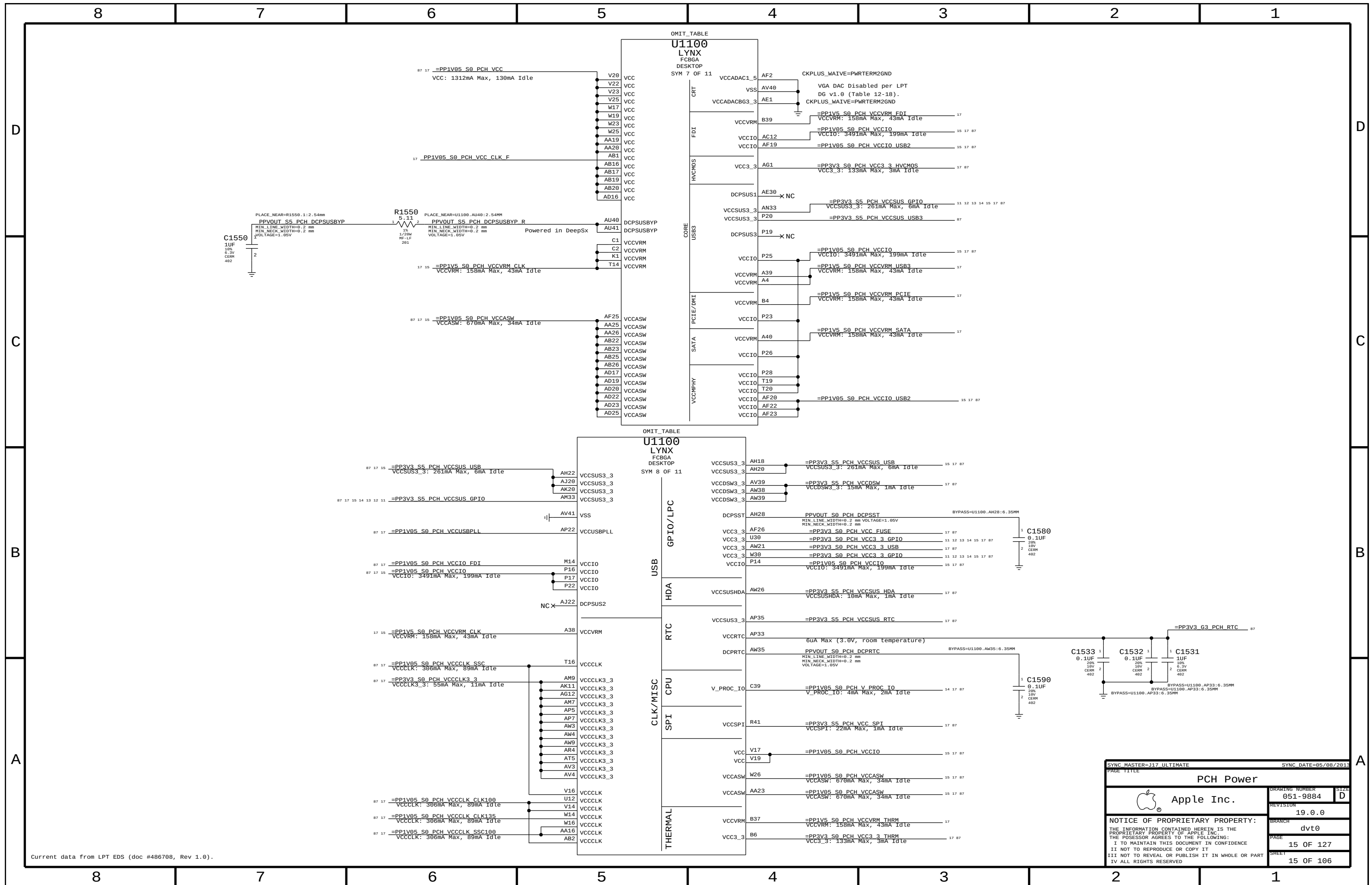
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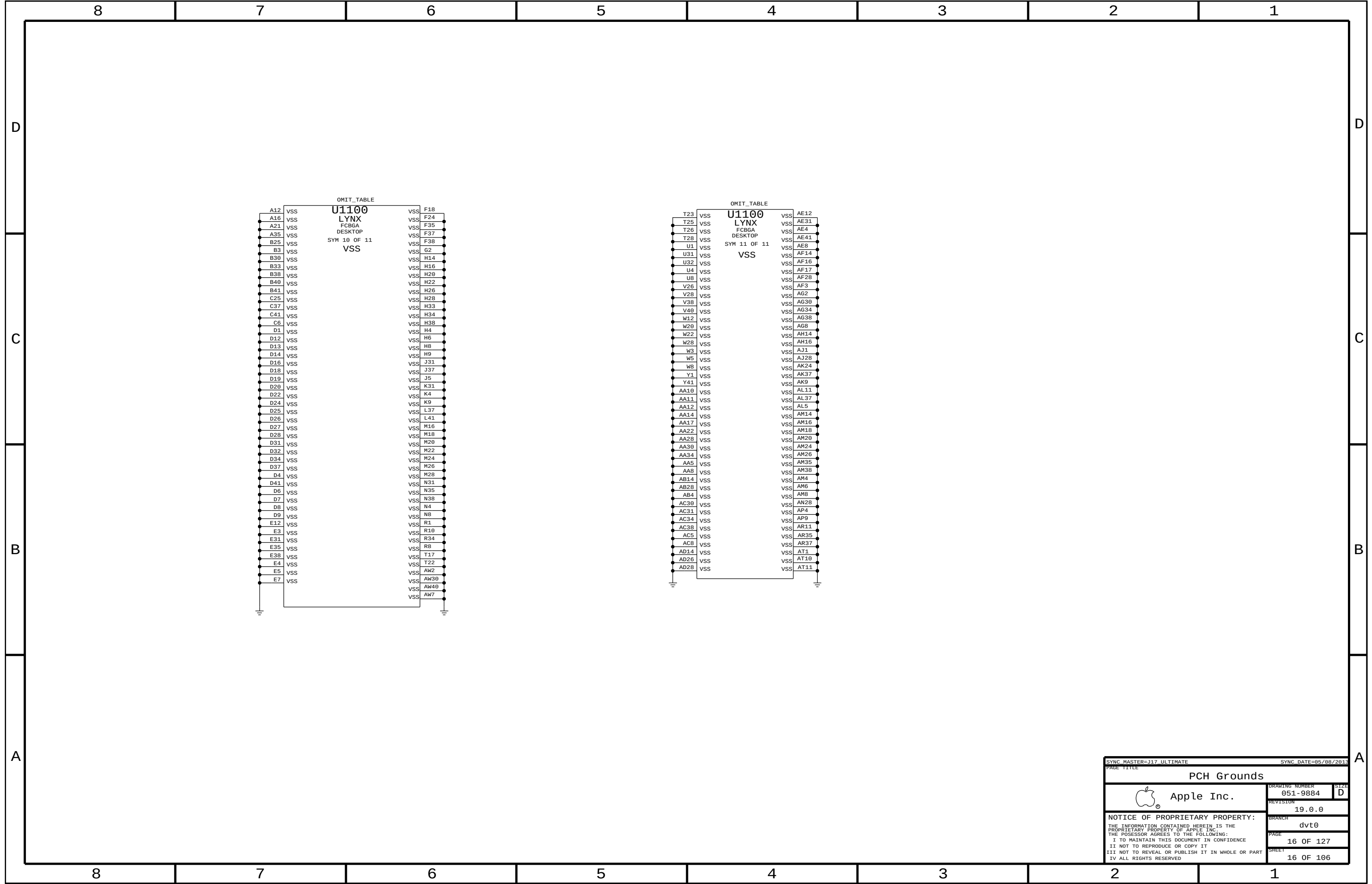
11 OF 127

11 OF 106

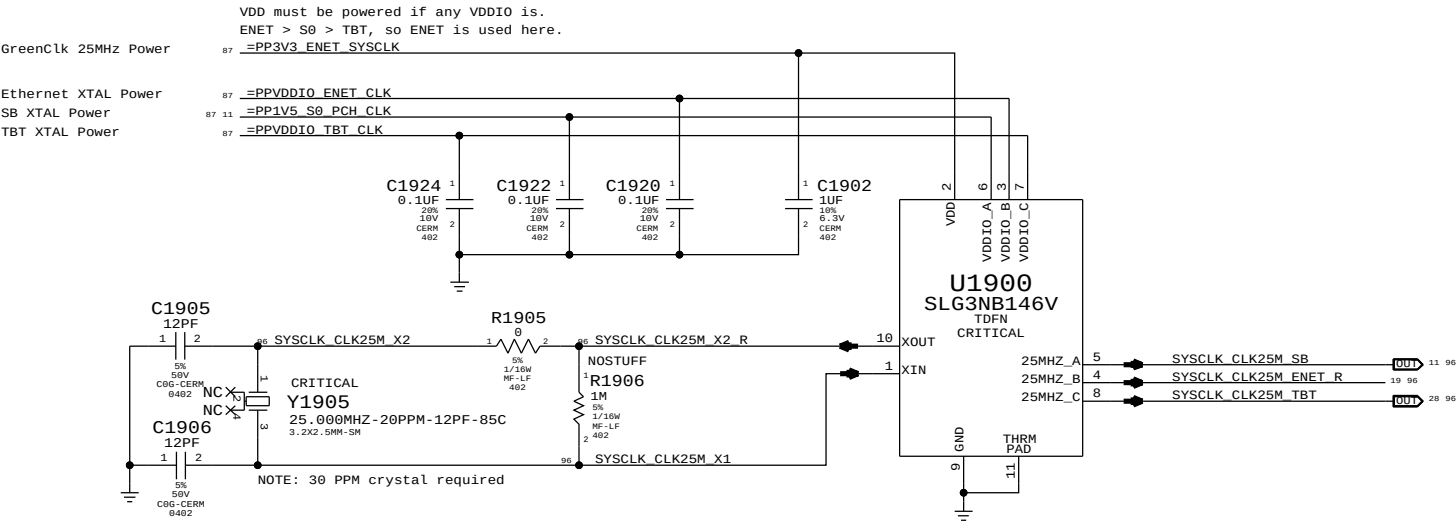




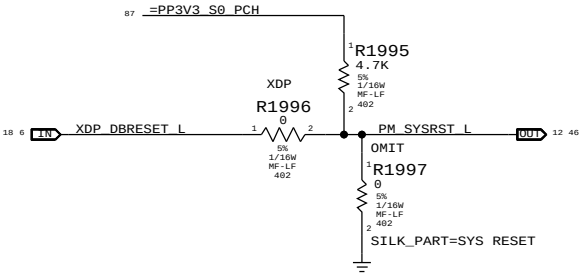




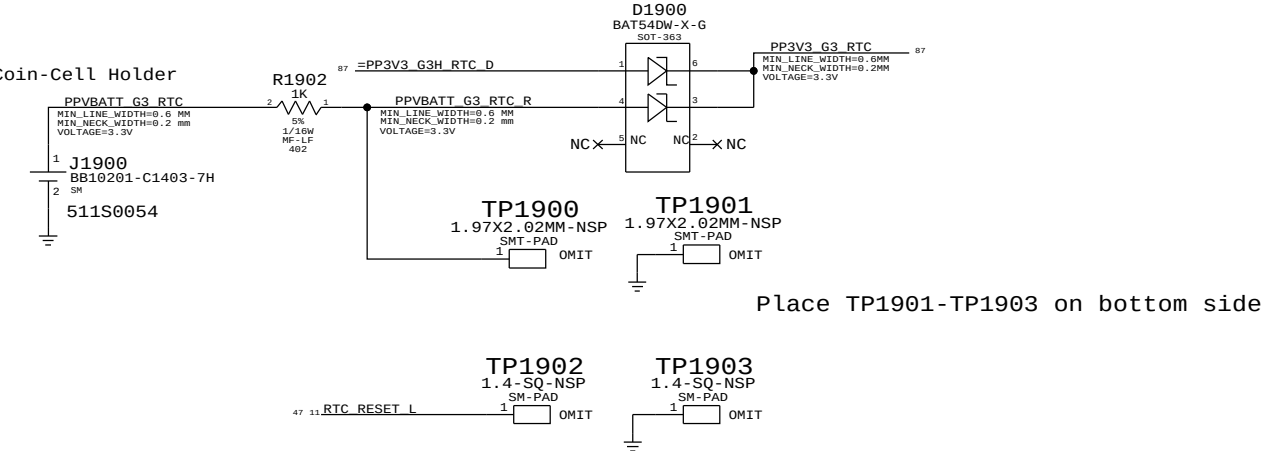
System 25MHz Clock Generator



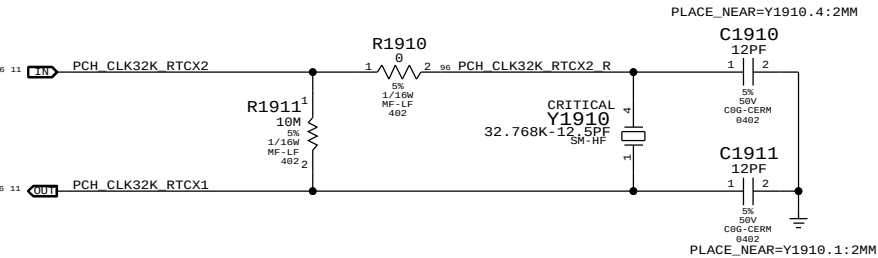
PCH Reset Button



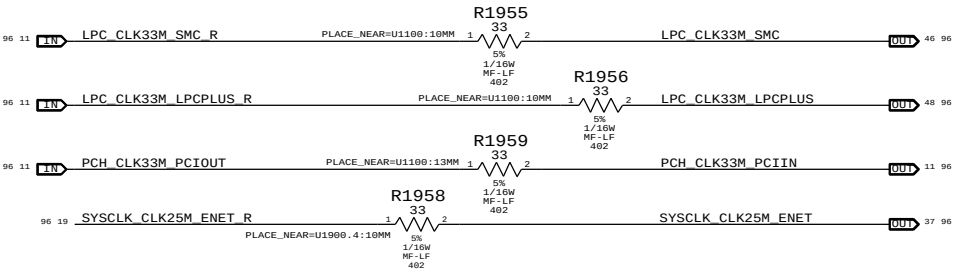
RTC Power Sources



PCH RTC Crystal

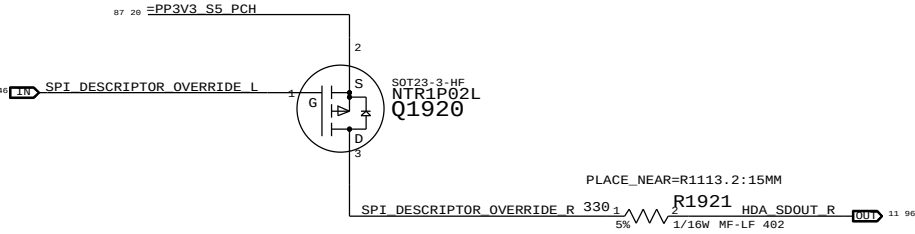


Clock series termination

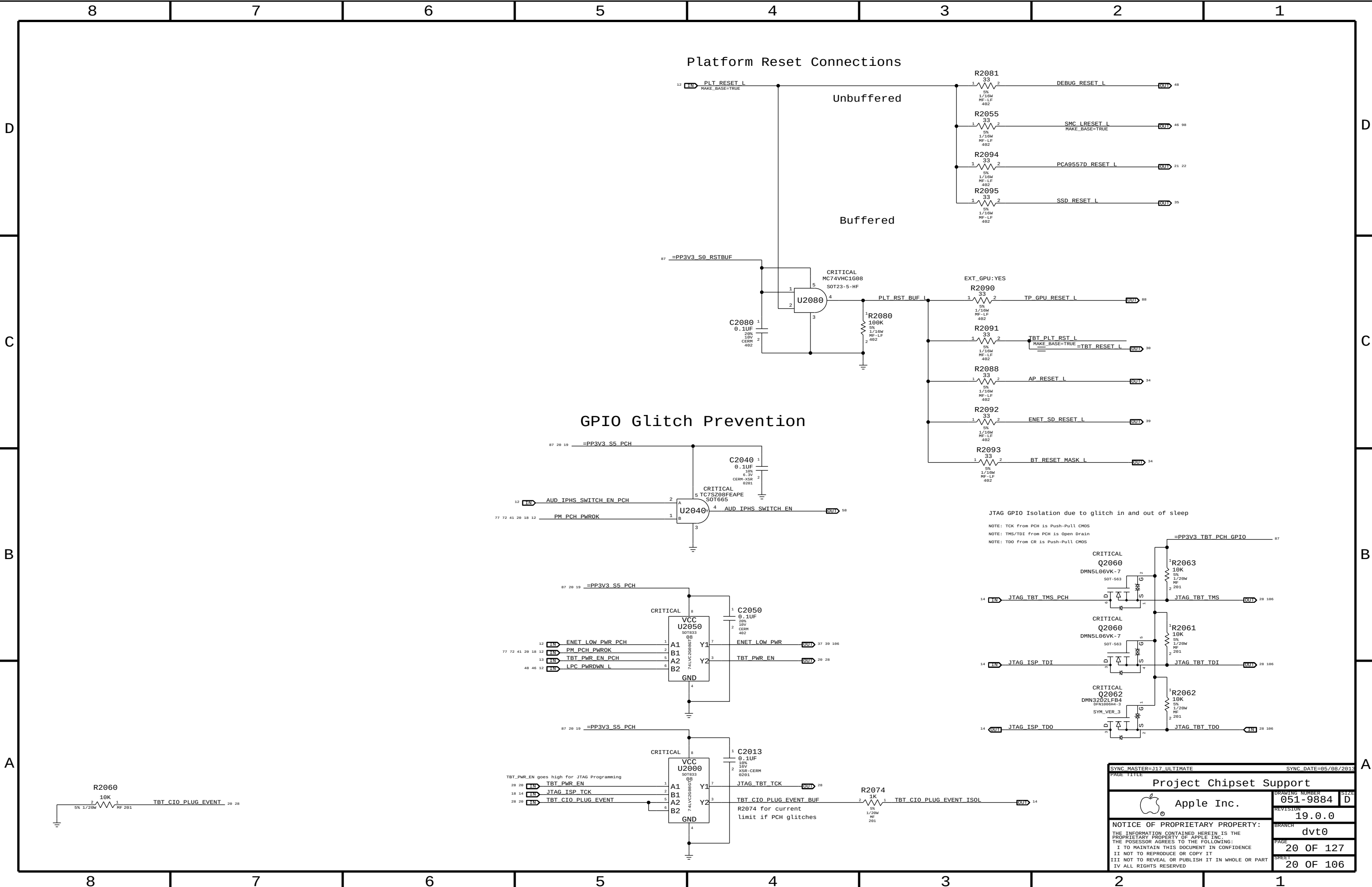


PCH ME Disable Strap

PCH uses HDA_SD0 as a power-up strap. If low, ME functions normally. If high, ME is disabled. This allows for full re-flashing of SPI ROM. SMC controls strap enable to allow in-field control of strap setting.



SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/08/2013	
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		SHEET	19 OF 106



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Project Chipset Support

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20 OF 127

SHEET
20 OF 106

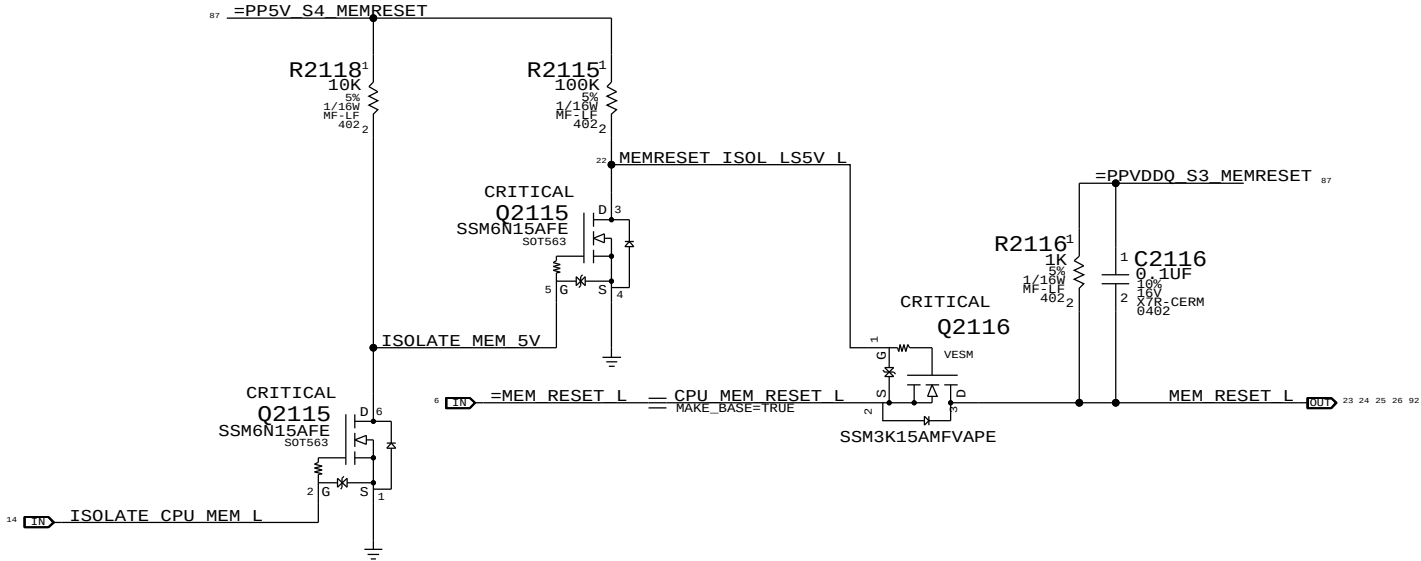
The circuit below handles CPU and VTT power during S0->S3->S0 transitions, as well as isolating the CPU's SM_DRAMRST# output from the S0-DIMMs when necessary.

ISOLATE_CPU_MEM_L GPIO state during S3->S0 transitions determines behavior of signals.

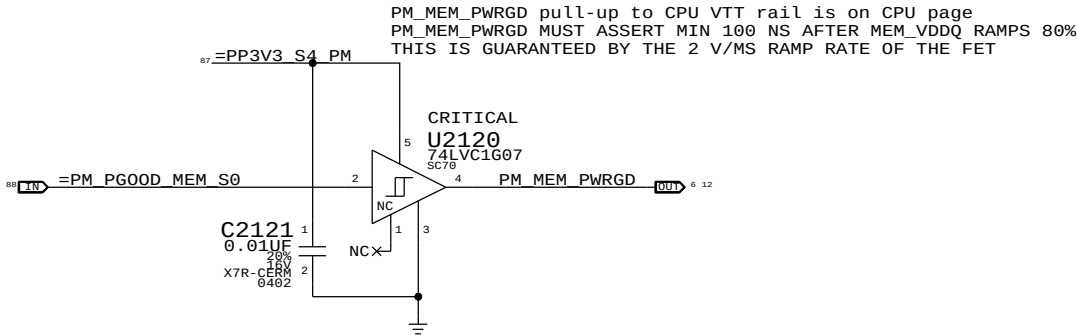
WHEN HIGH: CPU 1.5V remains powered in S3, VTT follows S0 rails, MEM_RESET_L not isolated.

WHEN LOW: CPU 1.5V follows S0 rails, VTT ensures clean CKE transition, MEM_RESET_L isolated.

MEMVTT_EN = PLT_RESET_L * PM_SLP_S3_L
MEM_RESET_L = !ISOLATE_CPU_MEM_L + CPU_MEM_RESET_L

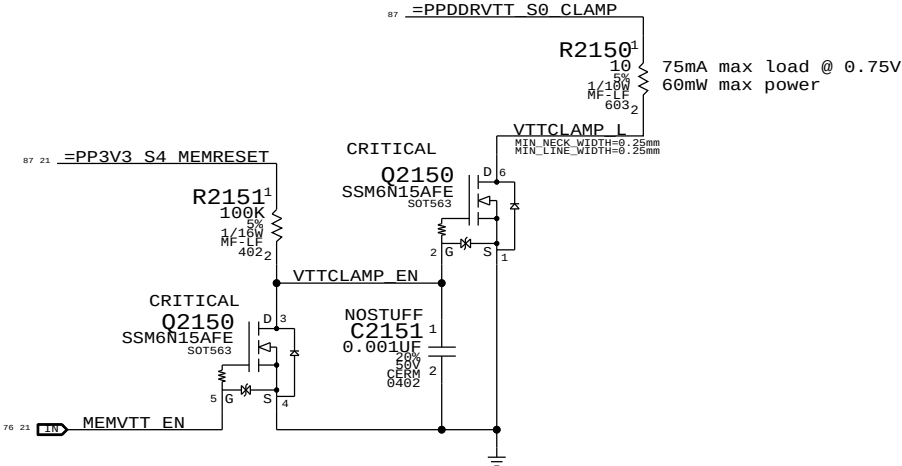
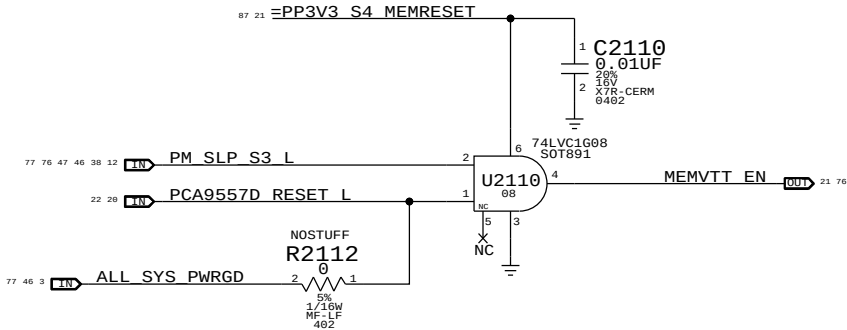


MEM S0 "PGOOD" FOR CPU



MEMVTT Clamp

Ensures CKE signals are held low in S3



Step	ISOLATE_CPU_MEM_L	PLT_RESET_L	PM_SLP_S3_L	CPU_MEM_RESET_L	MEM_RESET_L	MEMVTT_EN
S0	0	1	1	1	CPU_MEM_RESET_L	1
1	1	0	1	1	1	1
2	0	0	1	1	1	0
3	0	0	0	X	1	0
4	0	0	1	X	1	0
5	0	1	1	0 (*)	1	1
6	0	1	1	1	1	1
S0	7	1	1	1	CPU_MEM_RESET_L	1

(*) CPU_MEM_RESET_L asserts due to loss of PM_MEM_PWRGD, must wait for software to clear before deasserting ISOLATE_CPU_MEM_L GPIO.

NOTE: In the event of a S3->S5 transition ISOLATE_CPU_MEM_L will still be asserted on next S5->S0 transition. Rails will power-up as if from S3, but MEM_RESET_L will not properly assert. Software must de-assert ISOLATE_CPU_MEM_L and then generate a valid reset cycle on CPU_MEM_RESET_L.

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Page Notes

Power aliases required by this page:

- =PP3V3_S3_VREFMRGN
- =PPDDR_S3_MEMVREF

Signal aliases required by this page:

- =I2C_VREFDACS_SCL
- =I2C_VREFDACS_SDA
- =I2C_PCA9557D_SCL
- =I2C_PCA9557D_SDA

BOM options provided by this page:

- DDRVREF_DAC - Stuffs DAC margining circuit.

CPU-Based Margining

FETs for CPU isolation during S3

NOTE: CPU DAC output step sizes:
DDR3 (1.5V) 7.70mV per step
DDR3L (1.35V) 6.99mV per step

NOTE: CPU has single output for VREFCA. Split into two signals for independent DAC margining support. When DAC margining VREFCA ensure ISOLATE_CPU_MEM_L is low to remove short due to CPU.

DAC-Based Margining

DAC sets voltage level, PCA9557 & FETs enable outputs and disables margining after platform reset.

OMIT
R2218
SHORT

PP3V3_S3_VREFMRGN

VOLTAGE=3.3V

MIN_LINE_WIDTH=0.3 mm
MIN_NECK_WIDTH=0.2 mm

DDR3 (1.5V)

DDR3L (1.35V)

Addr=0x98(WR)/0x99(RD)

Addr=0x30(WR)/0x31(RD)

PCA9557D_RESET_L

RST* on 'platform reset' so that system watchdog will disable margining.

NOTE: Margining will be disabled across all soft-resets and sleep/wake cycles.

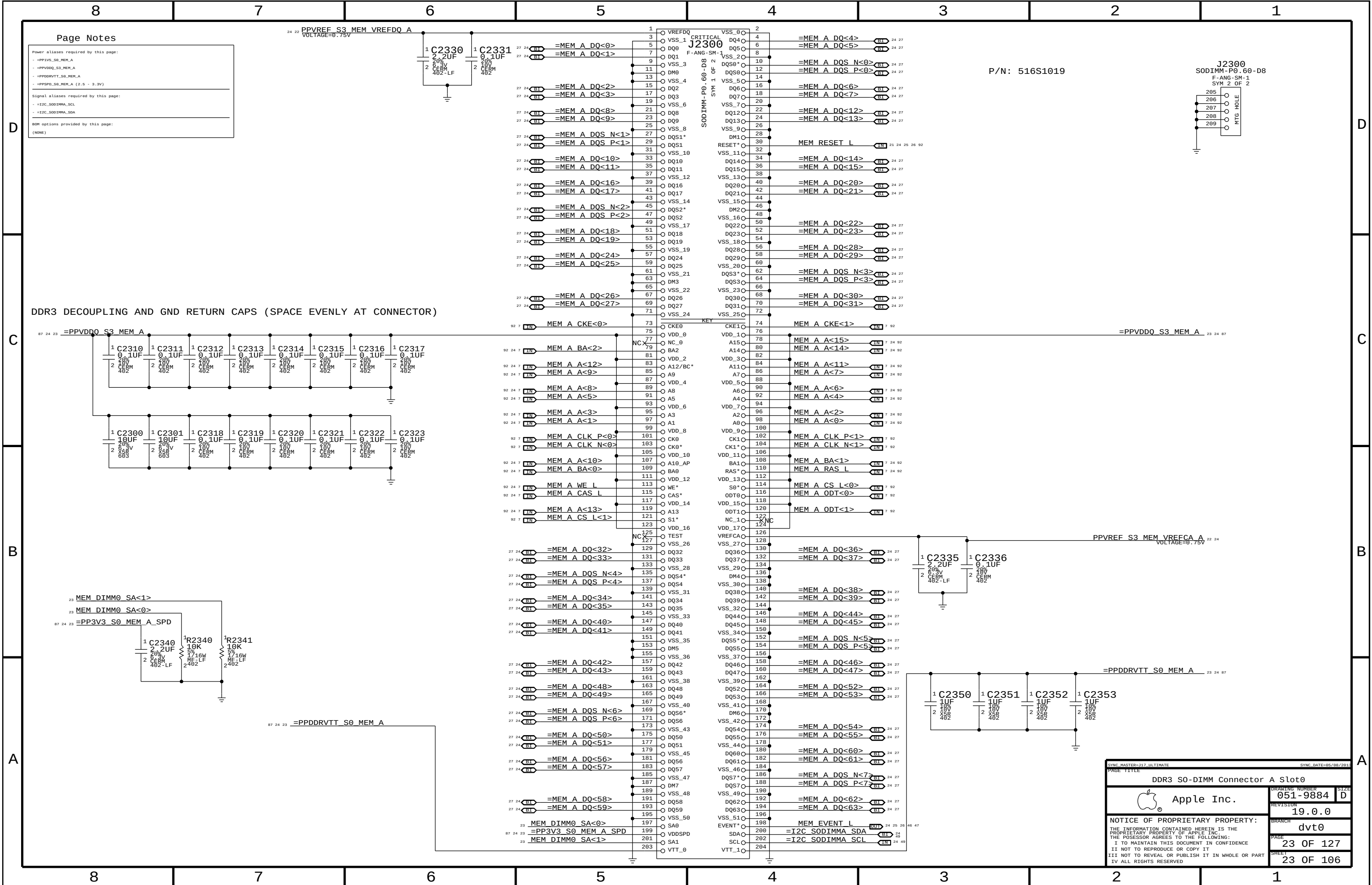
	MEM A VREF DQ	MEM B VREF DQ	MEM A VREF CA	MEM B VREF CA	MEM VREG
DAC Channel:	A	B	C	C	D
PCA9557D Pin:	1	2	3	4	5
	DDR3 (1.5V)		DDR3L (1.35V)		
Nominal value	0.750V (DAC: 0x3A = 0.747mV)		0.675V (DAC: 0x34 = 0.670mV)		1.500V (DAC: 0x74 = 1.495V)
Margined target:	0.300V - 1.200V (+/- 450mV)		0.275V - 1.075V (+/- 400mV)		1.200V - 1.800V (+/- 300mV)
DAC range:	0.000V - 1.508V (0x00 - 0x75)		0.000V - 1.354V (0x00 - 0x69)		0.000V - 3.004V (0x00 - 0xE9)
Margined range:	0.299V - 1.206V (+/- 453mV)		0.269V - 1.083V (+/- 406mV)		1.199V - 1.801V (+/- 301mV)
VRef current:	+901uA - -911uA (- = sourced)		+811uA - -816uA (- = sourced)		+36uA - -36uA (- = sourced)
DAC step size:	7.68mV / step @ output		7.67mV / step @ output		2.575mV / step @ output

NOTE: DDR3 assumes TPS51916 supply with 10.0k/49.9k divider
DDR3L assumes TPS51916 supply with 19.6k/57.6k divider

VRef Dividers

Always used, regardless of margining option.

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PAGE TITLE		DDR3 VREF MARGINING	
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Page Notes

Power aliases required by this page:

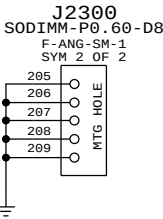
- =PP3V3_S0_MEM_A
- =PPVDDQ_S3_MEM_A
- =PPDDRVT_S0_MEM_A
- =PPSPD_S0_MEM_A (2.5 - 3.3V)

Signal aliases required by this page:

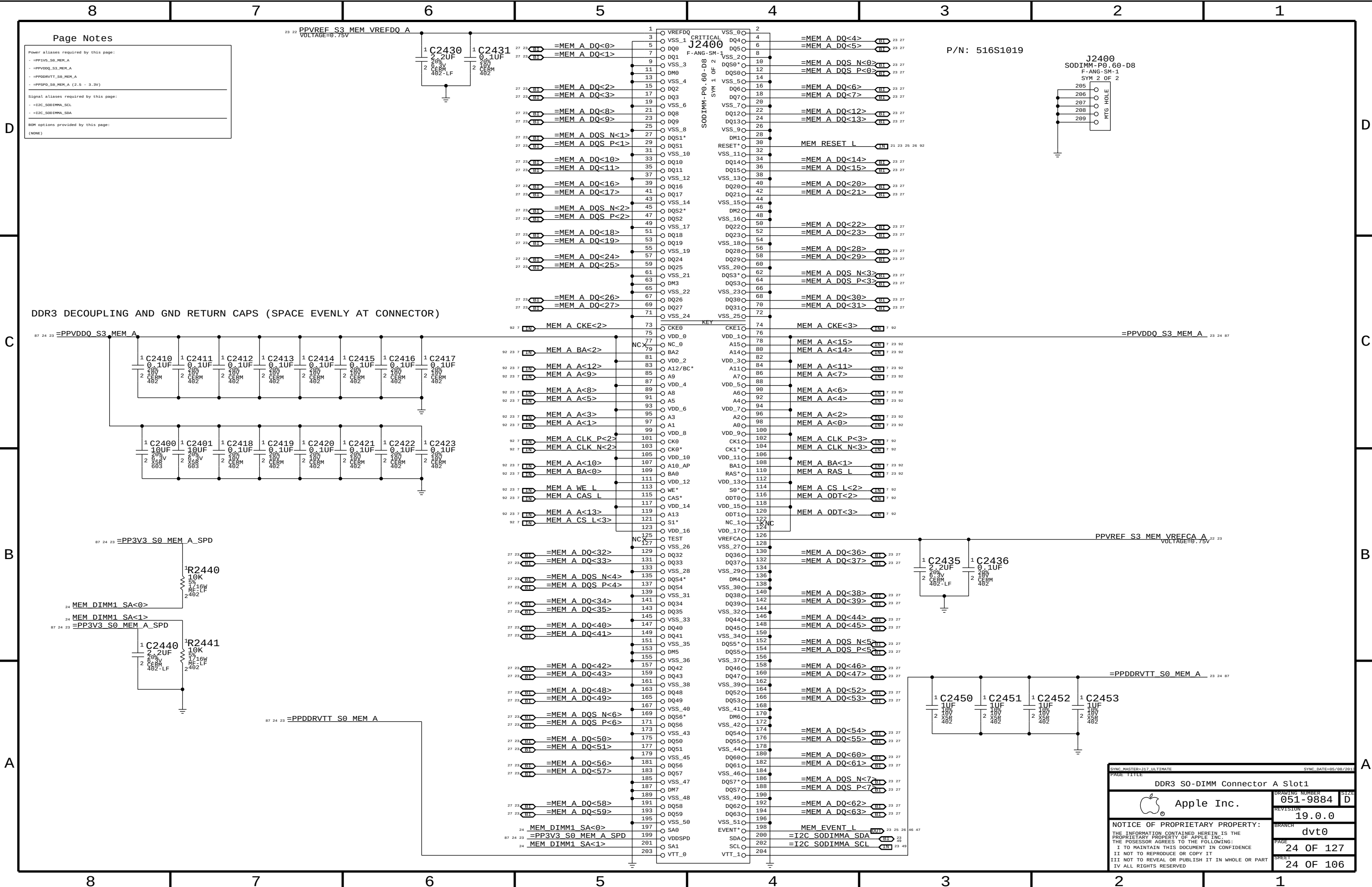
- =I2C_SODIMMA_SCL
- =I2C_SODIMMA_SDA

BOM options provided by this page:

(NONE)



SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/09/2011	
PAGE TITLE			
DDR3 S0-DIMM Connector A Slot0			
 Apple Inc.		DRAWING NUMBER	051-9884
		SIZE	D
		REVISION	19.0.0
		BRANCH	dvt0
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Page Notes

Power aliases required by this page:

- =PP3V3_S0_MEM_A
- =PPVDDQ_S3_MEM_A
- =PPDDRVT_S0_MEM_A
- =PPSPD_S0_MEM_A (2.5 - 3.3V)

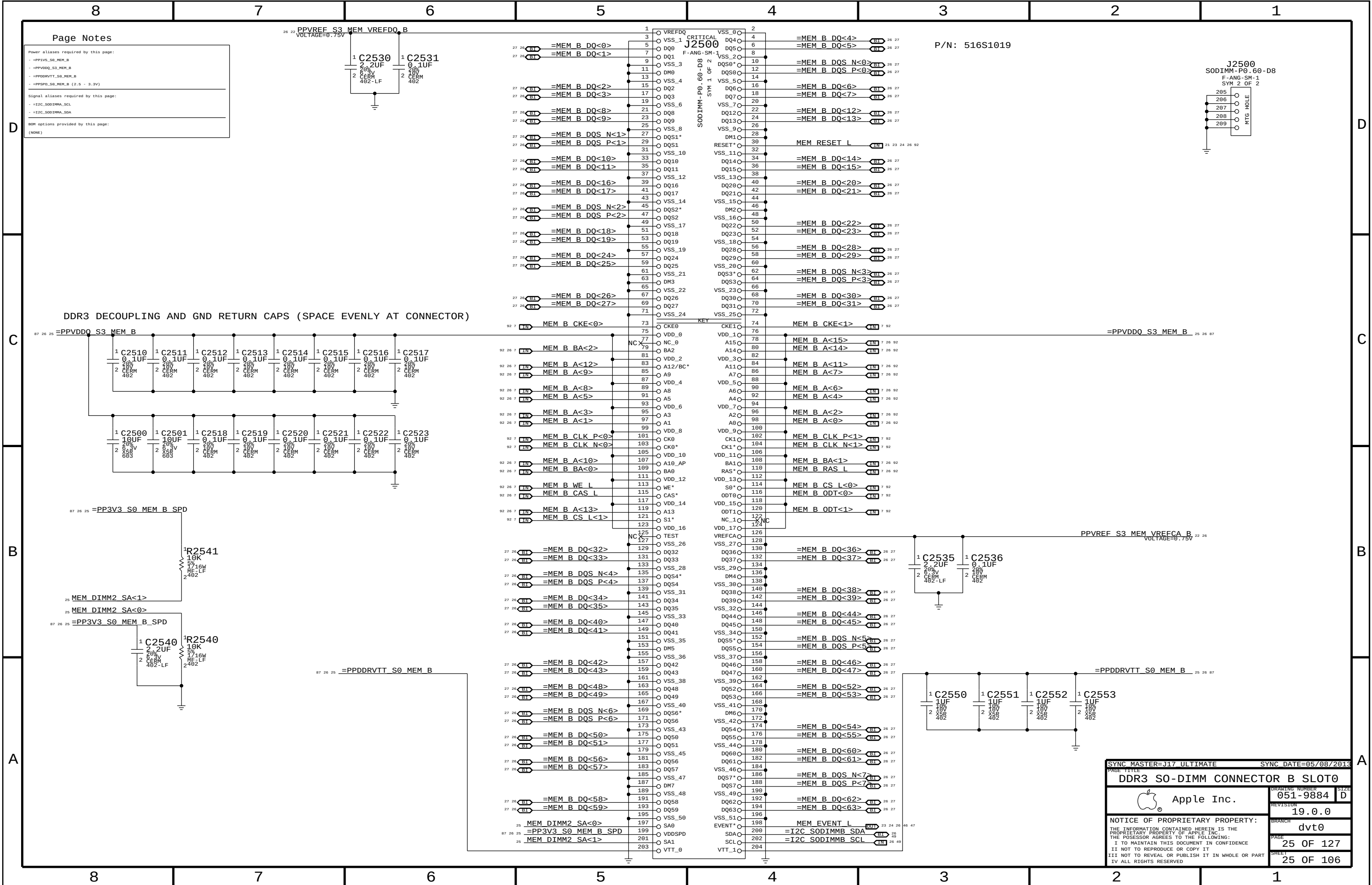
Signal aliases required by this page:

- =I2C_SODIMMA_SCL
- =I2C_SODIMMA_SDA

BOM options provided by this page:

(NONE)

SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/08/2011	
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DDR3 S0-DIMM Connector A Slot1			
	DRAWING NUMBER	051-9884	SIZE D
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PAGE	24	OF	127
SHEET	24	OF	106



Page Notes

Power aliases required by this page:

- =PP1V5_S0_MEM_B
- =PPVDDQ_S3_MEM_B
- =PPDDRVT_S0_MEM_B
- =PPSPD_S0_MEM_B (2.5 - 3.3V)

Signal aliases required by this page:

- =I2C_S0DIMM_SCL
- =I2C_S0DIMM_SDA

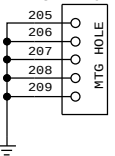
BOM options provided by this page:

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P/N: 516S1019

J2500
SODIMM-P0.60-D8

F-ANG-SM-1
SYM 2 OF 2



SYNC MASTER=J17 ULTIMATE SYNC DATE=05/08/2013

PAGE TITLE

DDR3 SO-DIMM CONNECTOR B SLOT0



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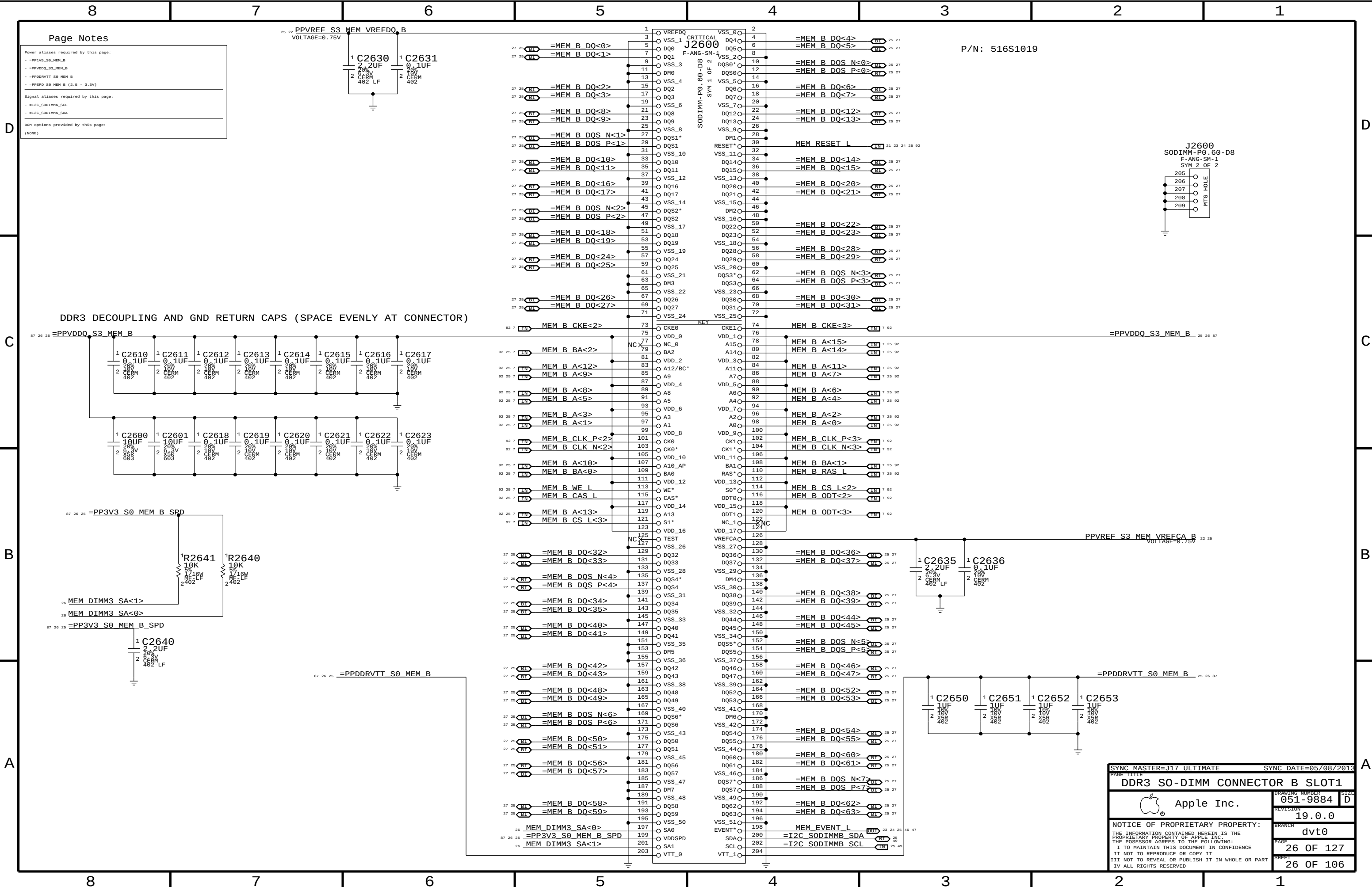
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PAGE

25 OF 127

SHEET

25 OF 106



Page Notes

Power aliases required by this page:

- =PP1V5_S0_MEM_B
- =PPVDDQ_S3_MEM_B
- =PPDDRVT_S0_MEM_B
- =PPSPD_S0_MEM_B (2.5 - 3.3V)

Signal aliases required by this page:

- =I2C_S0DIMM_SCL
- =I2C_S0DIMM_SDA

BOM options provided by this page:

(NONE)

P/N: 516S1019

J2600
SODIMM-P0.60-D8
F-ANG-SM-1
SYM 2 OF 2

SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/08/2013	
PAGE TITLE		DDR3 SO-DIMM CONNECTOR B SLOT1	
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		SHEET	26 OF 106
		SIZE	D

8	7	6	5	4	3	2	1	
THERE ARE NO PIN SWAPS								
D	MEM A DQS N<0>	==	=MEM A DQS N<0>	23	24	MEM B DQS N<0>	23	24
	MEM A DQS P<0>	==	=MEM A DQS P<0>	23	24	MEM B DQS P<0>	23	24
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	MEM A DQ<5>	==	=MEM A DQ<5>	23	24	MEM B DQ<5>	23	24
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	MEM A DQ<3>	==	=MEM A DQ<3>	23	24	MEM B DQ<3>	23	24
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	MEM A DQ<1>	==	=MEM A DQ<1>	23	24	MEM B DQ<1>	23	24
	MEM A DQ<0>	==	=MEM A DQ<0>	23	24	MEM B DQ<0>	23	24
C	MEM A DQS N<1>	==	=MEM A DQS N<1>	23	24	MEM B DQS N<1>	23	24
	MEM A DQS P<1>	==	=MEM A DQS P<1>	23	24	MEM B DQS P<1>	23	24
	MEM A DQ<15>	==	=MEM A DQ<15>	23	24	MEM B DQ<15>	23	24
	MEM A DQ<14>	==	=MEM A DQ<14>	23	24	MEM B DQ<14>	23	24
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	MEM A DQ<8>	==	=MEM A DQ<8>	23	24	MEM B DQ<8>	23	24
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	MEM A DQS P<7>	==	=MEM A DQS P<7>	23	24	MEM B DQS P<7>	23	24
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	MEM A DQ<60>	==	=MEM A DQ<60>	23	24	MEM B DQ<60>	23	24
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SYNC MASTER=J17 ULTIMATE

SYNC DATE=05/08/2013

DDR3 ALIASES AND BITSWAPS

 Apple Inc.

DRAWING NUMBER
051-9884

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D

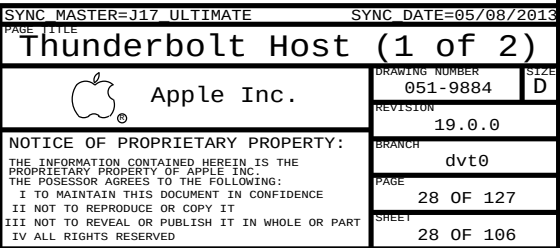
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BRANCH
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PAGE
27 OF 127

SHEET
27 OF 106



Page Notes

Power aliases required by this page:

- =PPVIN_SW_TBTBST (8-13V Boost Input)
- =PP15V_TBT_REG (15V Boost Output)
- =PP3V3_TBT_P3V3TBTFTET (3.3V FET Input)
- =PP3V3_TBT_FET (3.3V FET Output)
- =PP3V3_S0_TBTTPWRCTL
- =PP1V05_TBT_P1V05TBTFTET (1.05V FET Input)
- =PP1V05_TBT_FET (1.05V FET Output)

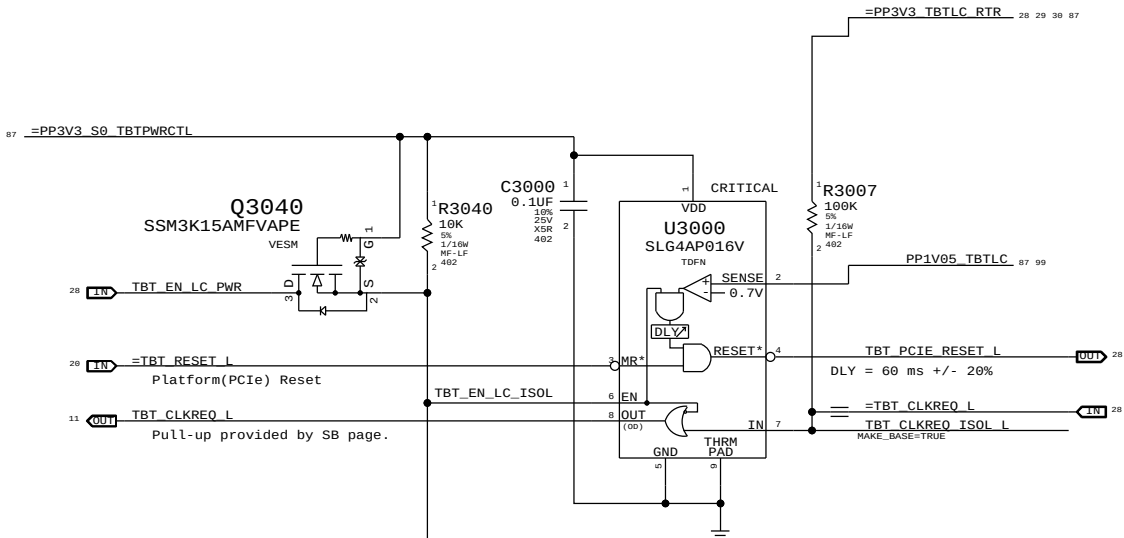
Signal aliases required by this page:

- =TBT_CLKREQ_L
- =TBT_RESET_L

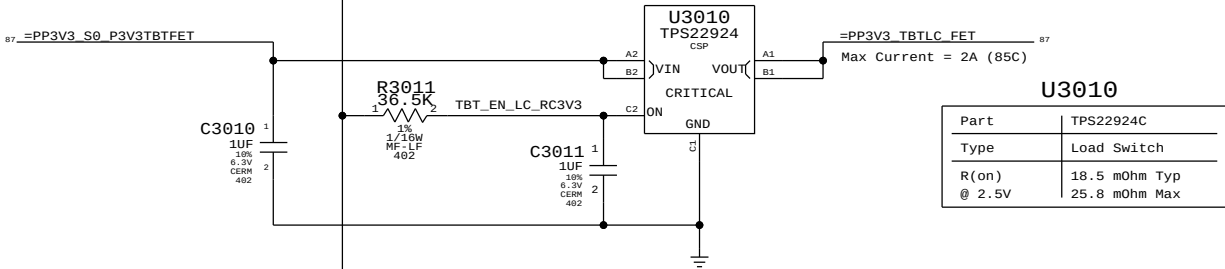
BOM options provided by this page:

TBTBST:Y - Stuffs 15V boost circuitry.

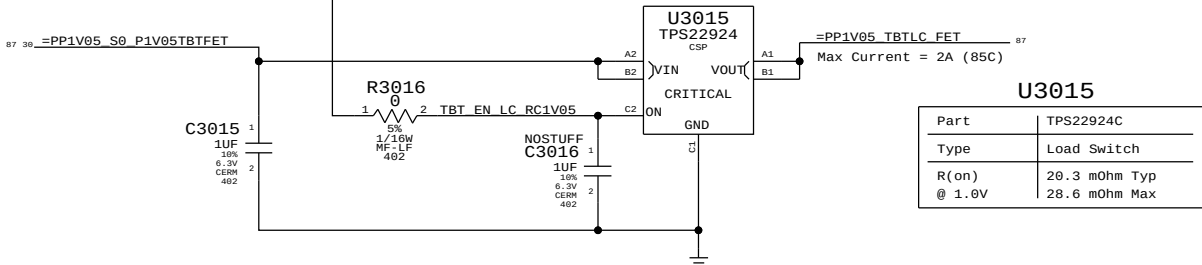
Supervisor & CLKREQ# Isolation



3.3V TBT "LC" Switch

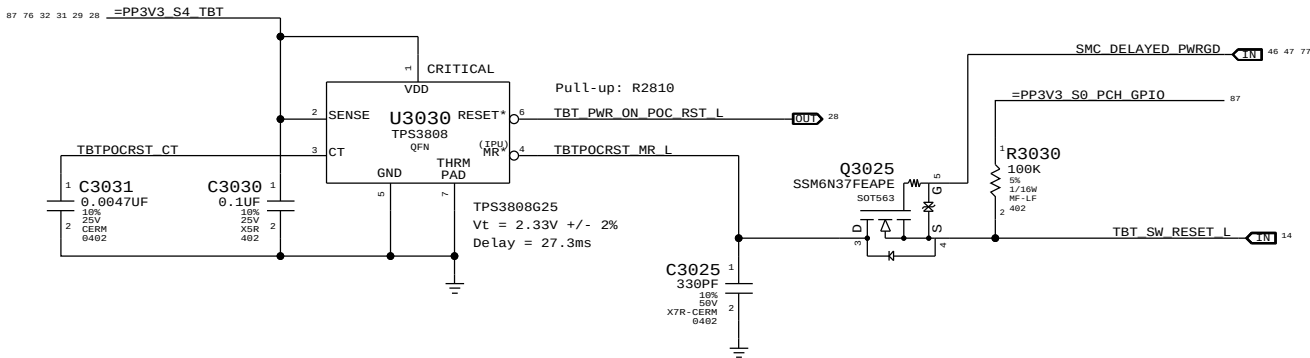


1.05V TBT "LC" Switch

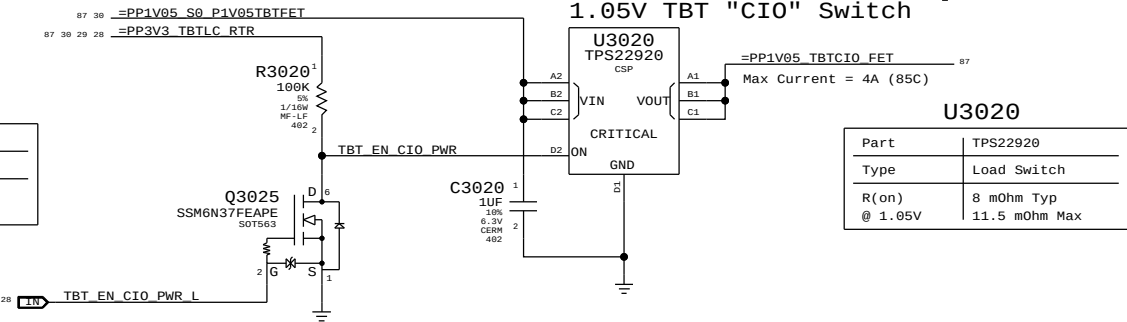


TBT "POC" Power-up Reset

Intel investigating whether RC is sufficient.



1.05V TBT "CIO" Switch



SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/08/2013	
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8	7	6	5	4	3	2	1
---	---	---	---	---	---	---	---



PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
311S0596	311S0593		ALL	TI Alternate
128S0398	128S0220		ALL	3.3V INPUT CAP



Thunderbolt Connector A

B

DP Source must pull down HPD input with greater than or equal to 100K (DPv1.1a).

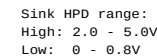
Sink HPD range:
High: 2.0 - 5.0V
Low: 0 - 0.8V

B

8	7	6	5	4	3	2	1
---	---	---	---	---	---	---	---

CB

A

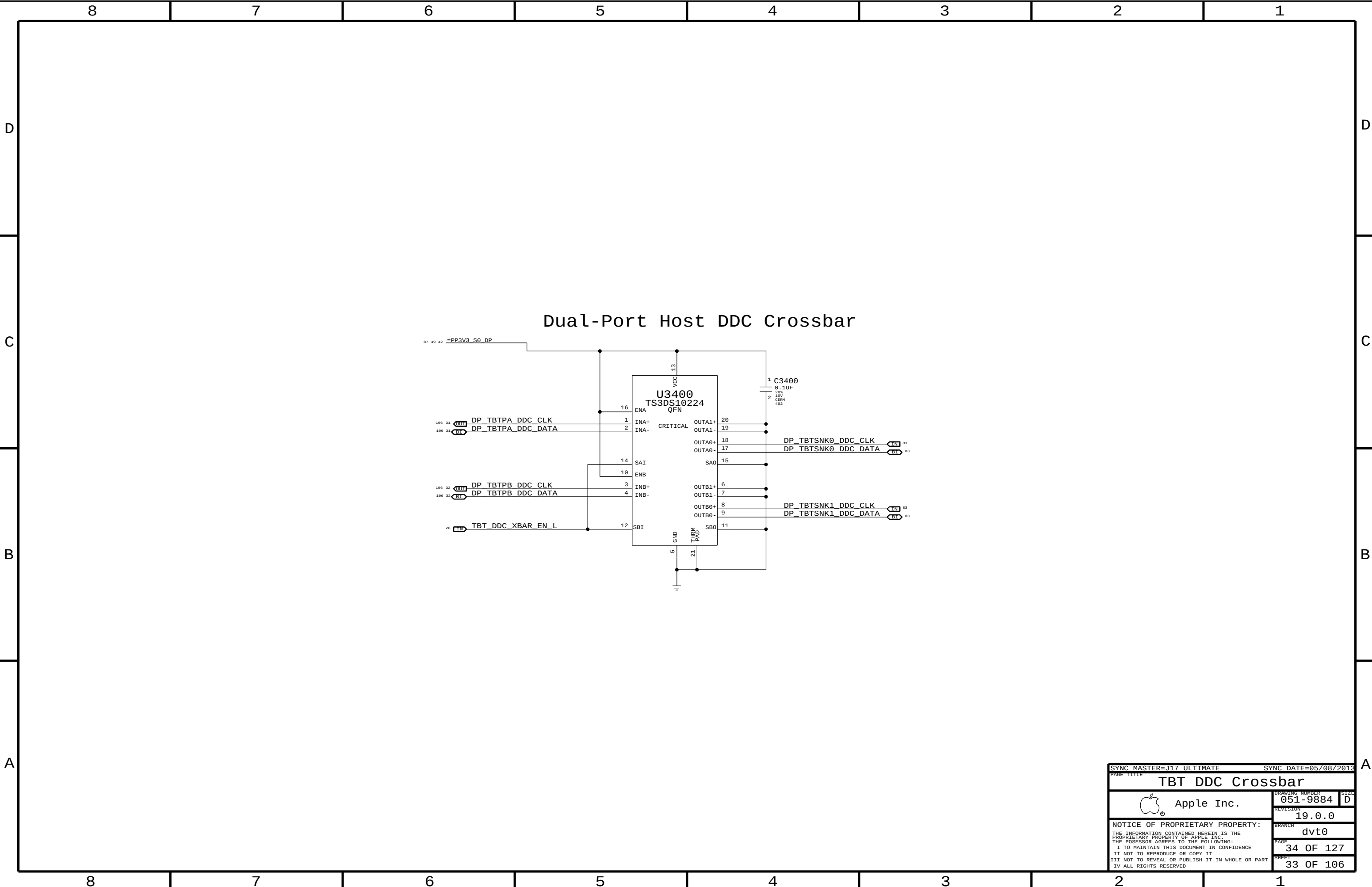


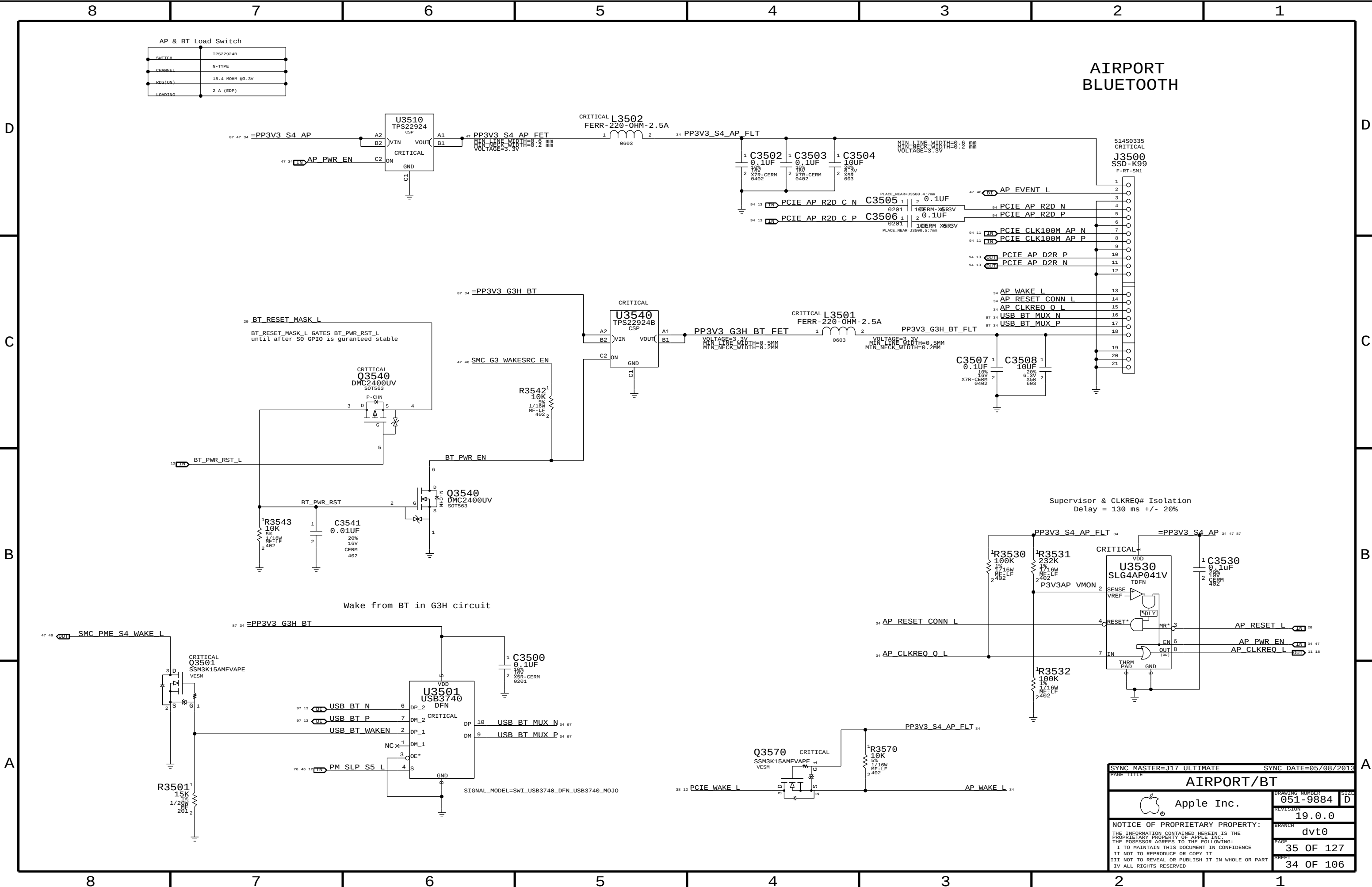
C



A

SYNCH MASTER=J17 ULTIMATE		SYNCH DATE=05/08/2013	
PAGE TITLE			
Thunderbolt Connector B			
 Apple Inc.		DRAWING NUMBER 051-9884	
		SIZE D	
		REVISION 19.0.0	
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		SHEET 32 OF 106	

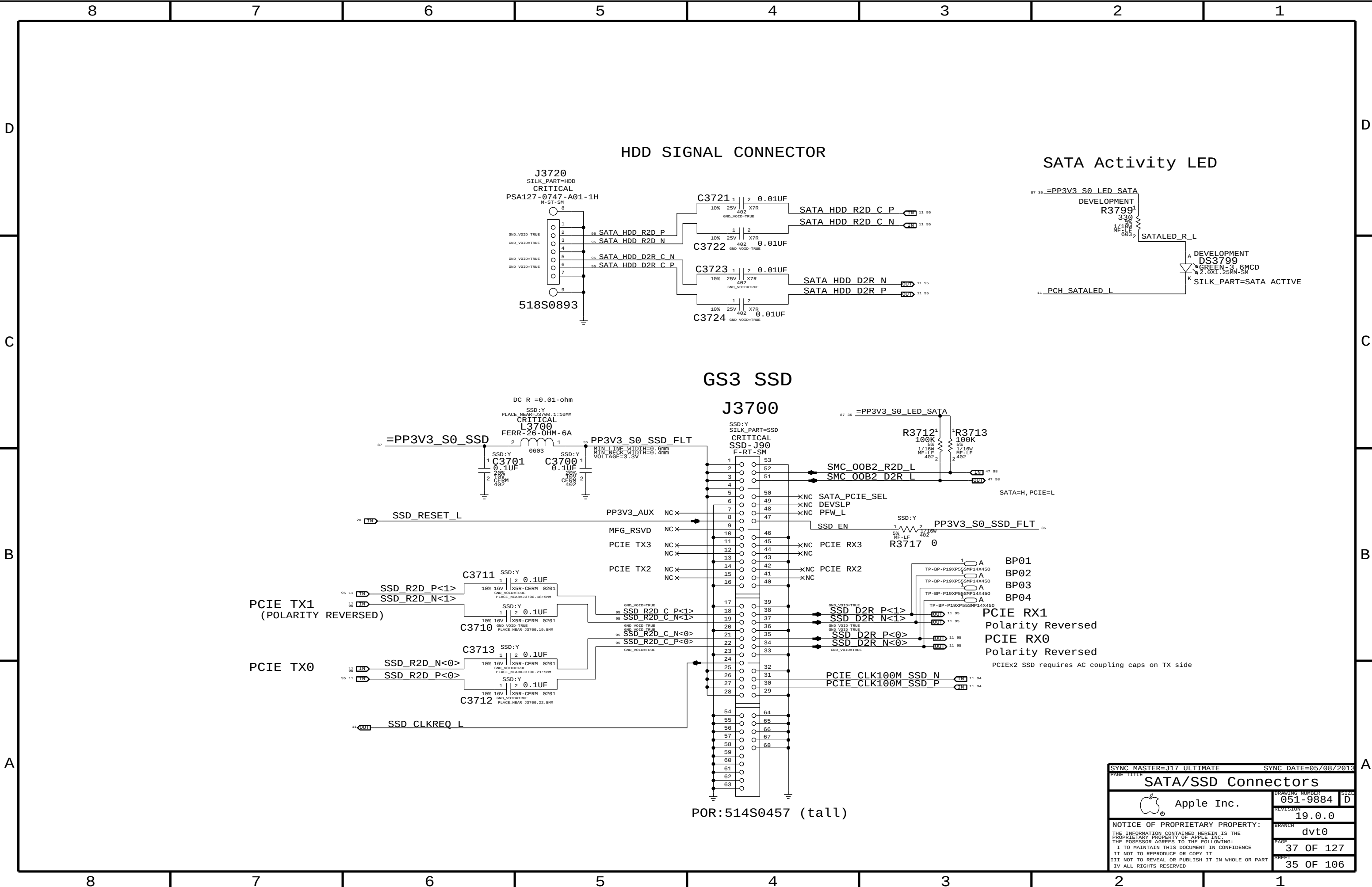




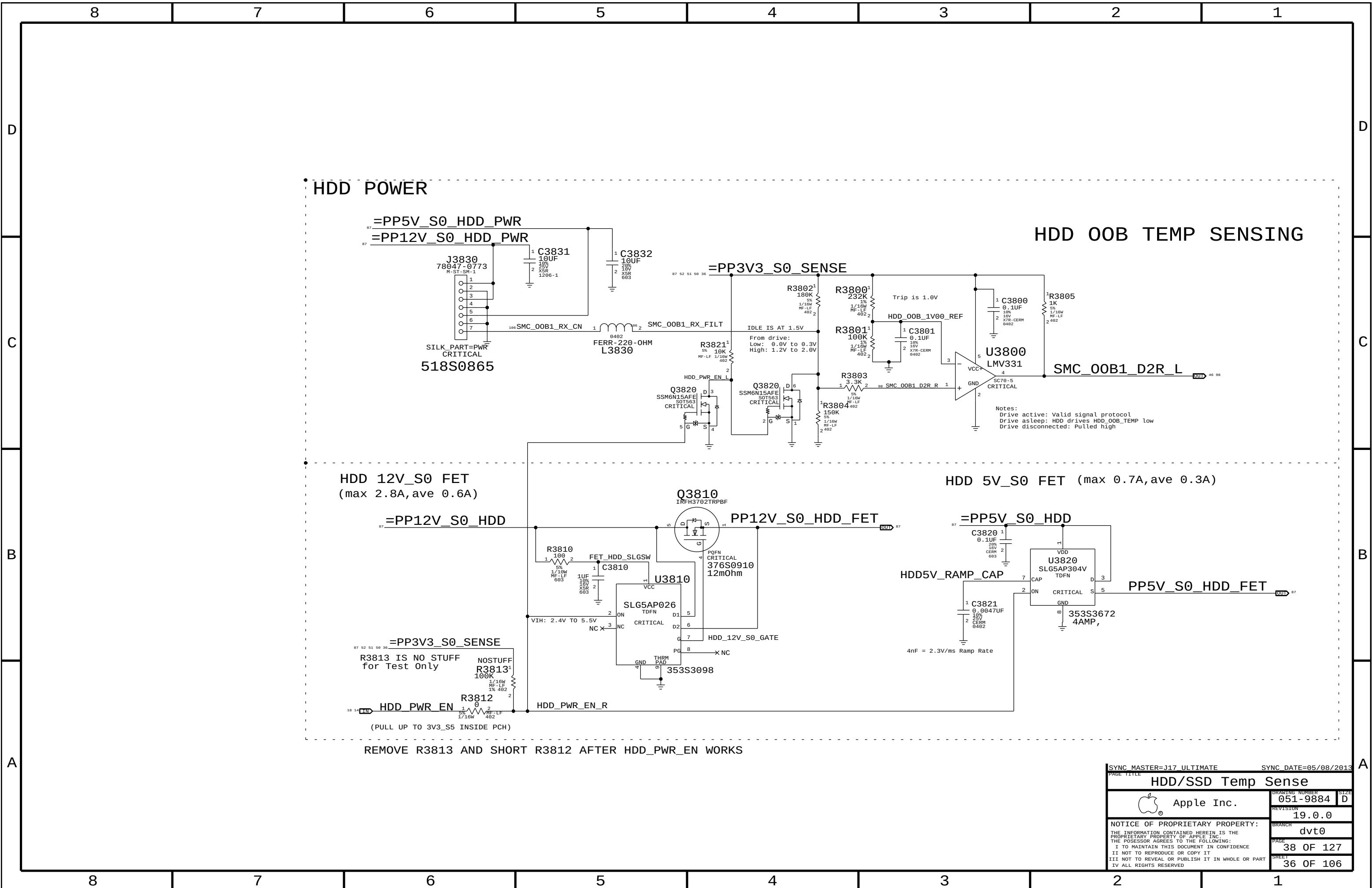
AIRPORT BLUETOOTH

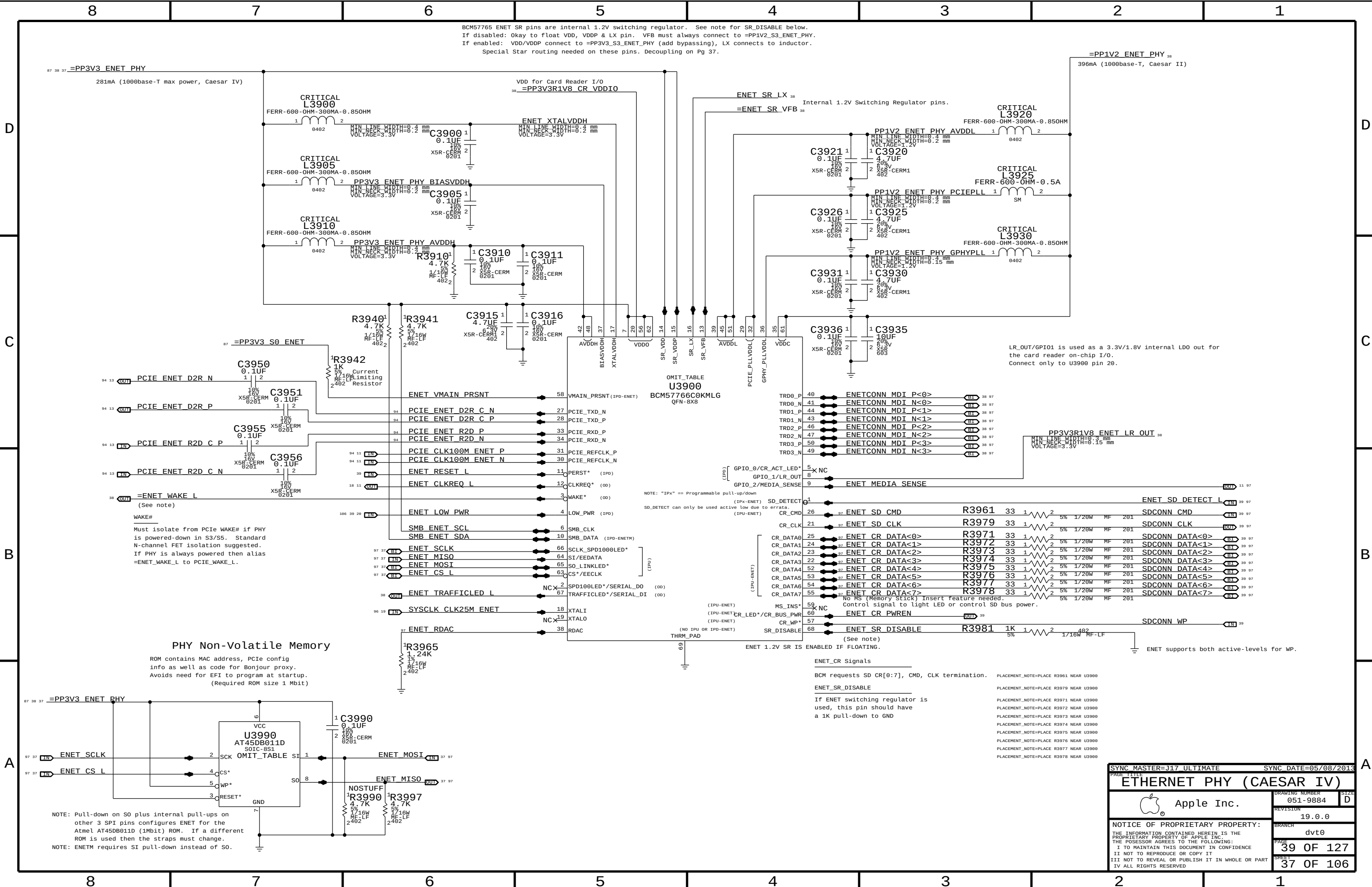
Supervisor & CLKREQ# Isolation
Delay = 130 ms +/- 20%

SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/08/2013	
PAGE TITLE			
AIRPORT/BT			
		DRAWING NUMBER	051-9884
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		BRANCH	dvt0
		PAGE	35 OF 127
		SHEET	34 OF 106



SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/08/2013	
PAGE TITLE			
SATA/SSD Connectors			
 Apple Inc.	DRAWING NUMBER		051-9884
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BCM57765 ENET SR pins are internal 1.2V switching regulator. See note for SR_DISABLE below.
If disabled: Okay to float VDD, VDDP & LX pin. VFB must always connect to =PP1V2_S3_ENET_PHY.
If enabled: VDD/VDDP connect to =PP3V3_S3_ENET_PHY (add bypassing), LX connects to inductor.
Special Star routing needed on these pins. Decoupling on Pg 37.

=PP1V2 ENET PHY 38
396mA (1000base-T, Caesar II)

87 38 37 =PP3V3 ENET PHY
281mA (1000base-T max power, Caesar IV)

ENET SR LX 38
=ENET SR VFB 38
Internal 1.2V Switching Regulator pins.

LR_OUT/GPIO1 is used as a 3.3V/1.8V internal LDO out for the card reader on-chip I/O.
Connect only to U3900 pin 20.

WAKE#
Must isolate from PCIe WAKE# if PHY is powered-down in S3/S5. Standard N-channel FET isolation suggested.
If PHY is always powered then alias =ENET_WAKE_L to PCIe_WAKE_L.

PHY Non-Volatile Memory

ROM contains MAC address, PCIe config info as well as code for Bonjour proxy. Avoids need for EFI to program at startup. (Required ROM size 1 Mbit)

ENET_CR Signals

BCM requests SD CR[0:7], CMD, CLK termination.

ENET_SR_DISABLE

If ENET switching regulator is used, this pin should have a 1K pull-down to GND

PLACEMENT_NOTE=PLACE R3961 NEAR U3900

PLACEMENT_NOTE=PLACE R3979 NEAR U3900

PLACEMENT_NOTE=PLACE R3971 NEAR U3900

PLACEMENT_NOTE=PLACE R3972 NEAR U3900

PLACEMENT_NOTE=PLACE R3973 NEAR U3900

PLACEMENT_NOTE=PLACE R3974 NEAR U3900

PLACEMENT_NOTE=PLACE R3975 NEAR U3900

PLACEMENT_NOTE=PLACE R3976 NEAR U3900

PLACEMENT_NOTE=PLACE R3977 NEAR U3900

PLACEMENT_NOTE=PLACE R3978 NEAR U3900

SYNC MASTER=J17 ULTIMATE

SYNC DATE=05/08/2013

ETHERNET PHY (CAESAR IV)

Apple Inc.

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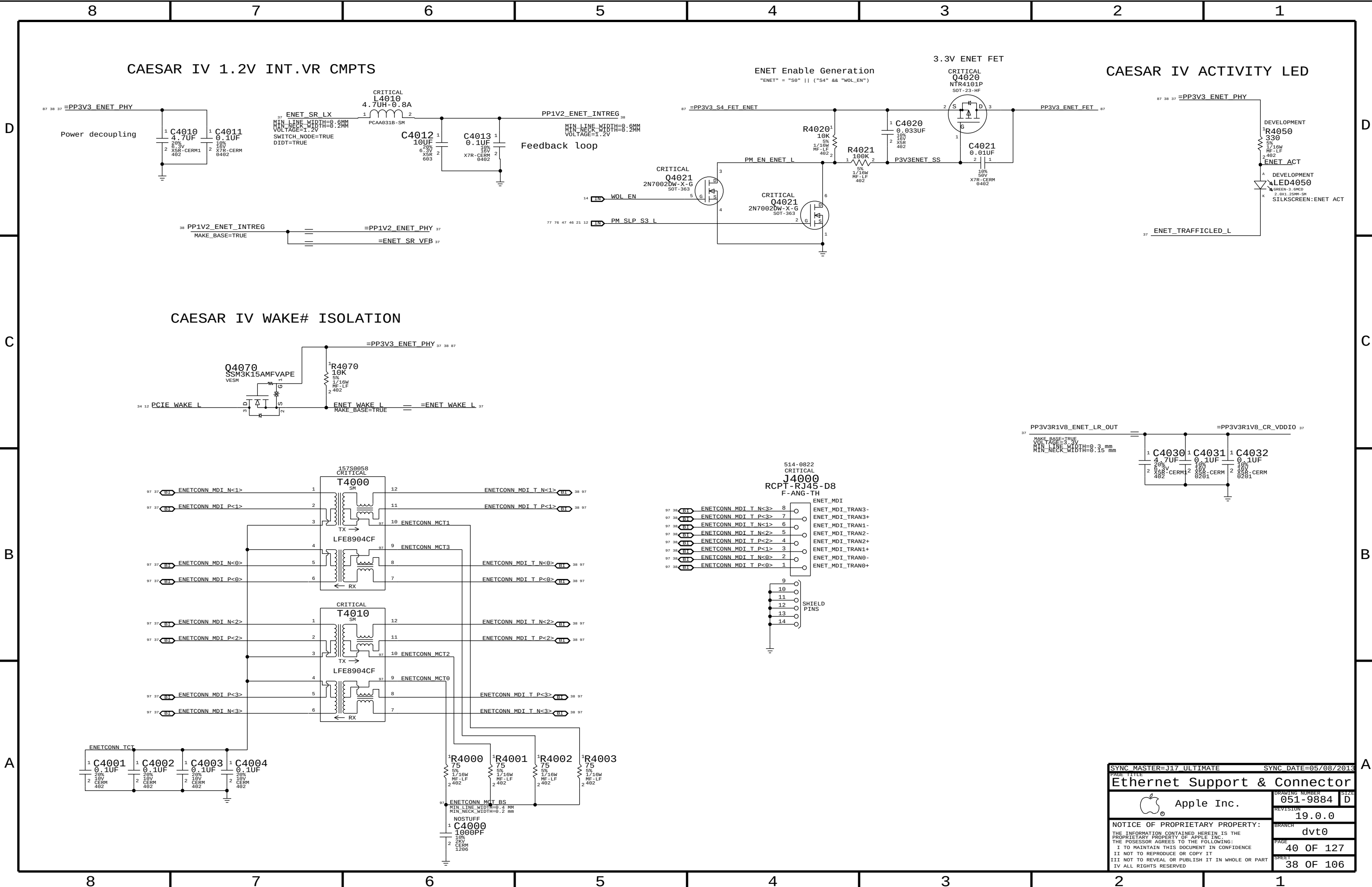
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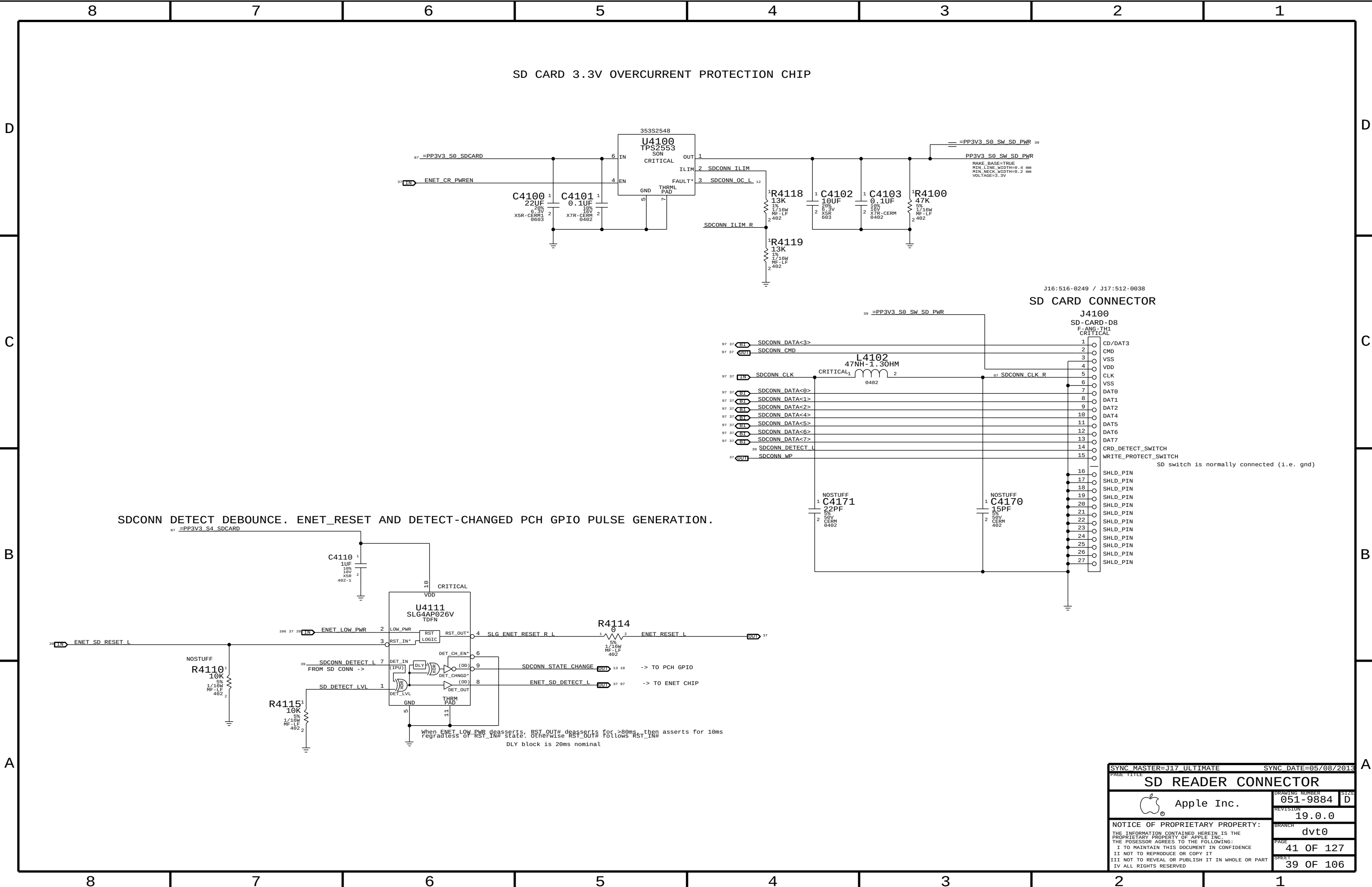
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39 OF 127

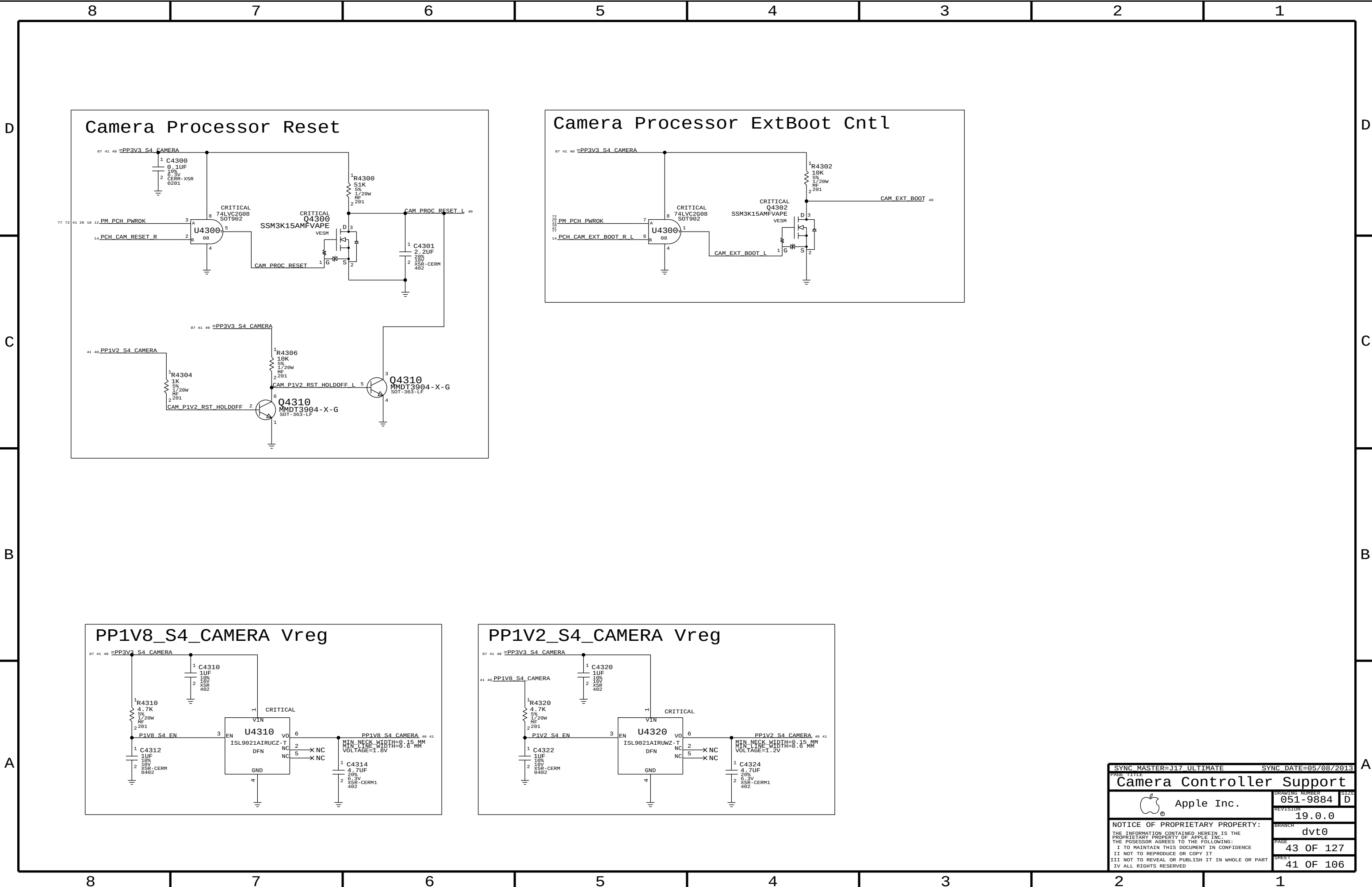
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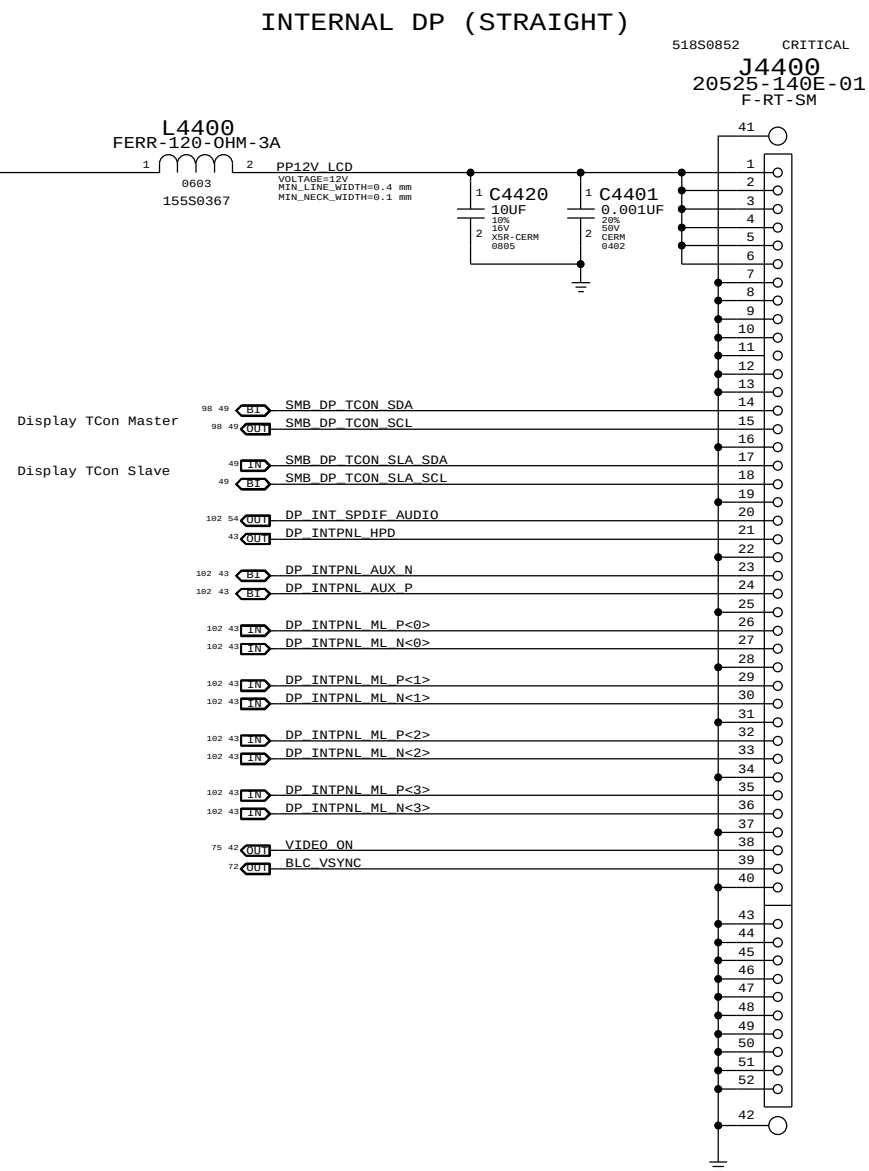
37 OF 106





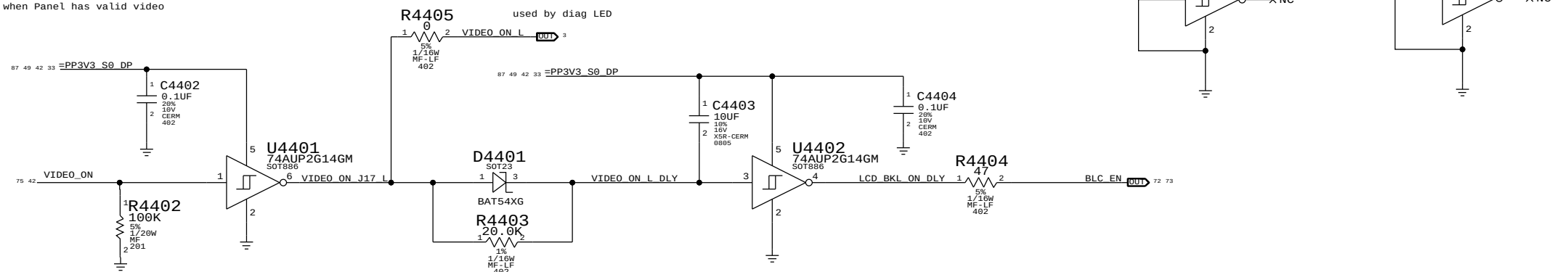
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SD READER CONNECTOR		SD READER CONNECTOR	
Apple Inc.		DRAWING NUMBER	051-9884
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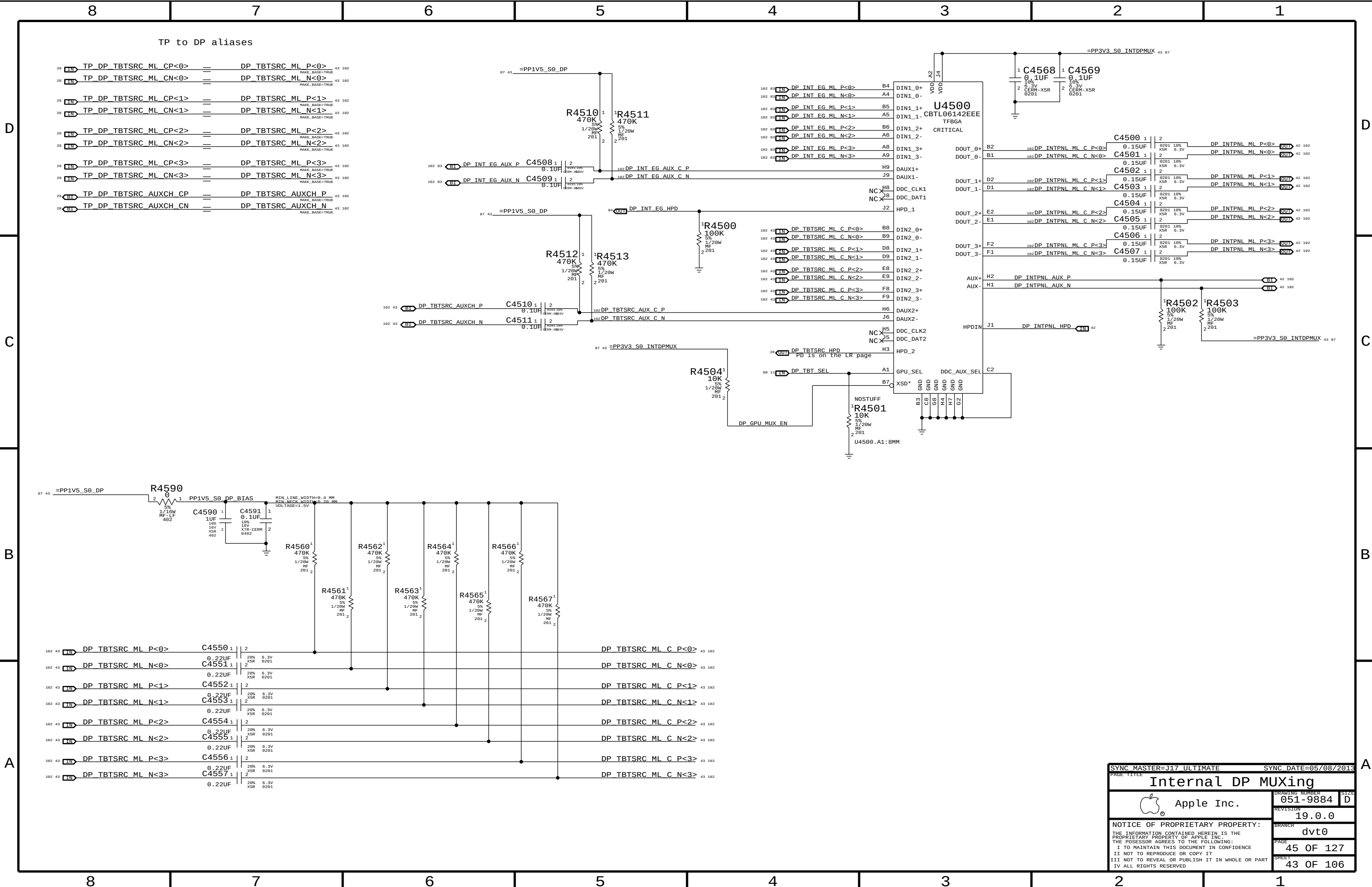


BACKLIGHT CONTROL SUPPORT

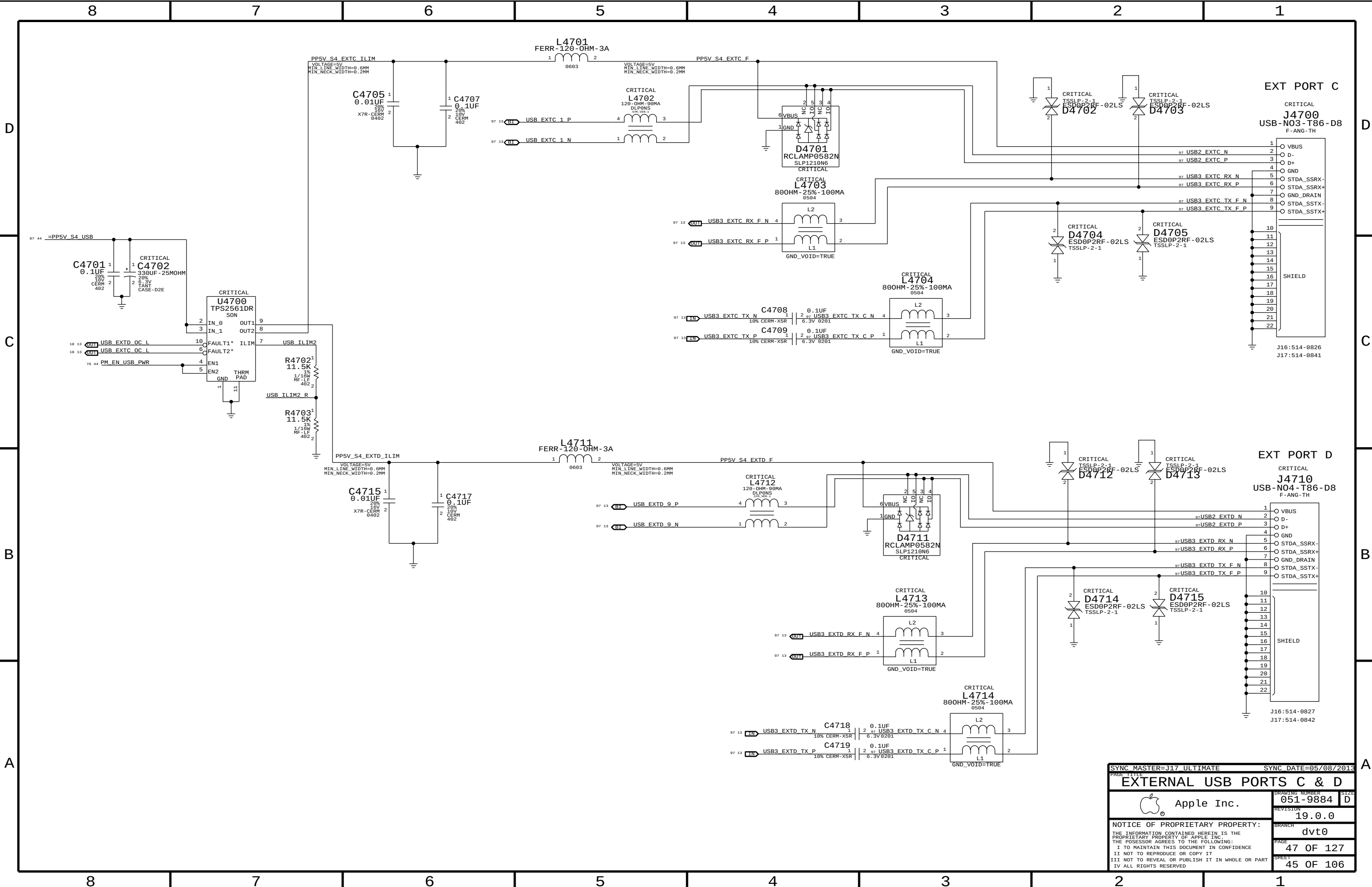
guarantee backlight is
only on when Panel has valid video



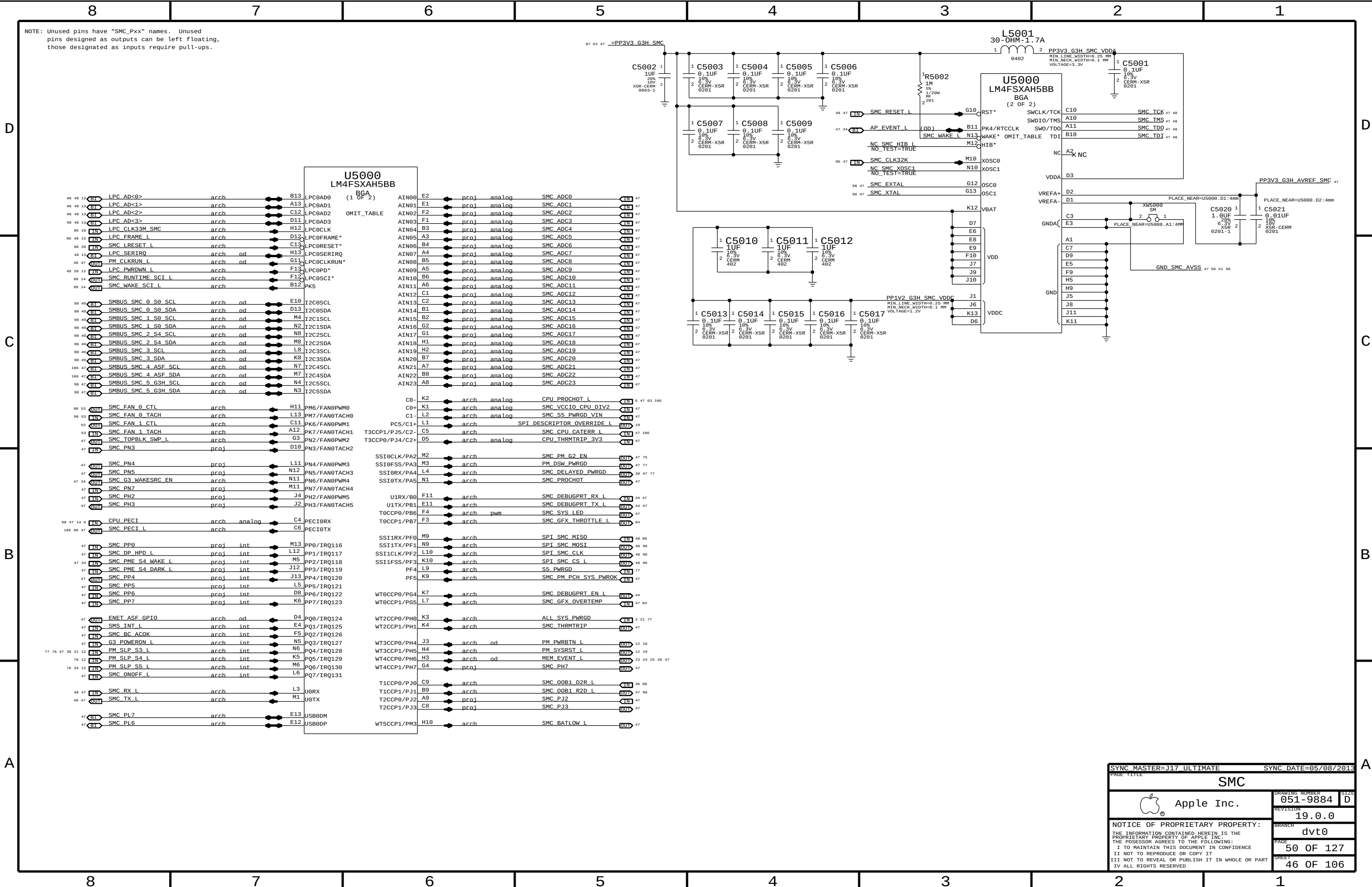
SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/08/2013	
PAGE TITLE			
Internal DP Support			
 Apple Inc.		DRAWING NUMBER 051-9884	
		SIZE D	
		REVISION 19.0.0	
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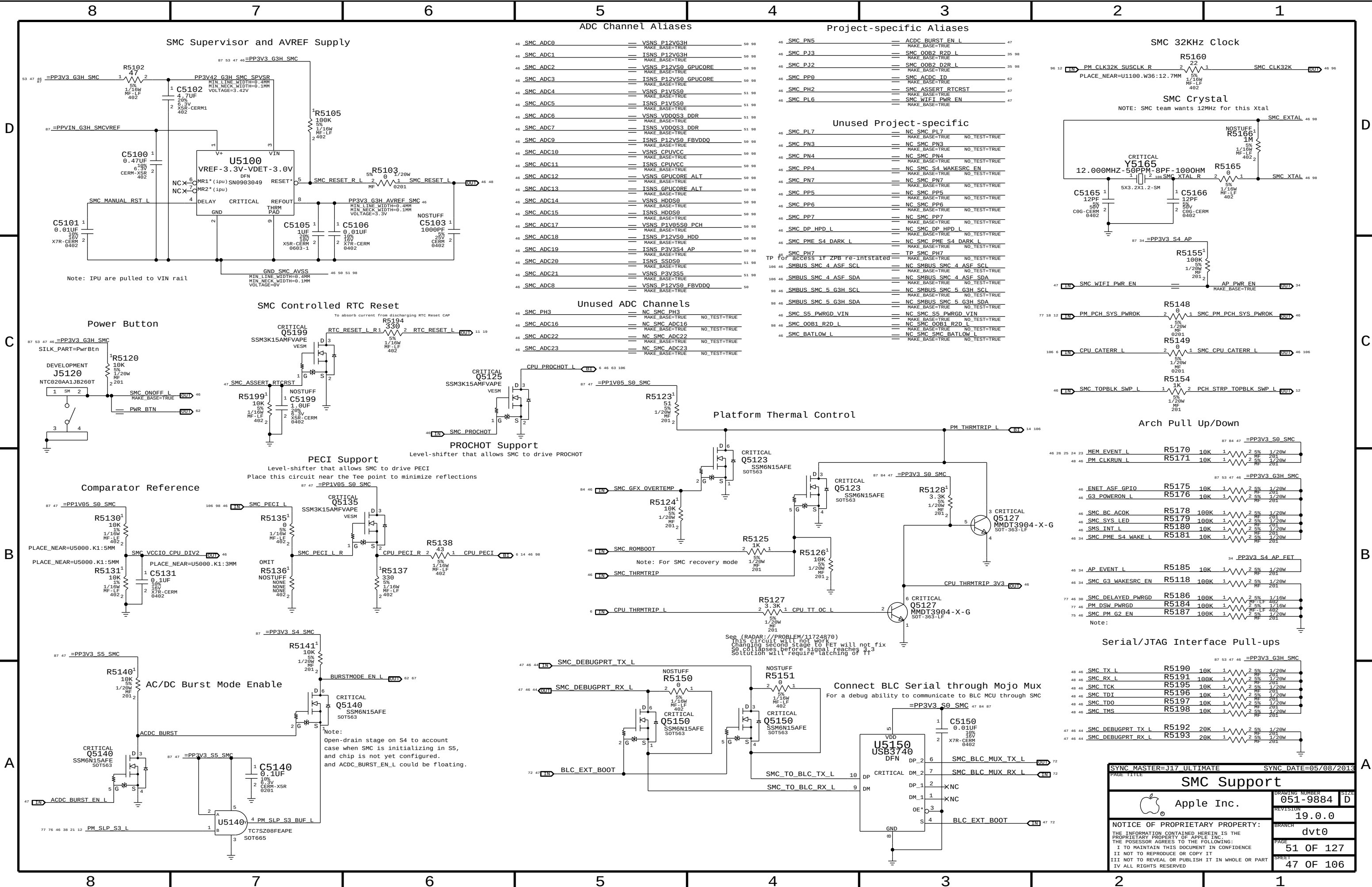


SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/08/2013	
PAGE TITLE		Internal DP MUXing	
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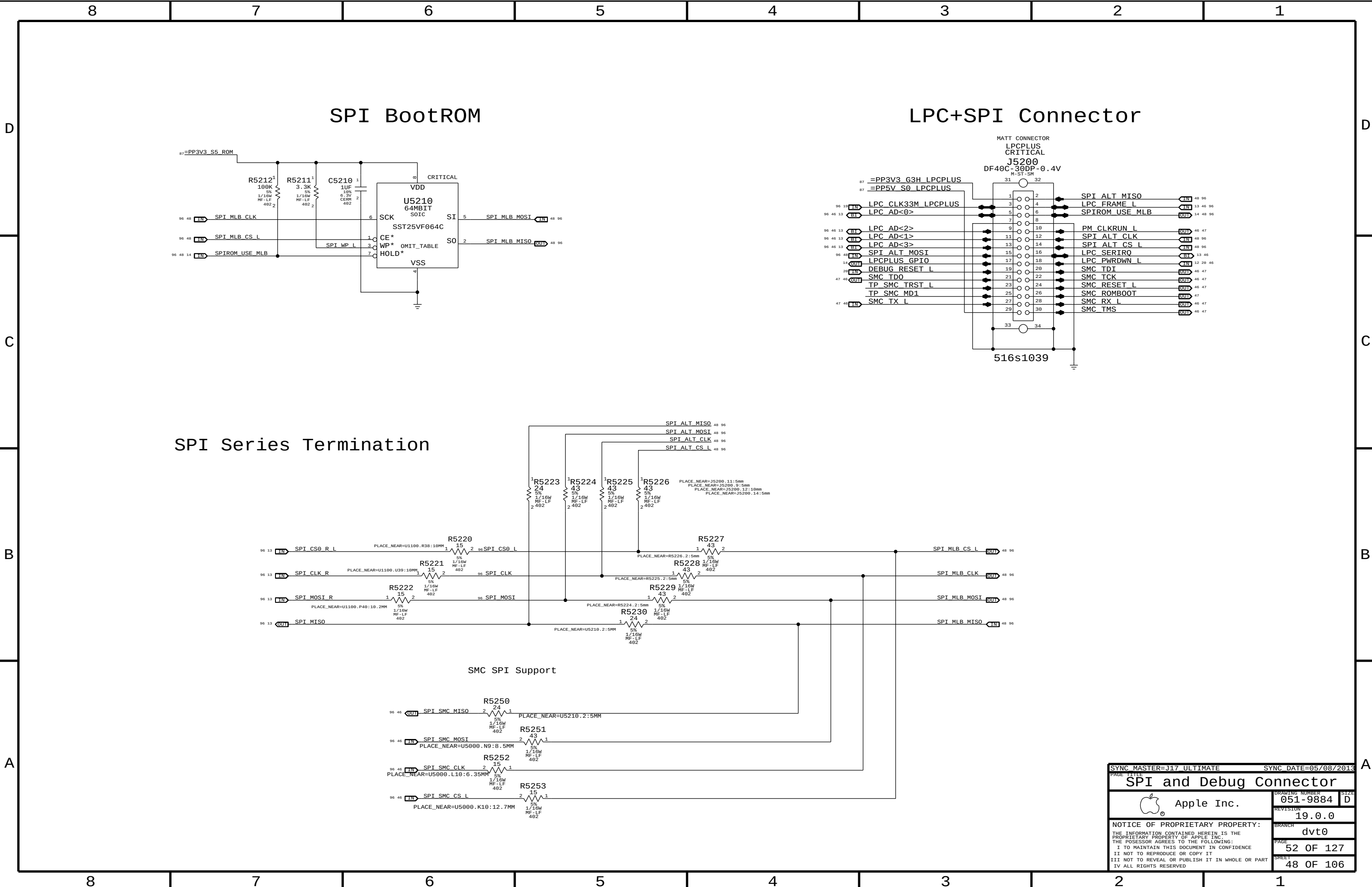


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PAGE TITLE		EXTERNAL USB PORTS C & D	
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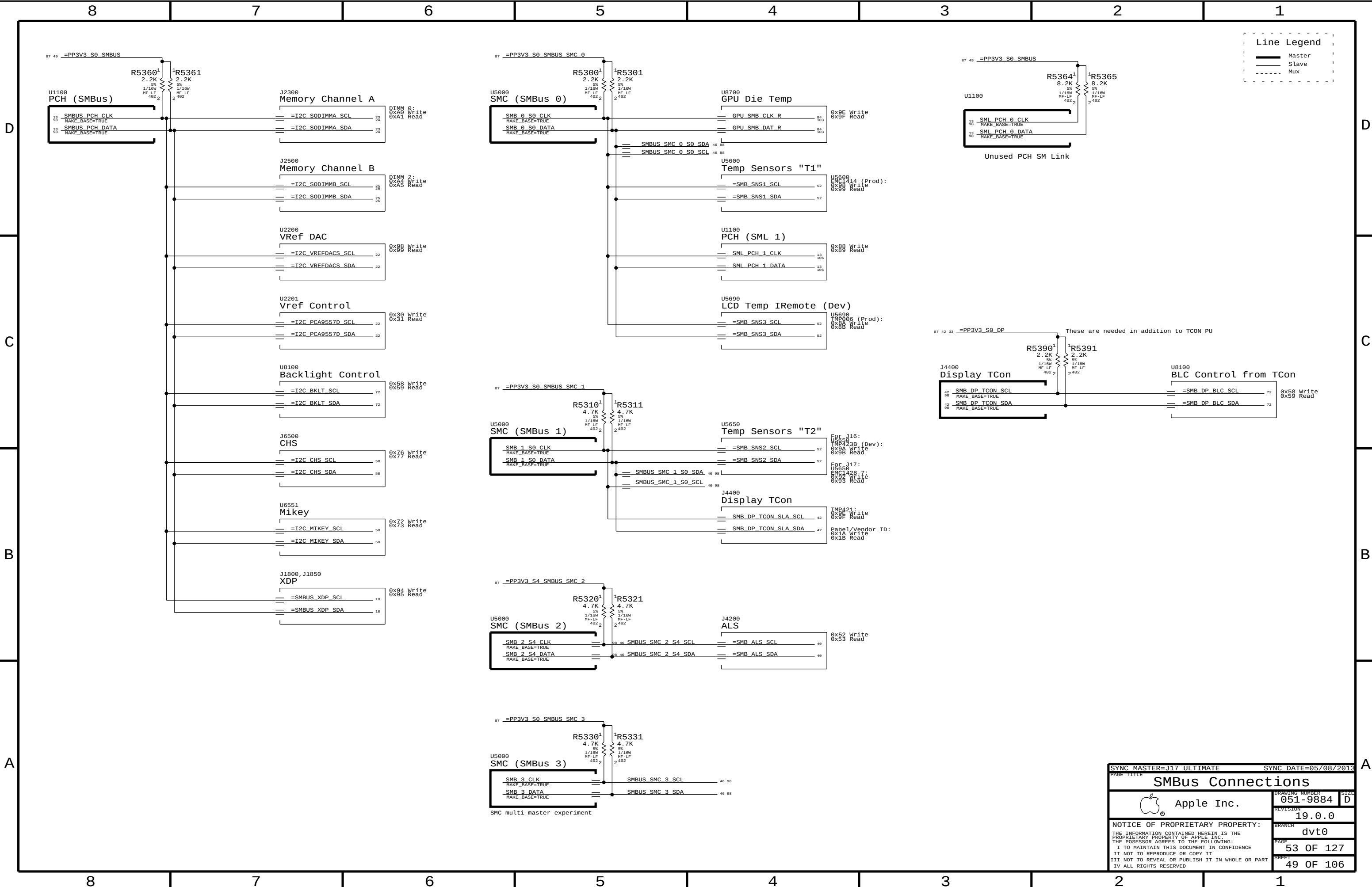




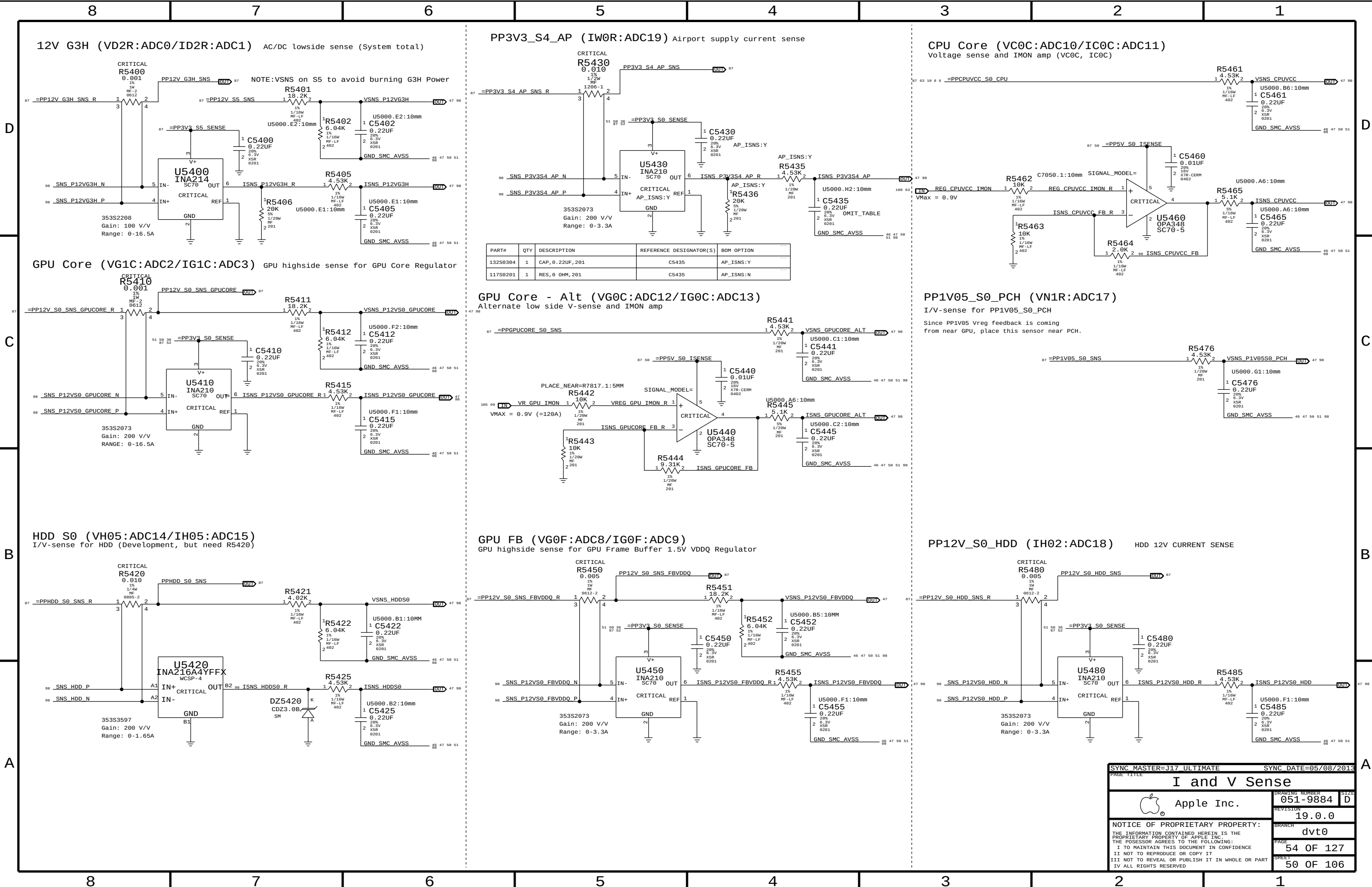
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PAGE TITLE		SMC Support	
 Apple Inc.		DRAWING NUMBER	051-9884
		REVISION	19.0.0
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PAGE TITLE			
SPI and Debug Connector			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-9884	D
		REVISION	
		19.0.0	
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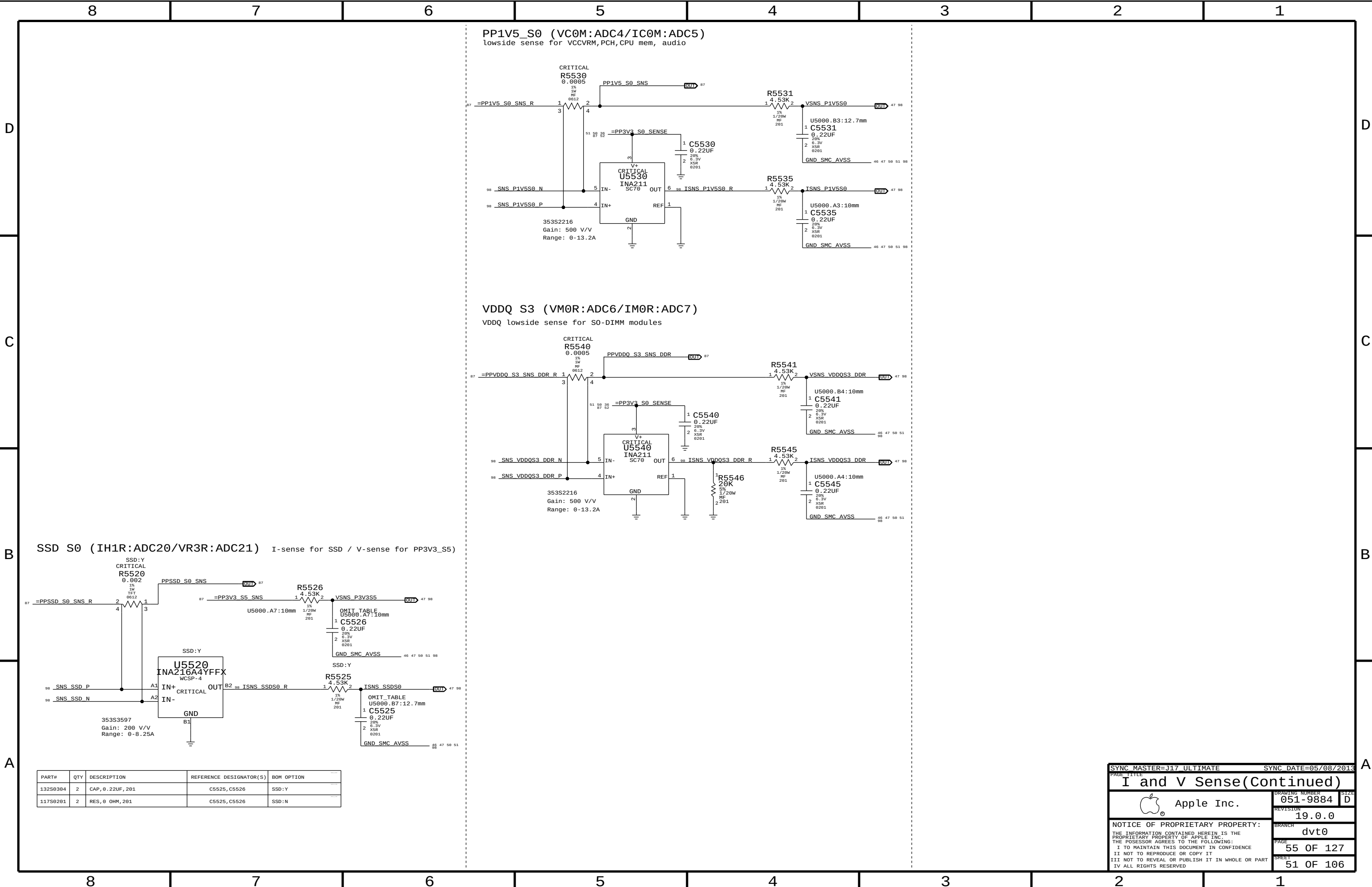


SMC multi-master experiment

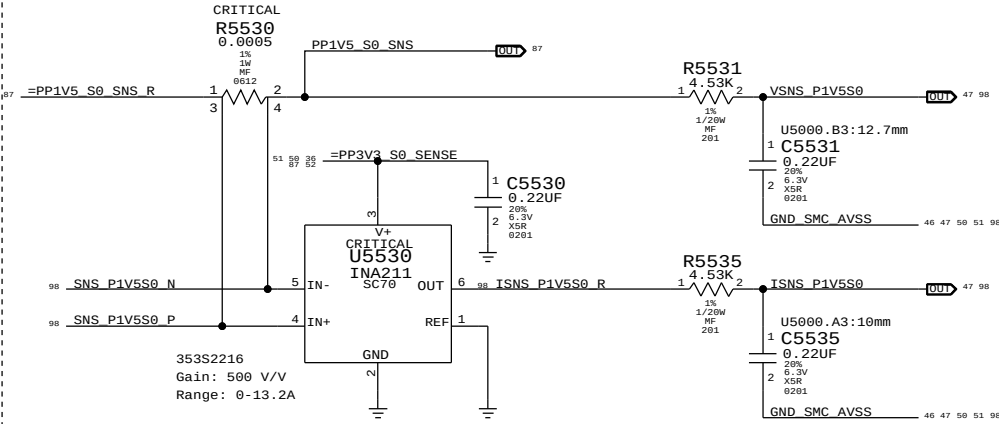


PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
13250304	1	CAP, 0.22UF, 201	C5435	AP_ISNS:Y
11750201	1	RES, 0 OHM, 201	C5435	AP_ISNS:N

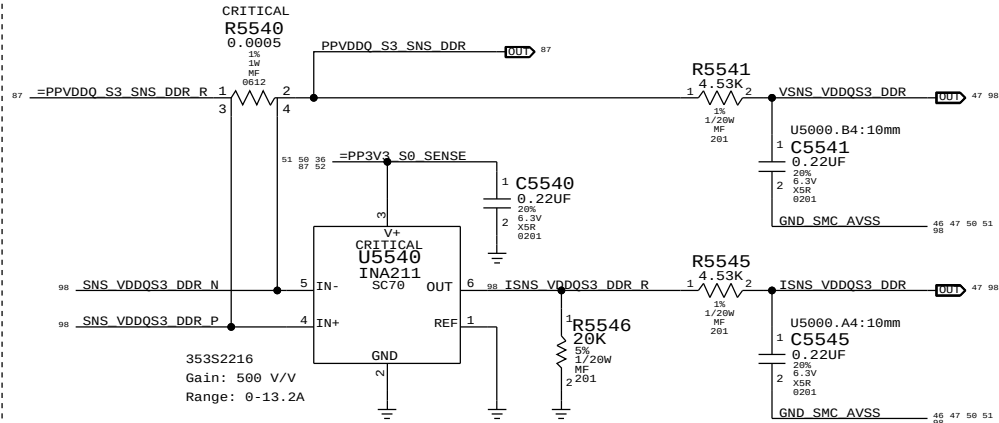
SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/08/2013	
PAGE TITLE		I and V Sense	
Apple Inc.		DRAWING NUMBER	051-9884
		REVISION	19.0.0
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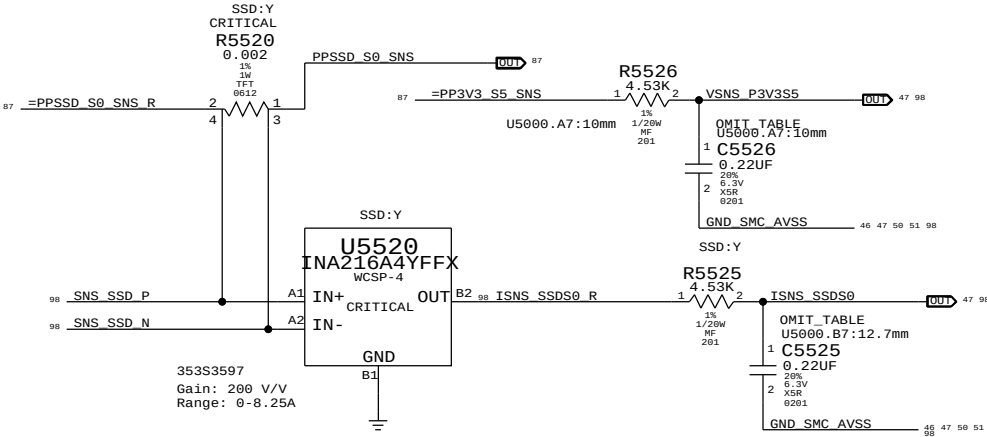
PP1V5_S0 (VC0M:ADC4/IC0M:ADC5)
lowside sense for VCCVRM,PCH,CPU mem, audio



VDDQ_S3 (VM0R:ADC6/IM0R:ADC7)
VDDQ lowside sense for S0-DIMM modules



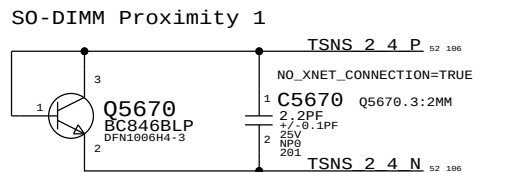
SSD_S0 (IH1R:ADC20/VR3R:ADC21) I-sense for SSD / V-sense for PP3V3_S5)



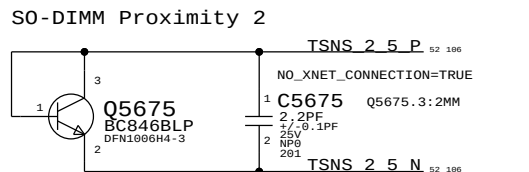
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
132S0304	2	CAP, 0.22UF, 201	C5525, C5526	SSD:Y
117S0201	2	RES, 0 OHM, 201	C5525, C5526	SSD:N

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
372S0186	372S0185		ALL	Alternate Temp Diode

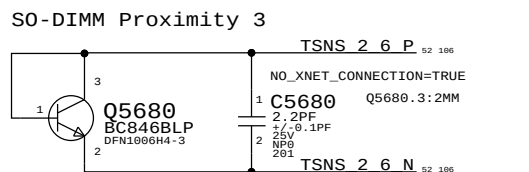
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87 52 51 50 36 =PP3V3 S0 SENSE
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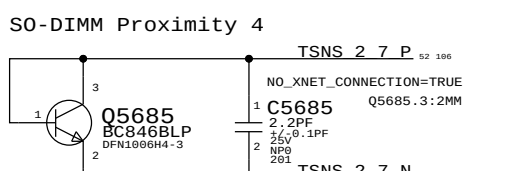
PLACEMENT_NOTE=Place Q5570 near S0-DIMM connectors (top left)



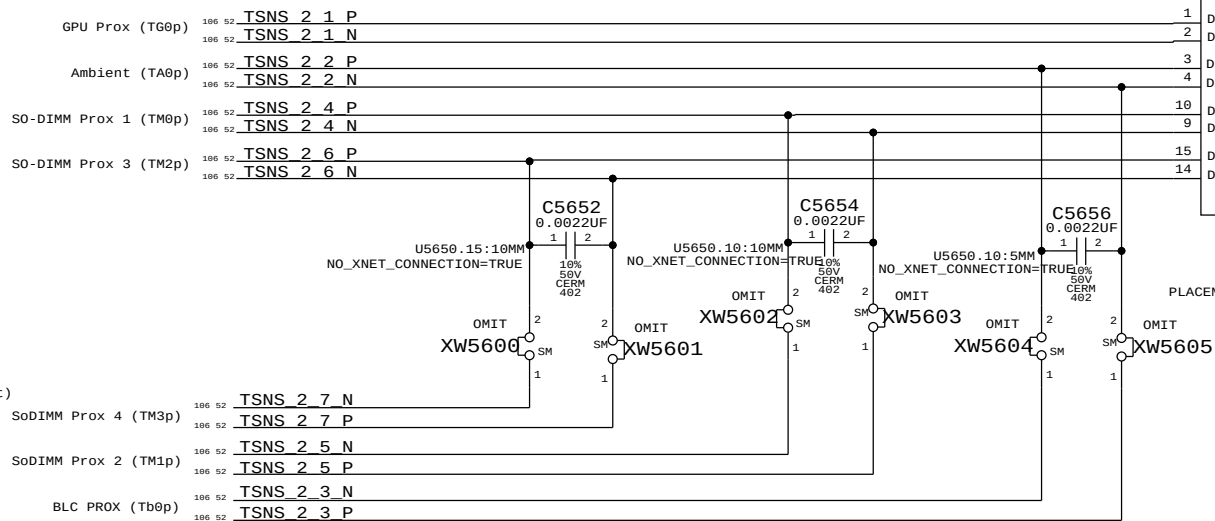
PLACEMENT_NOTE=Place Q5575 near S0-DIMM connectors (top right



PLACEMENT_NOTE=Place Q5580 near SO-DIMM connectors (bottom left)

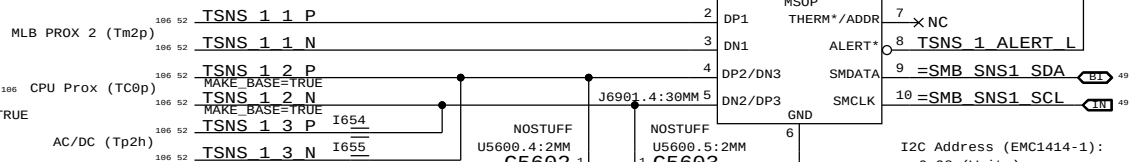


PLACEMENT_NOTE=Place Q5585 near SO-DIMM connectors (bottom ri



EMC1428-7: 6.8K PULL UP: I2C ADDRESS: WRITE: 0x92, READ: 0x93

```
87 52 51 50 36 =PP3V3_S0_SENSE
```



AC/DC (Tp2h) TSNS 1 3 N I655 U5600.4:2MM C5602 1 47PF 5% 50V CERM 482 U5600.5:2MM C5603 1 47PF 5% 50V CERM 482 2 I2C Address (EMC1414-1): 0x98 (Write) 0x99 (Read)

AC/DC Via connector to diode inside PSU

J6901.5:30MM
L5610
FERR-220-OHM

1 2

0402

J6901.4:30MM
L5611
FERR-220-OHM

1 2

0402

SNS ACDC P

SNS ACDC N

TSNS 1 3 P

TSNS 1 3 N

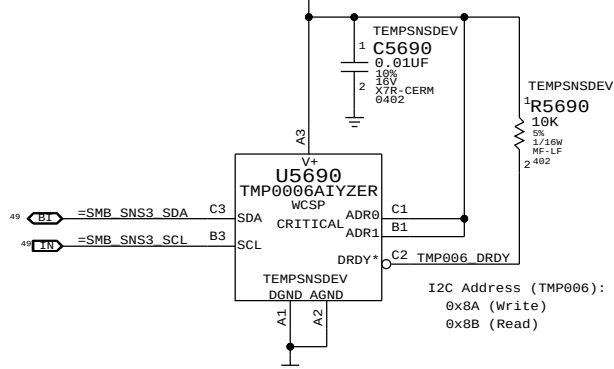
L5610.2:10MM
C5610
0.0022UF
10%
10V
CERM
402

Make sure these caps are OK with U5600 vendor! Internal sensor of the EMC 1414 will be used as MLB sensor.

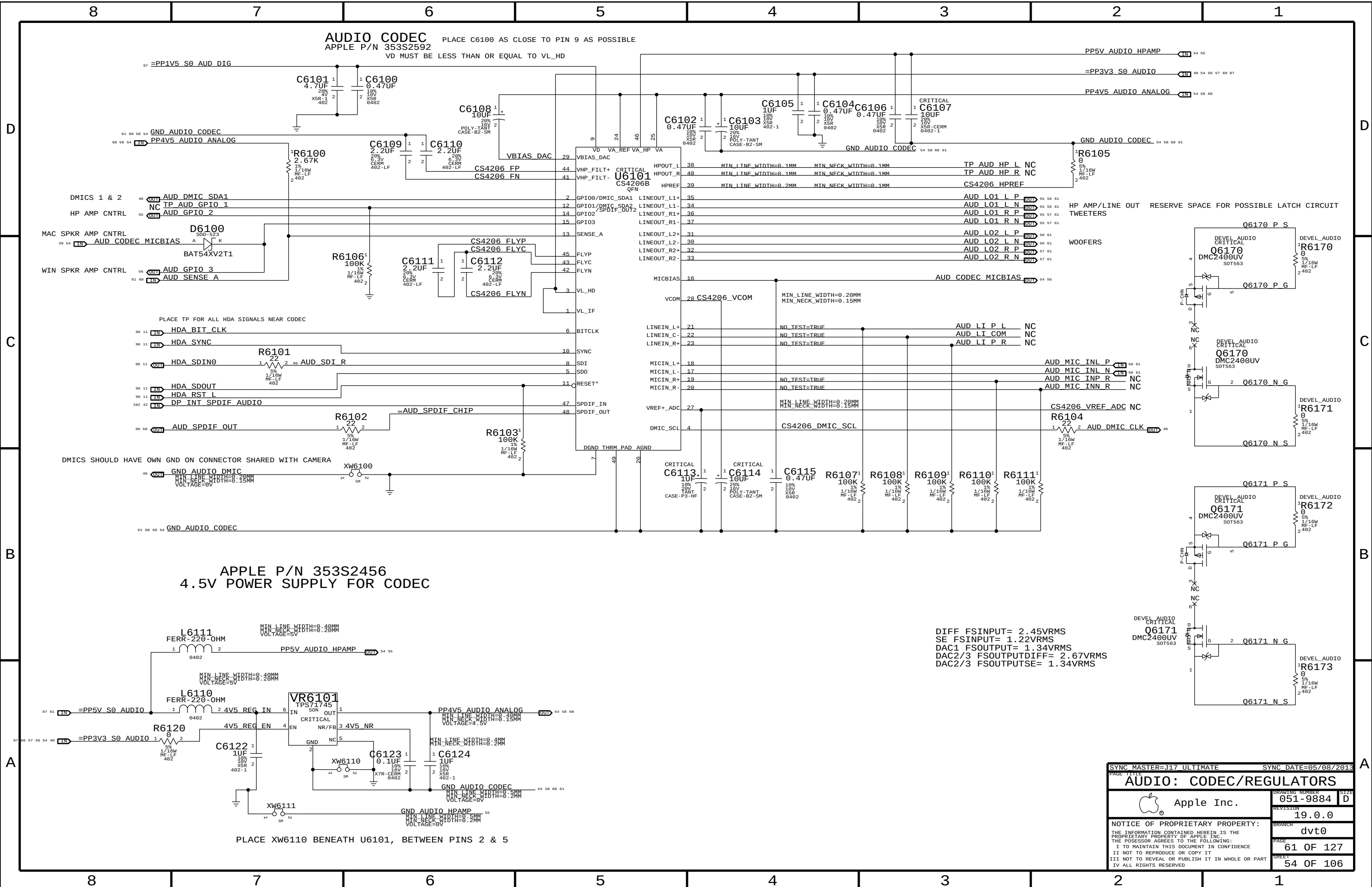
PLACEMENT NOTE=PLACE U5600 NEAR PSU CONNECTOR MLB PROX 0 (Tm0p)

PLACEMENT_NOTE=PLACE U5600 NEAR PSU CONNECTOR

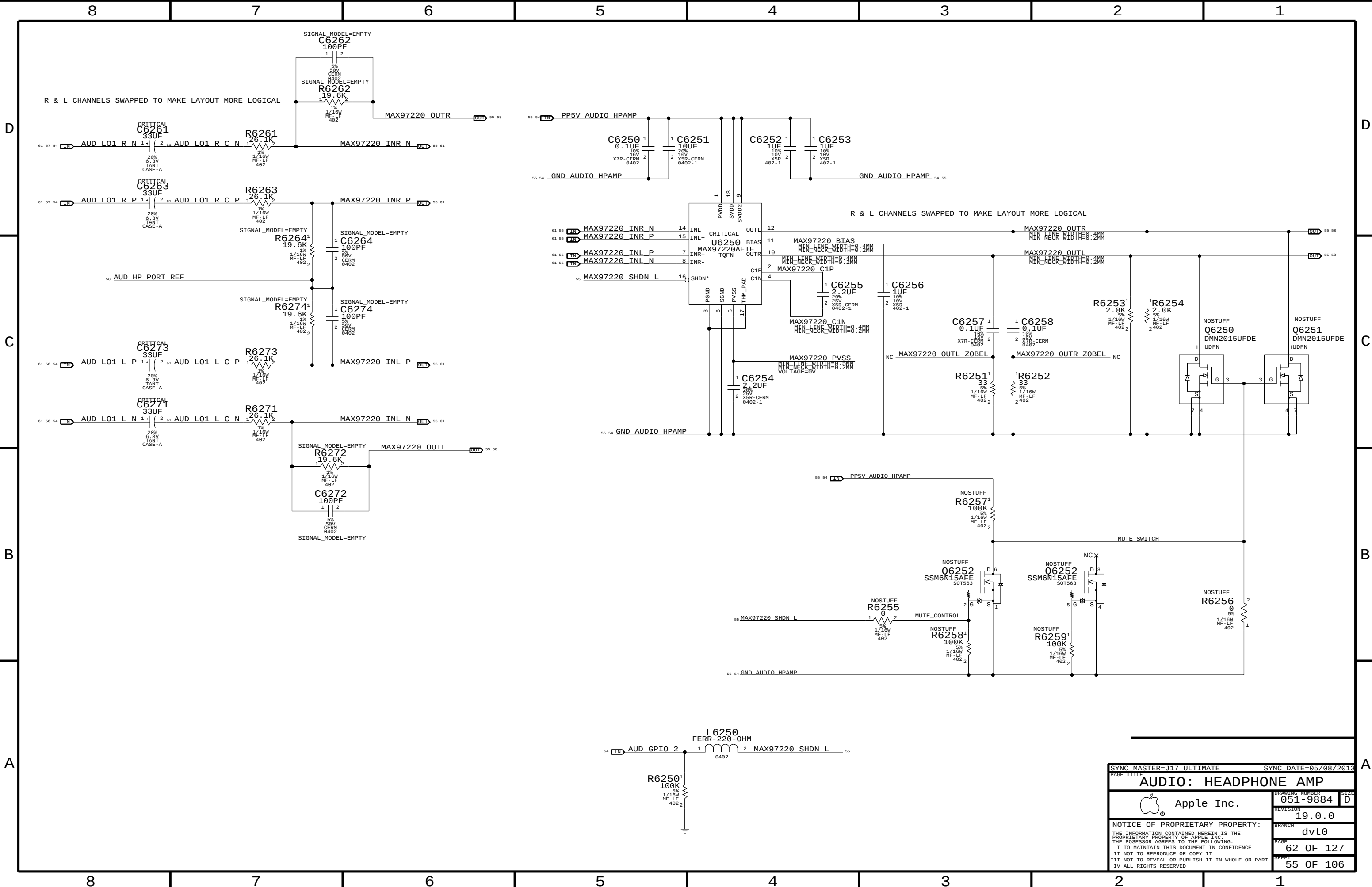
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87 52 51 50 36 =PP3V3 S0 SENSE
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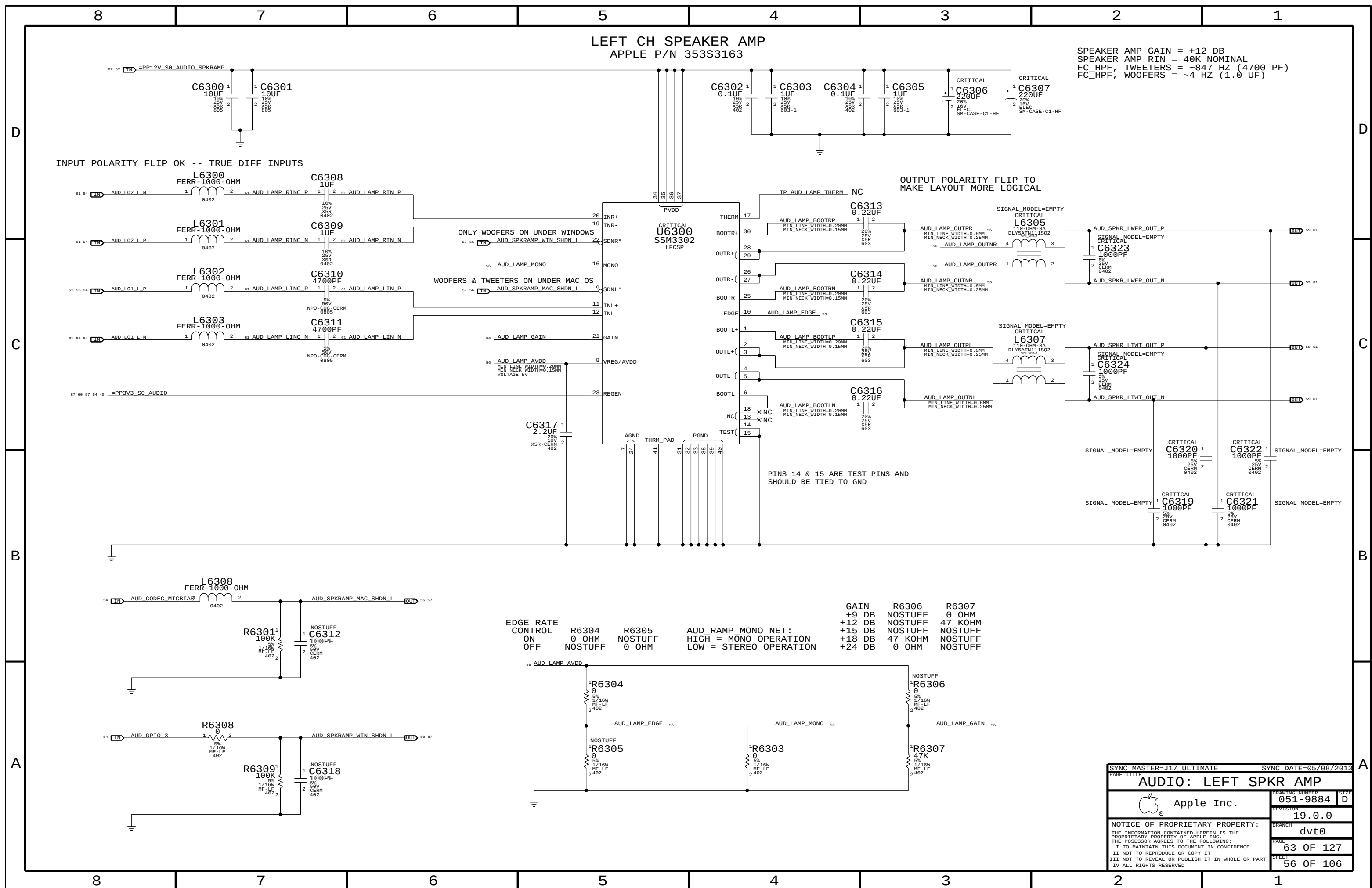
NOTE - Follow TI layout guide (SB0U108.pdf) for this part!!!



PAGE TITLE		SYNC DATE=05/08/2013	
AUDIO: CODEC/REGULATORS		DRAWING NUMBER	
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		SHEET	
		54 OF 106	



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PAGE TITLE		AUDIO: HEADPHONE AMP	
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		REVISION	19.0.0
		BRANCH	dvt0
		PAGE	62 OF 127
		SHEET	55 OF 106
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RIGHT CH SPEAKER AMP
APPLE P/N 353S3163

SPEAKER AMP GAIN = +12 DB
SPEAKER AMP RIN = 40K NOMINAL
FC_HPF, TWEETERS = ~847 HZ (4700 PF)
FC_HPF, WOOFERS = ~4 HZ (1.0 UF)

D

D

C

C

B

B

A

A

INPUT POLARITY FLIP OK -- TRUE DIFF INPUTS

OUTPUT POLARITY FLIP TO
MAKE LAYOUT MORE LOGICAL

PINS 14 & 15 ARE TEST PINS AND
SHOULD BE TIED TO GND

EDGE RATE CONTROL	ON OFF	R6404	R6405	AUD_RAMP_MONO NET: HIGH = MONO OPERATION LOW = STEREO OPERATION	GAIN	R6406	R6407
		0 OHM	NOSTUFF		+9 DB	NOSTUFF	0 OHM
		NOSTUFF	0 OHM		+12 DB	NOSTUFF	47 KOHM
		NOSTUFF	0 OHM		+15 DB	NOSTUFF	NOSTUFF
					+18 DB	47 KOHM	NOSTUFF
					+24 DB	0 OHM	NOSTUFF

SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/08/2013	
PAGE TITLE			
AUDIO: RIGHT SPKR AMP			
Apple Inc.		DRAWING NUMBER	051-9884
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MIKEY RECEIVER CKT

WRITE: 0X72 READ: 0X73 APN 353S3231

I2C ADDRESSES

MIKEY	U6551	READ	0111 0011	0X73
MIKEY	U6551	WRITE	0111 0010	0X72

D

D

C

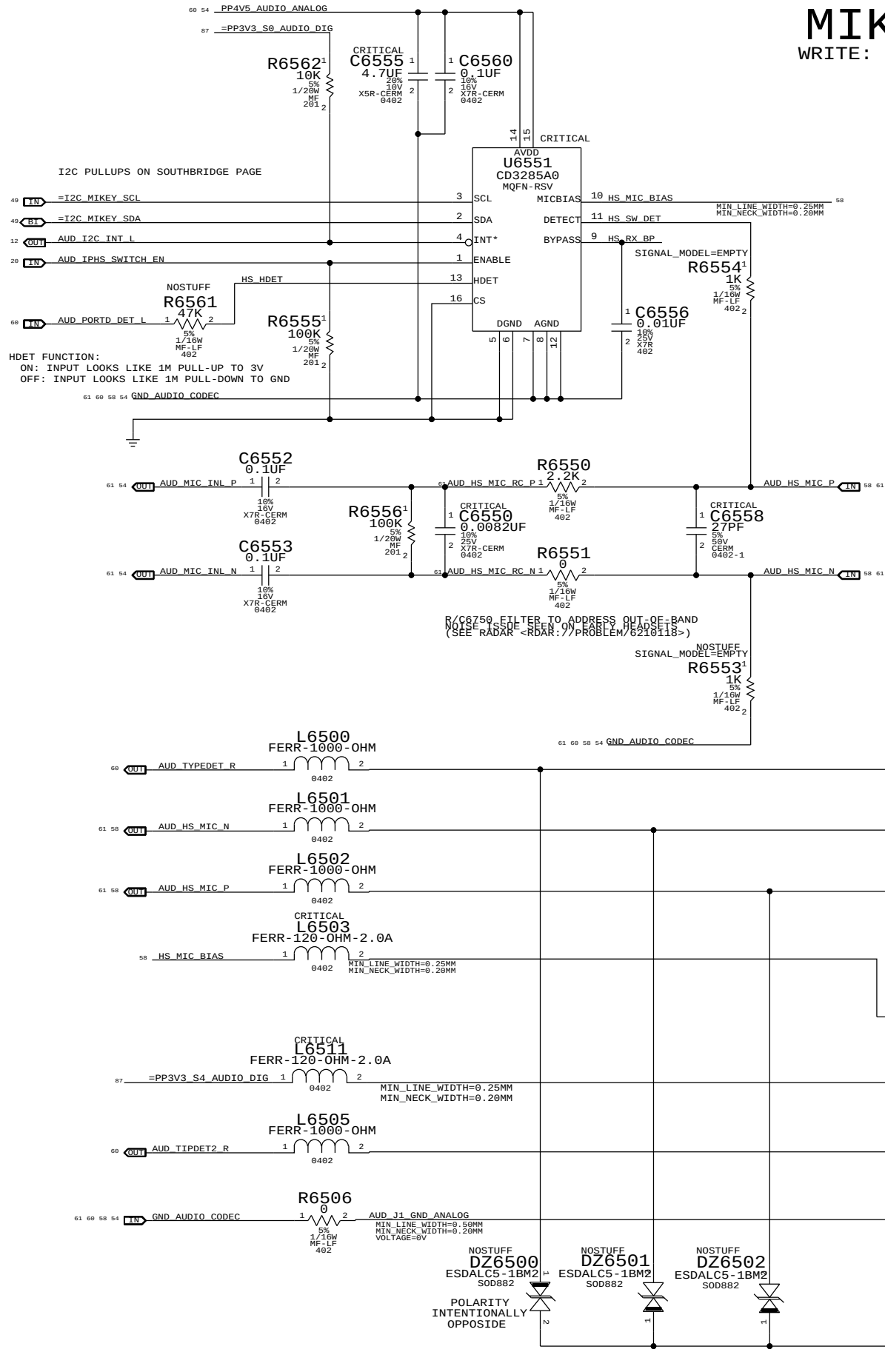
C

B

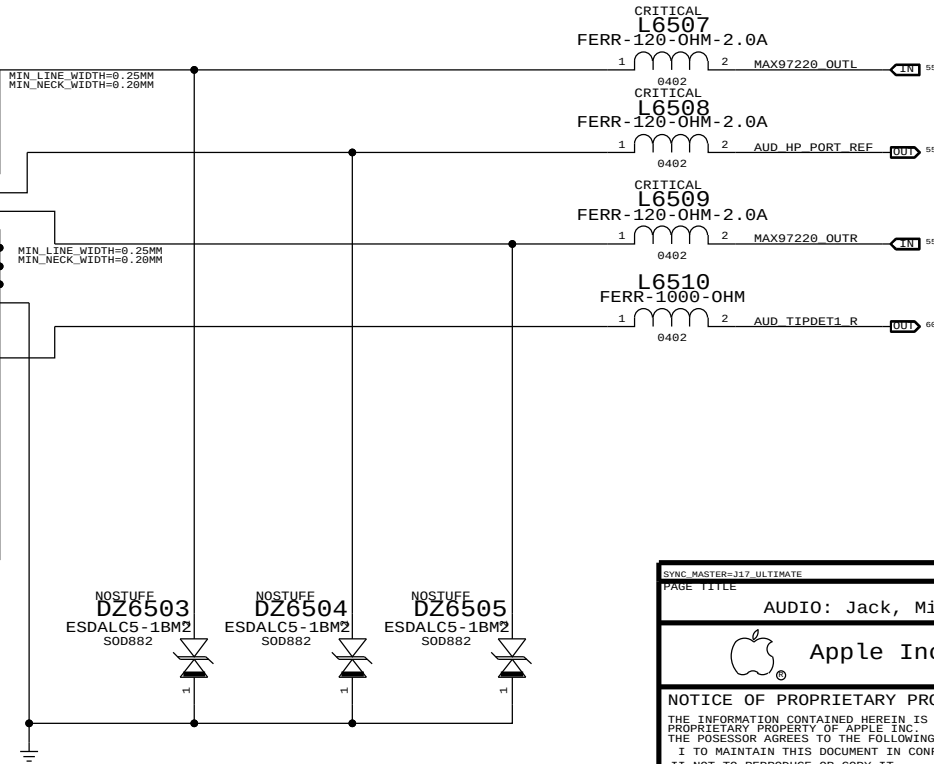
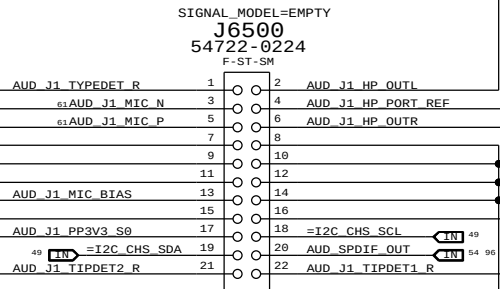
B

A

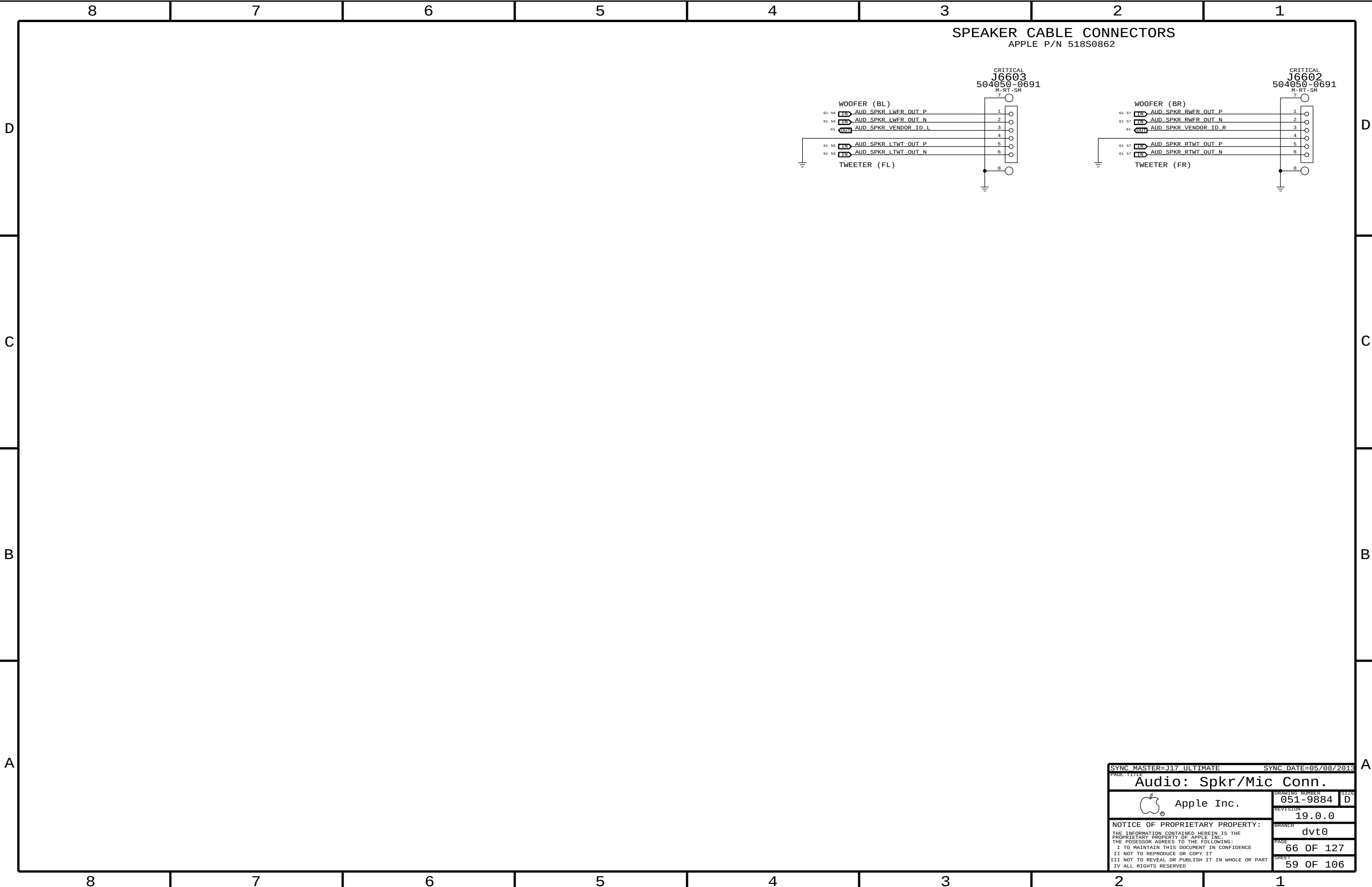
A



APPLE P/N 518S0687



PAGE TITLE		PAGE TITLE	
AUDIO: Jack, Mikey, CHS Switch		DRAWING NUMBER	
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REVISION		19.0.0	
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8	7	6	5	4	3	2	1
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C

8	7	6	5	4	3	2	1
---	---	---	---	---	---	---	---

A

CODEC OUTPUT SIGNAL PATHS

FUNCTION	VOLUME/MUTE	CONVERTER	PIN COMPLEX	MAC SHDN	WIN SHDN	DET ASSIGNMENT
HP/LINE OUT	0X03 (3)	0X03 (3)	0X0A (10,D)	GPIO_2	GPIO_2	0X0A (DET D)
PRIMARY SPKRS (WFR)	0X04 (4)	0X04 (4)	0X0B (11)	MICBIAS	GPIO_3	N/A
SECONDARY SPKRS (TWT)	0X03 (3)	0X03 (3)	0X0A (10,V24)	MICBIAS	N/A	N/A
SPDIF OUT	N/A	0X08 (8)	0x10 (16)	N/A	N/A	0X0D (DET B)

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
AUDIO	*	0.1 MM	?
SPKROUT	*	0.2 MM	?

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
AUDI001FF	*	Y	0.1 MM	0.1 MM	10 MM	0.1 MM	0.1 MM
SPKR002DIFF	*	Y	0.6 MM	0.25 MM	10 MM	0.2 MM	0.2 MM

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
AUDIODIFF	*	AUDIODIFF
SPKROUTDIFF	*	SPKROUTDIFF

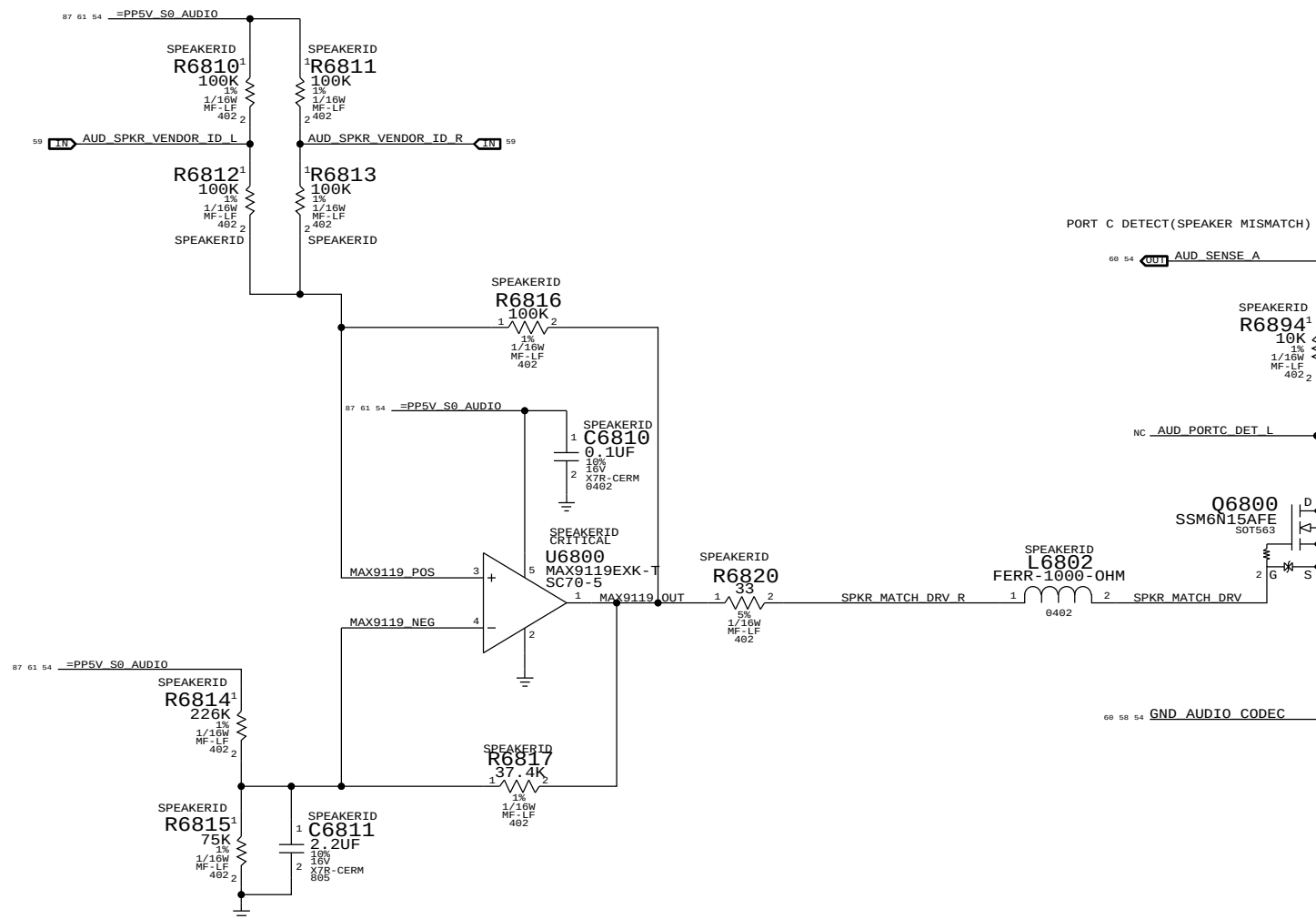
CODEC INPUT SIGNAL PATHS

FUNCTION	CONVERTER	PTIN	COMPLEX	ENABLE/CONTROL	DET ASSIGNMENT
SPI0F IN	0X07 (7)	0X0F (15)		N/A	0X09 (DET A)
INTERNAL MIC ARRAY	0X08 (8)	0X1F (18, LEFT, & RIGHT)		N/A	N/A
EXTERNAL MIC	0X06 (6)	0X0D (13, V22, B, LEFT)		Lynx POINT GPIO 16	Lynx POINT GPIO 5 (BCVR INT)

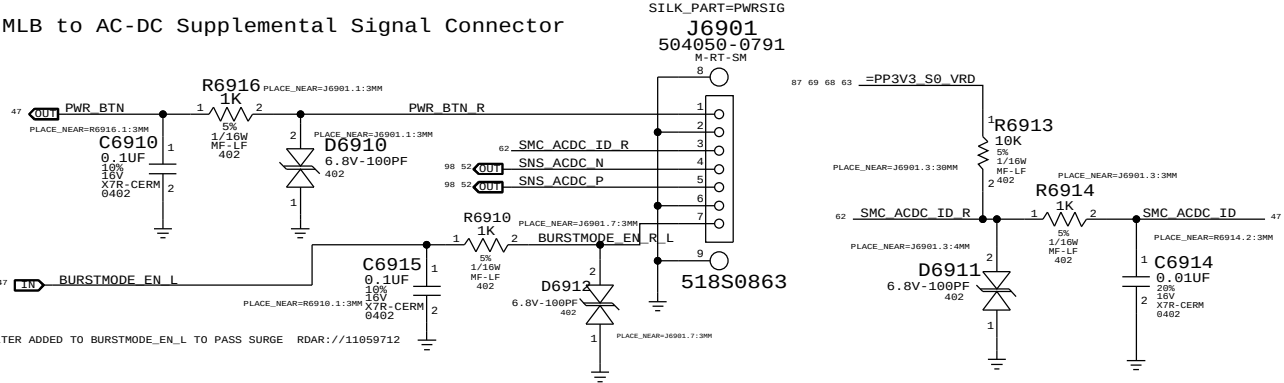
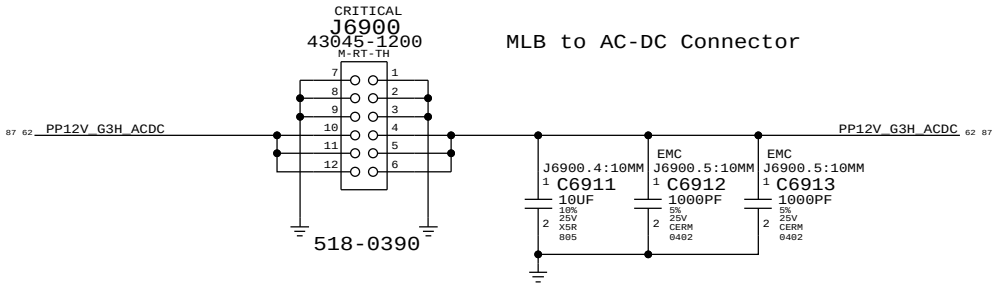
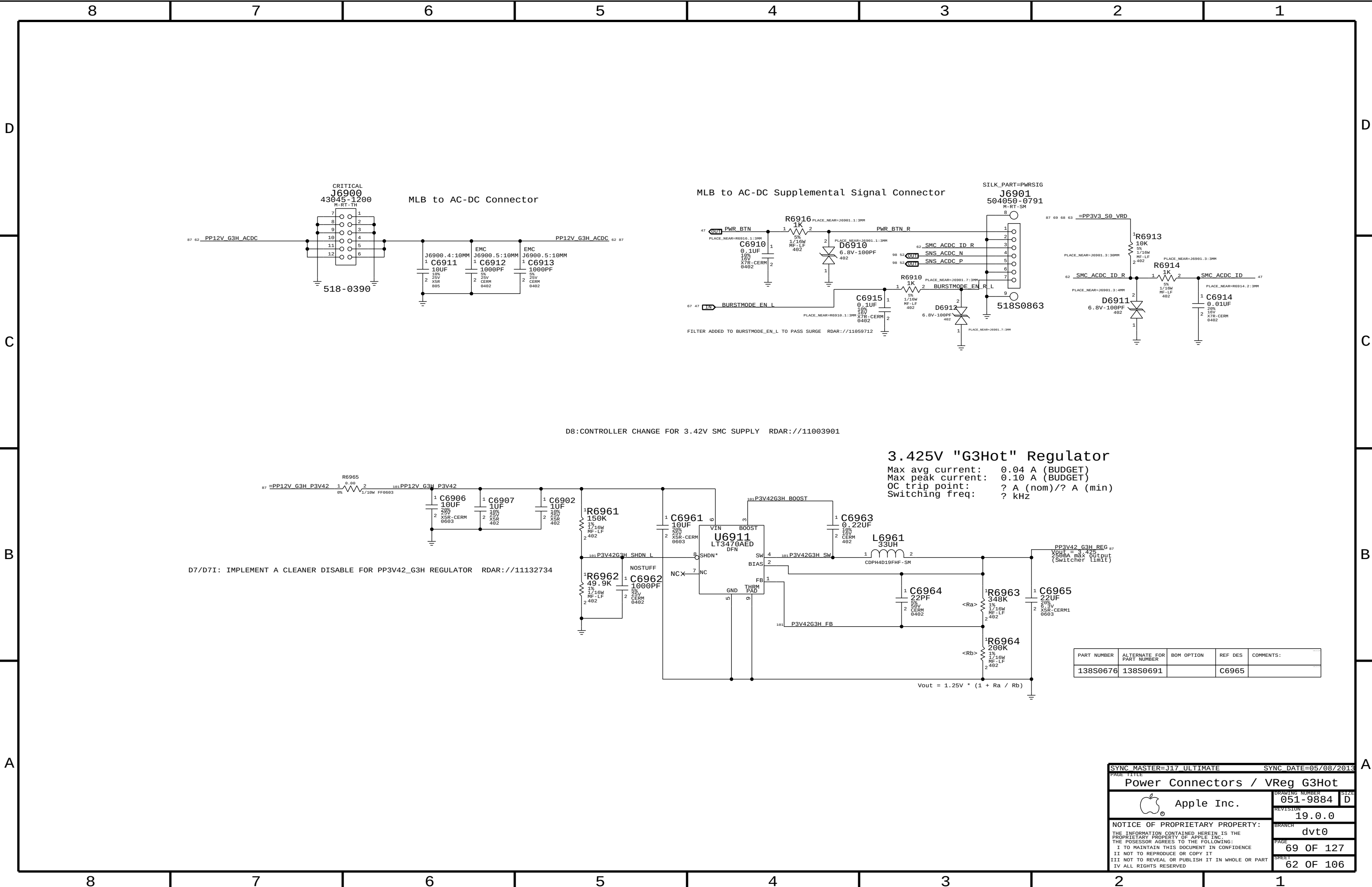
OTHER DETECT

FUNCTION	CONVERTER	PIN	COMPLEX	ENABLE/CONTROL	DET ASSIGNMENT
MULTIPLE SPKR VENDORS	N/A	N/A		N/A	0X0C (DET C)

CIRCUIT THEORY OF OPERATION AVAILABLE IN <RDAR://PROBLEM/9776522>

[illegible]

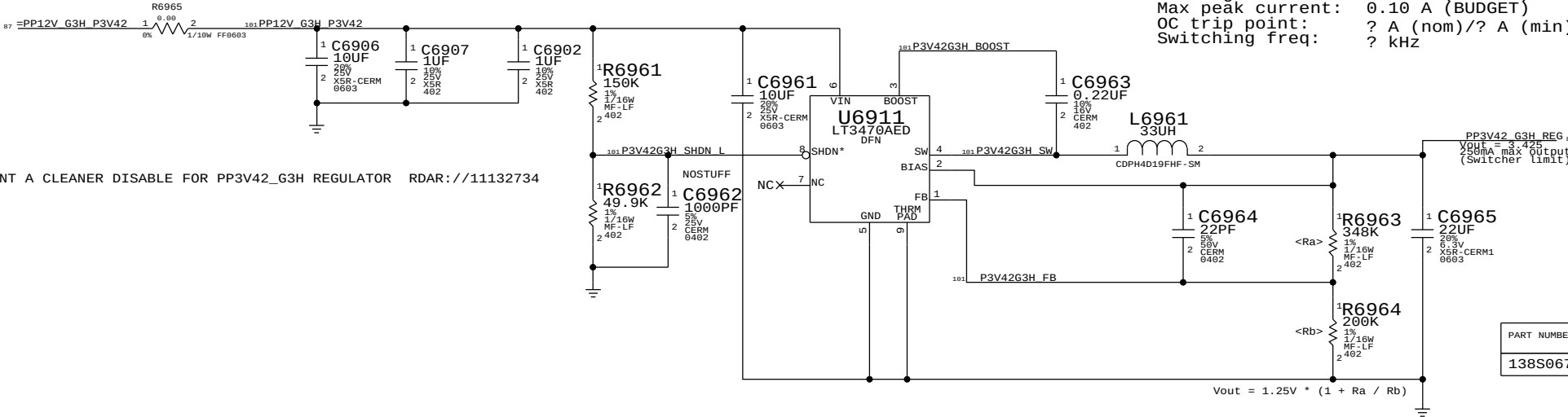
SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/08/2013	
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AUDIO: Speaker ID			
 Apple Inc.	DRAWING NUMBER	051-9884	
	SIZE	D	
	REVISION	19.0.0	
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D8:CONTROLLER CHANGE FOR 3.42V SMC SUPPLY RDAR://11003901

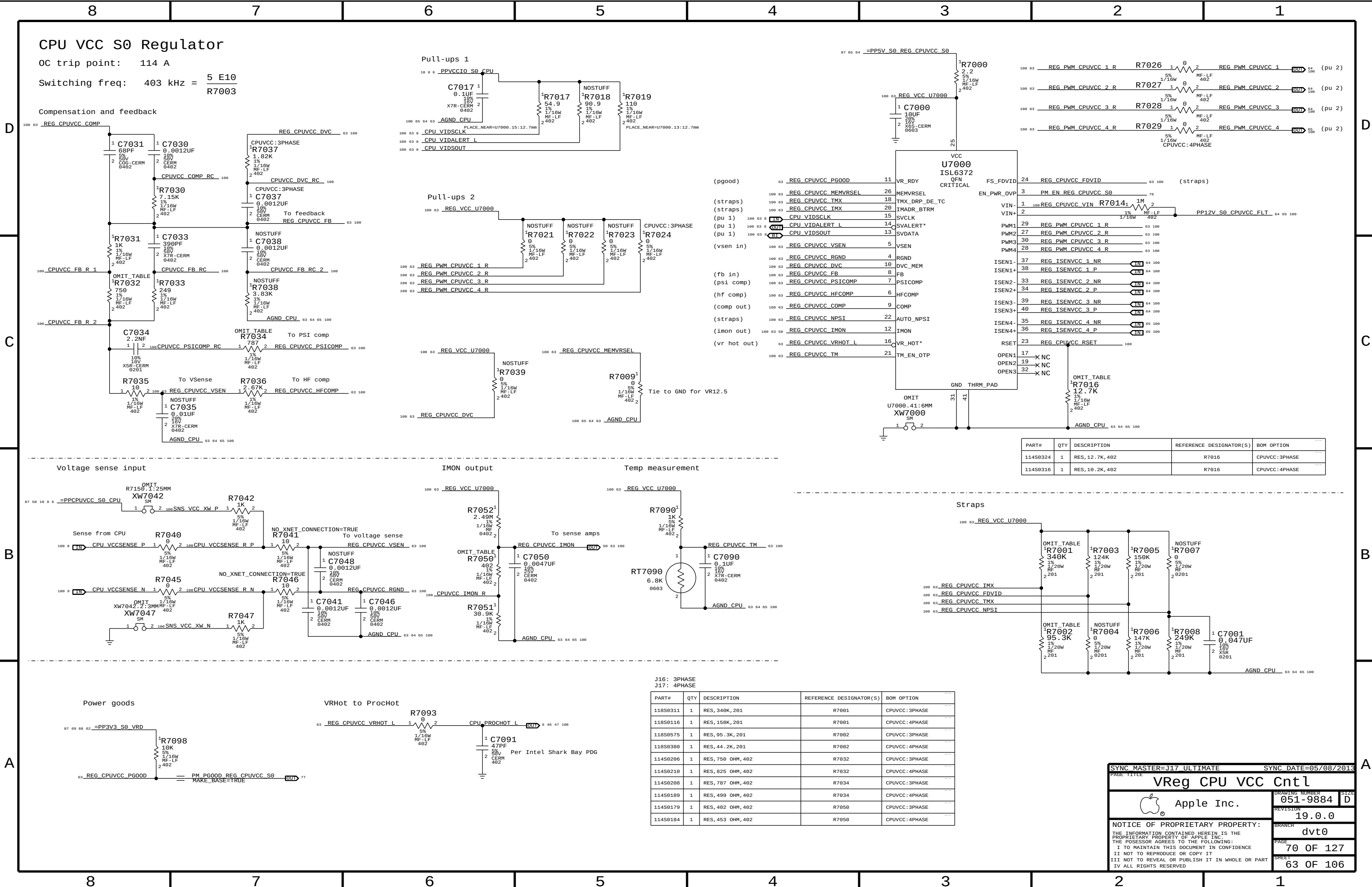
3.425V "G3Hot" Regulator

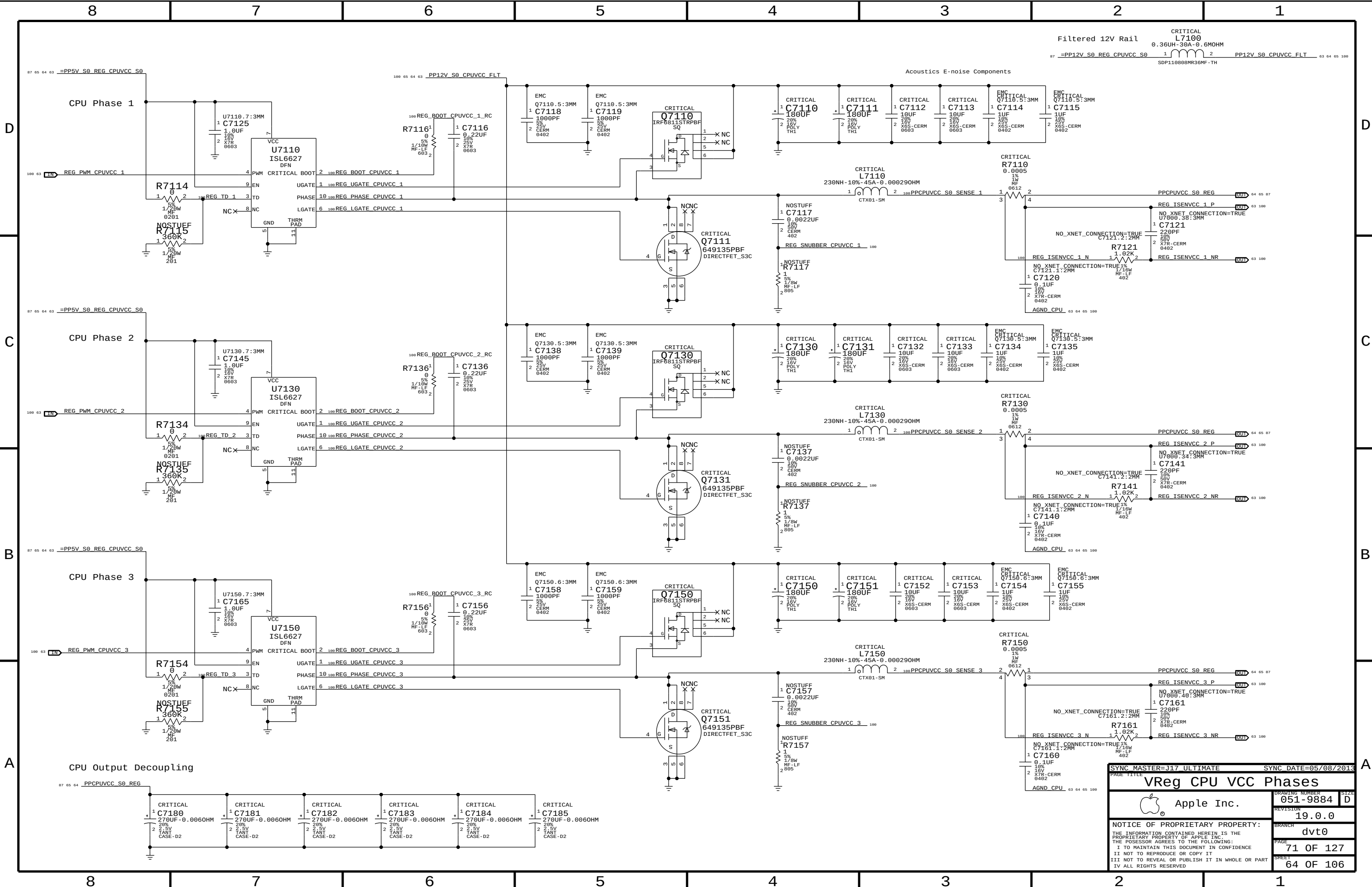
Max avg current: 0.04 A (BUDGET)
Max peak current: 0.10 A (BUDGET)
OC trip point: ? A (nom)/? A (min)
Switching freq: ? kHz



PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S0676	138S0691		C6965	

SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/08/2013	
PAGE TITLE			
Power Connectors / VReg G3Hot			
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VReg CPU VCC Phases

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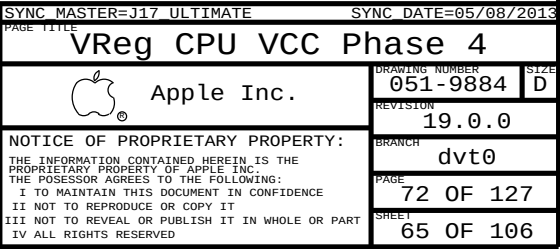
dvt0

PAGE

71 OF 127

SHEET

64 OF 106

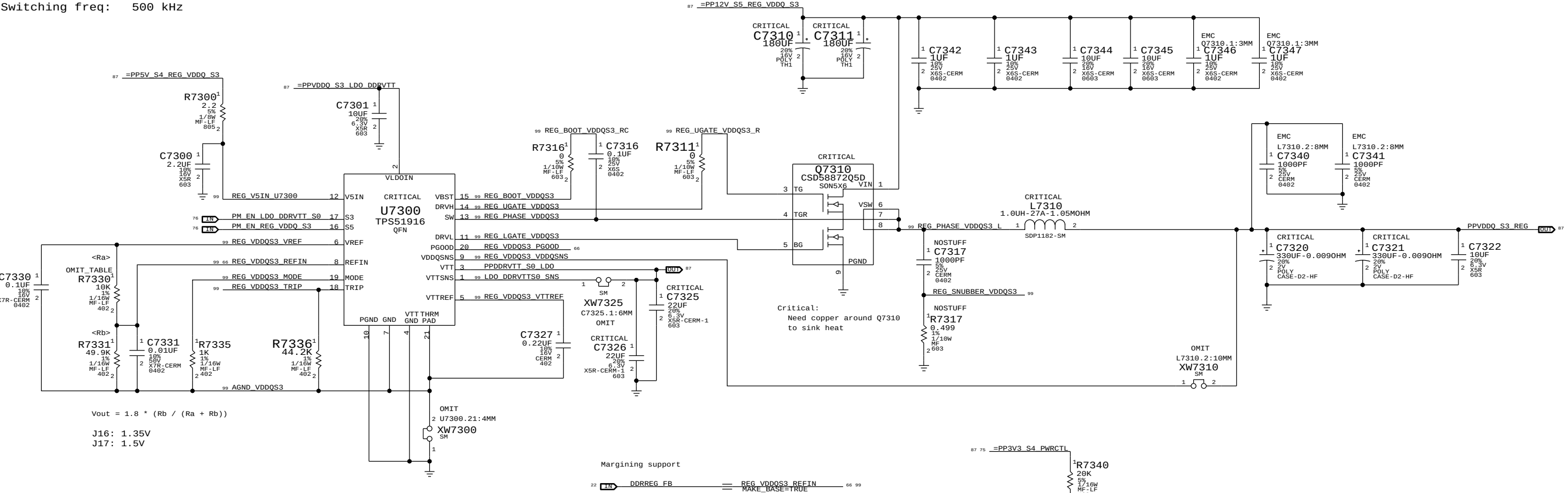


VDDQ (1.5V / 1.35V) S3 Regulator

OC trip point: $30.4 \text{ A VDDQ} = \frac{R7336}{8 \text{ E5} * Rds(Q7310)} + \frac{0.65625}{L7310 * f(\text{switch})}$

3 A VTT (FIXED)
10 mA VTTREF (FIXED)

Switching freq: 500 kHz



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
11450335	1	RES, 16.5K, 402	R7330	VDDQ:P1V35
11450315	1	RES, 10K, 402	R7330	VDDQ:P1V5

3.3V S5 Regulator

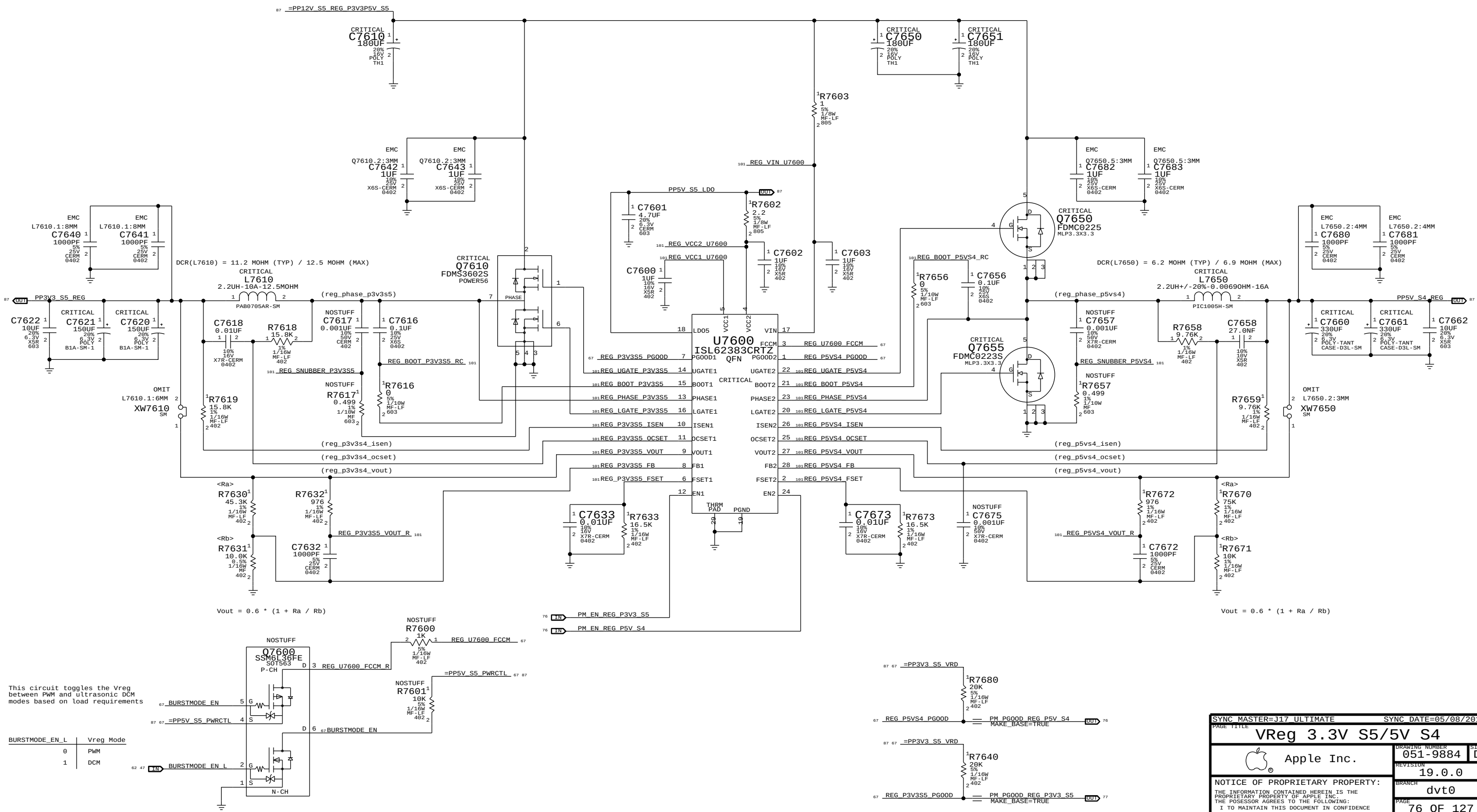
OC trip point: $12.5 \text{ A} = \frac{R7618 * 10 \text{ E-6}}{\text{DCR}(\text{L7610})}$

Switching freq: $356 \text{ kHz} = \frac{1}{170 \text{ E-12} * R7633}$

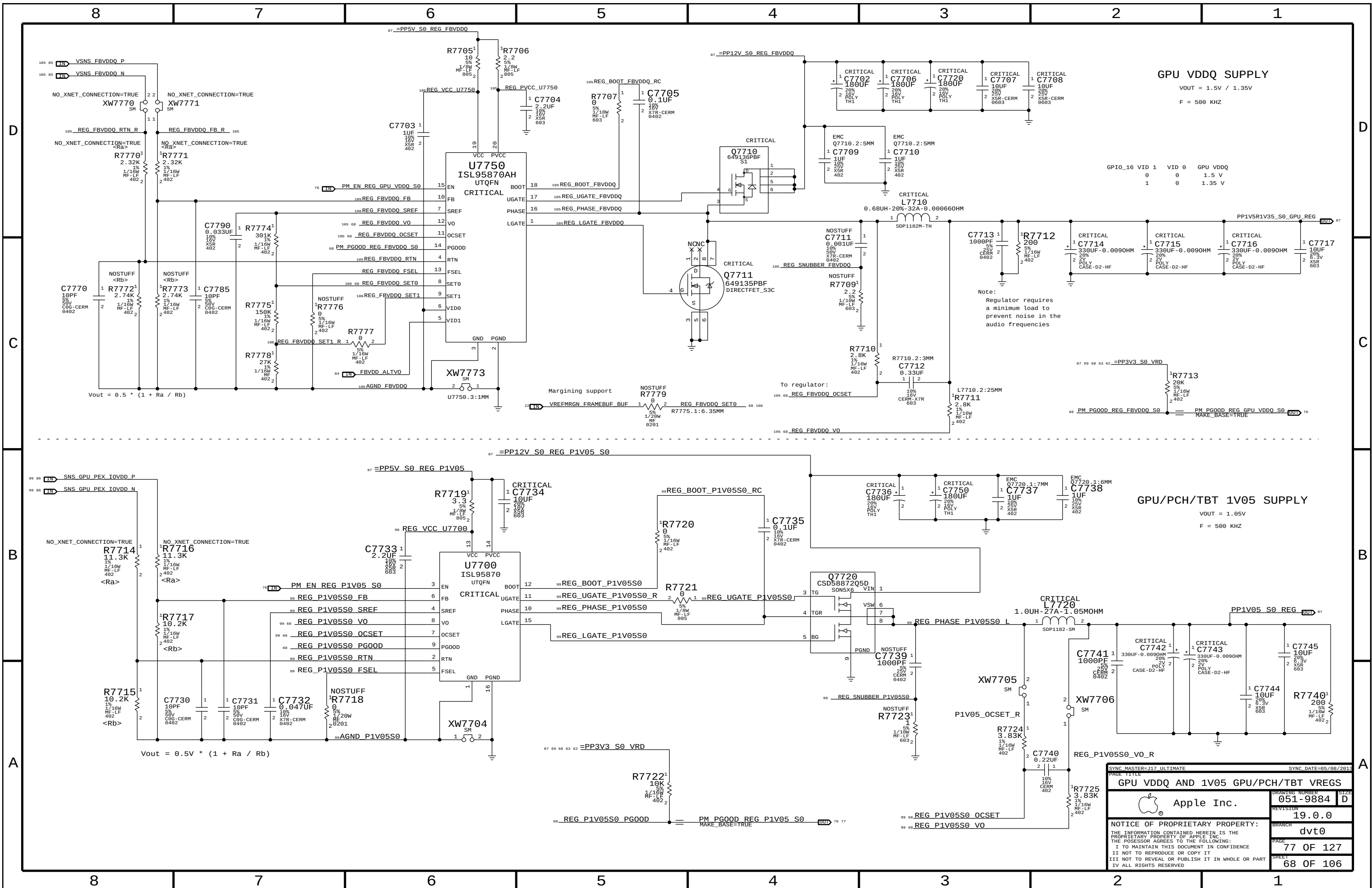
5V S4 Regulator

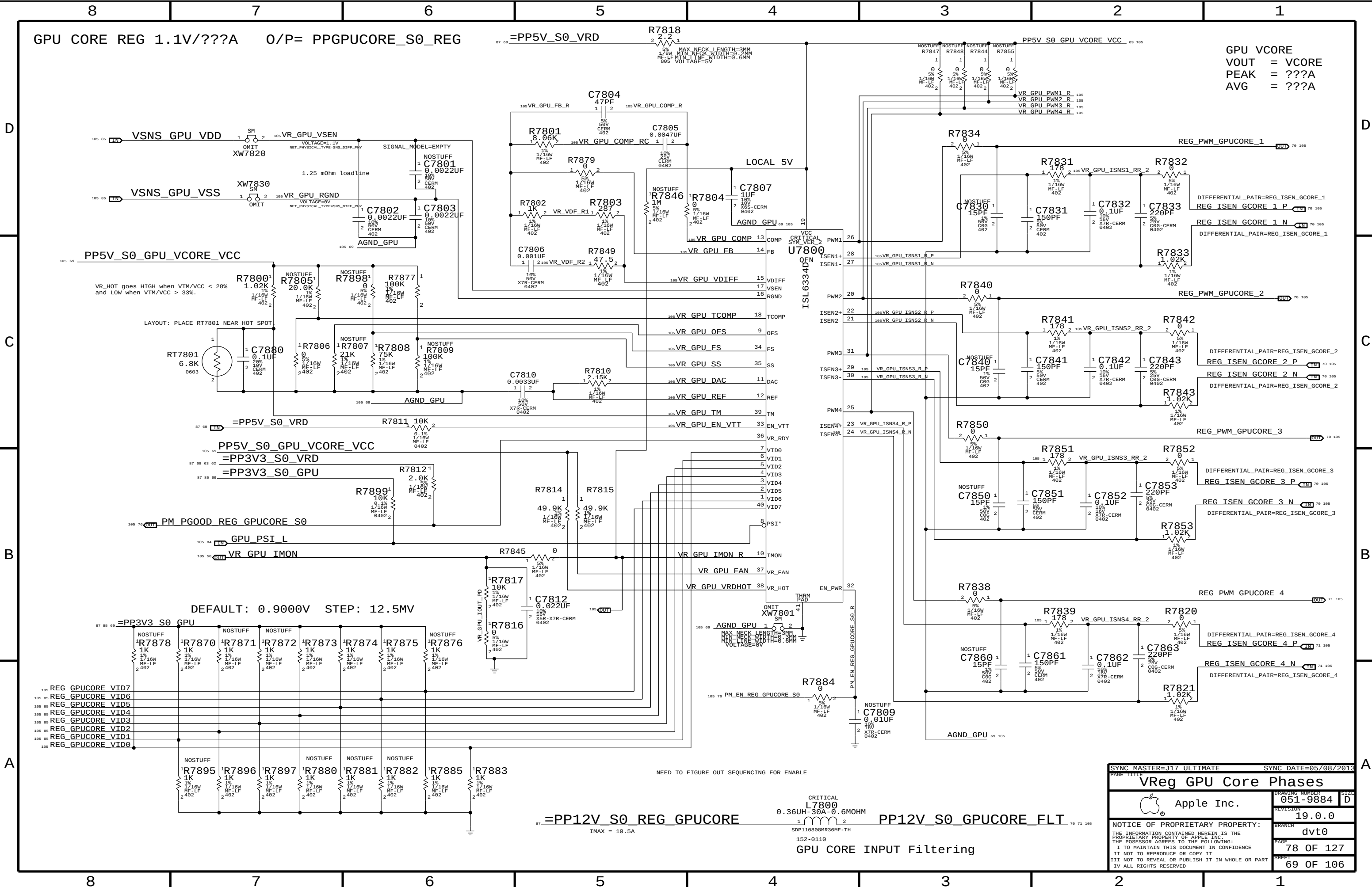
OC trip point: $14.1 \text{ A} = \frac{R7658 * 10 \text{ E-6}}{\text{DCR}(\text{L7650})}$

Switching freq: $356 \text{ kHz} = \frac{1}{170 \text{ E-12} * R7673}$



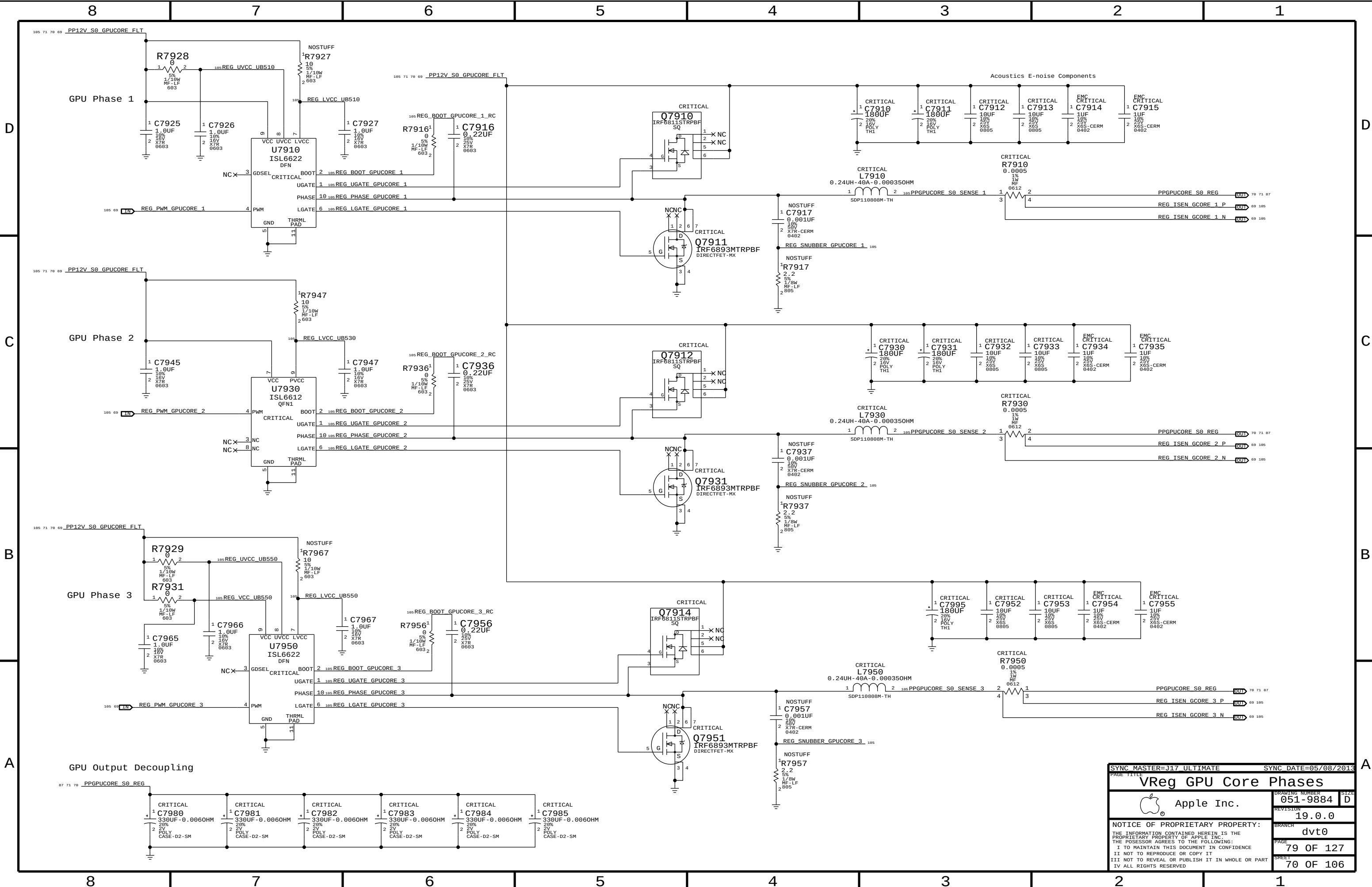
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Apple Inc.		DRAWING NUMBER	051-9884
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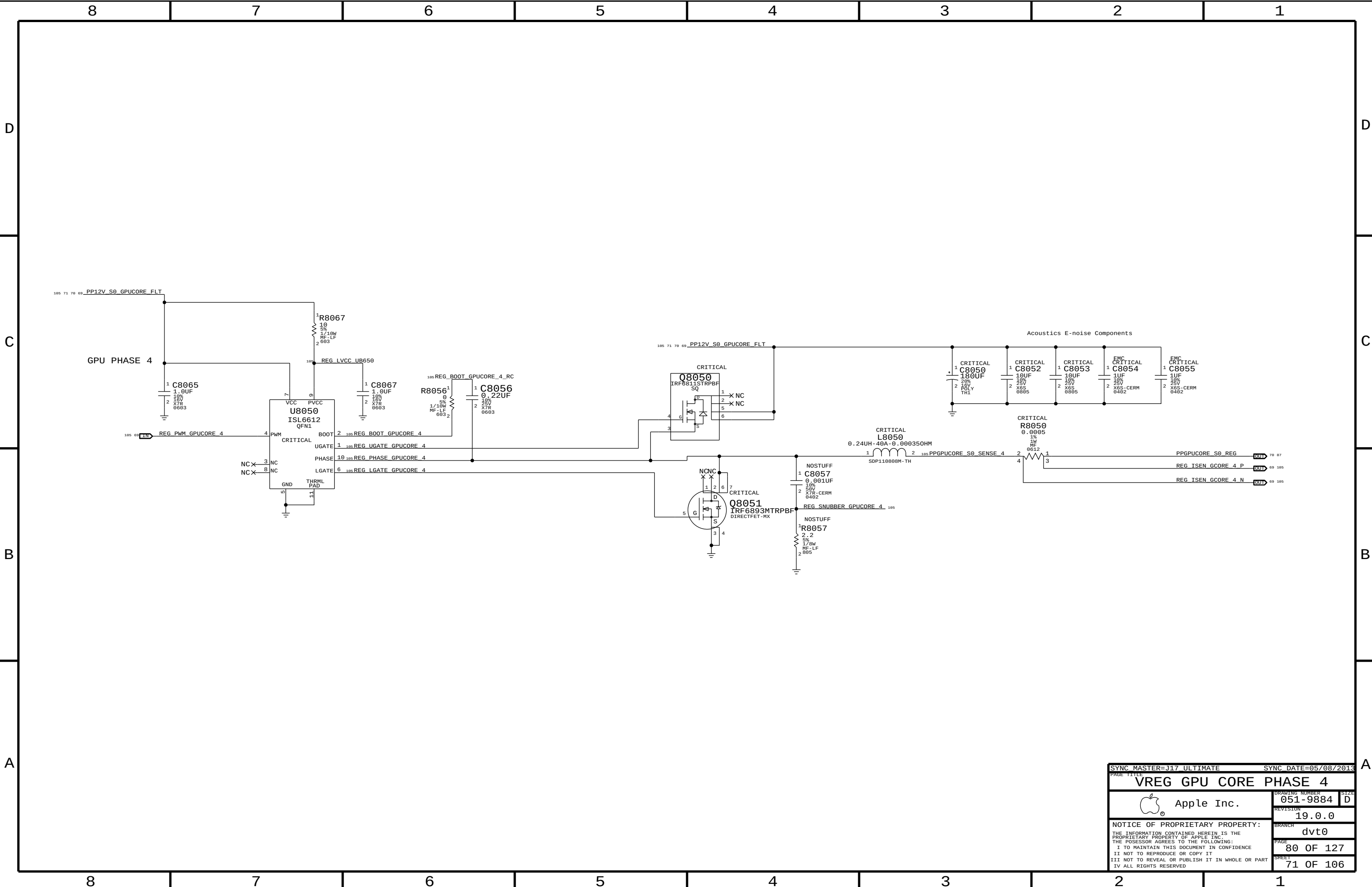


GPU VCORE
VOUT = VCORE
PEAK = ???A
AVG = ???A

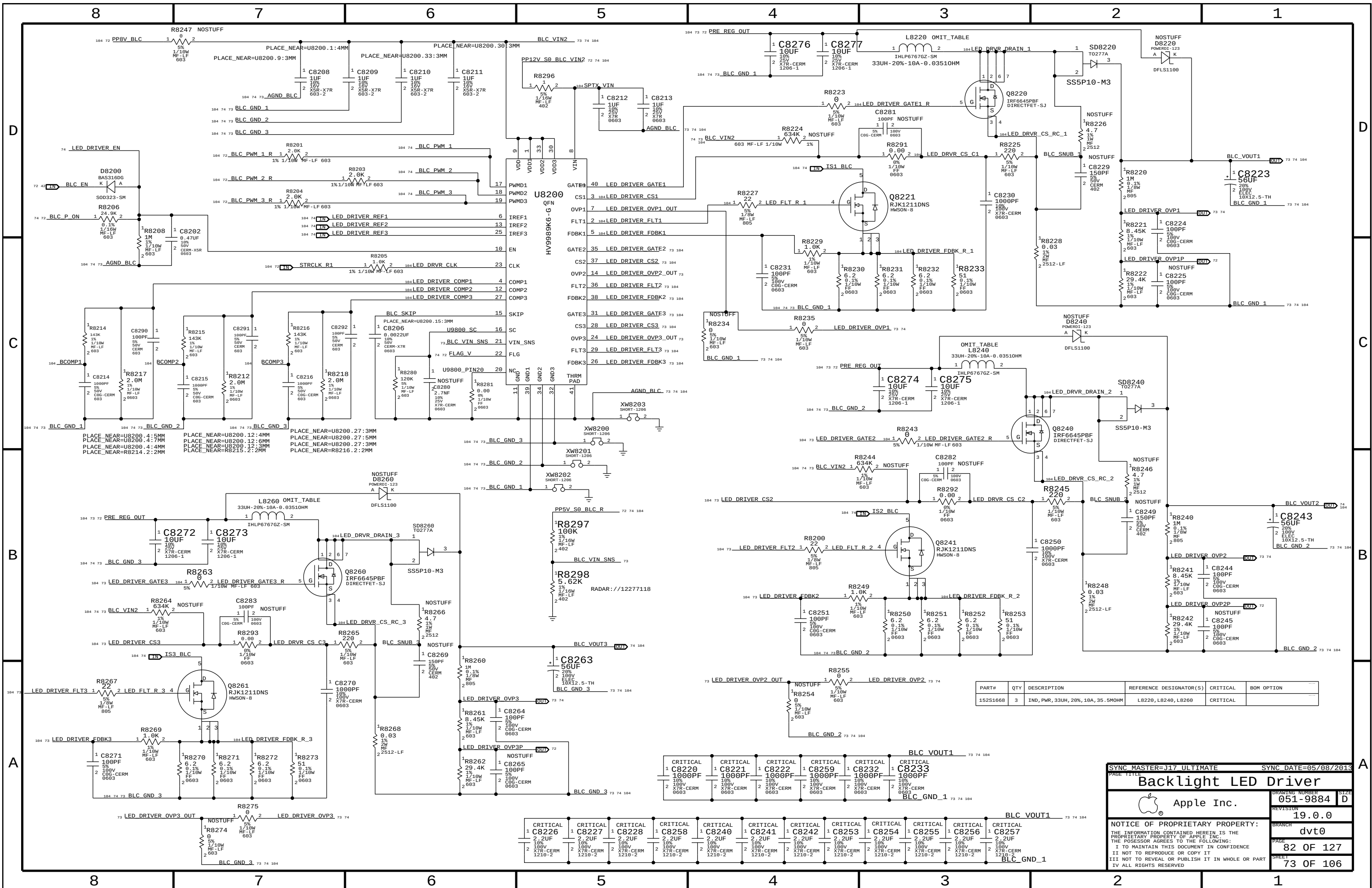
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SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/08/2013	
PAGE TITLE		VReg GPU Core Phases	
Apple Inc.		DRAWING NUMBER	051-9884
		REVISION	19.0.0
		BRANCH	dvt0
		PAGE	79 OF 127
		SHEET	70 OF 106
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		PAGE	80 OF 127
		SHEET	71 OF 106



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SYNC DATE=05/08/2013

Backlight LED Driver

Apple Inc.

051-9884

19.0.0

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82 OF 127

73 OF 106

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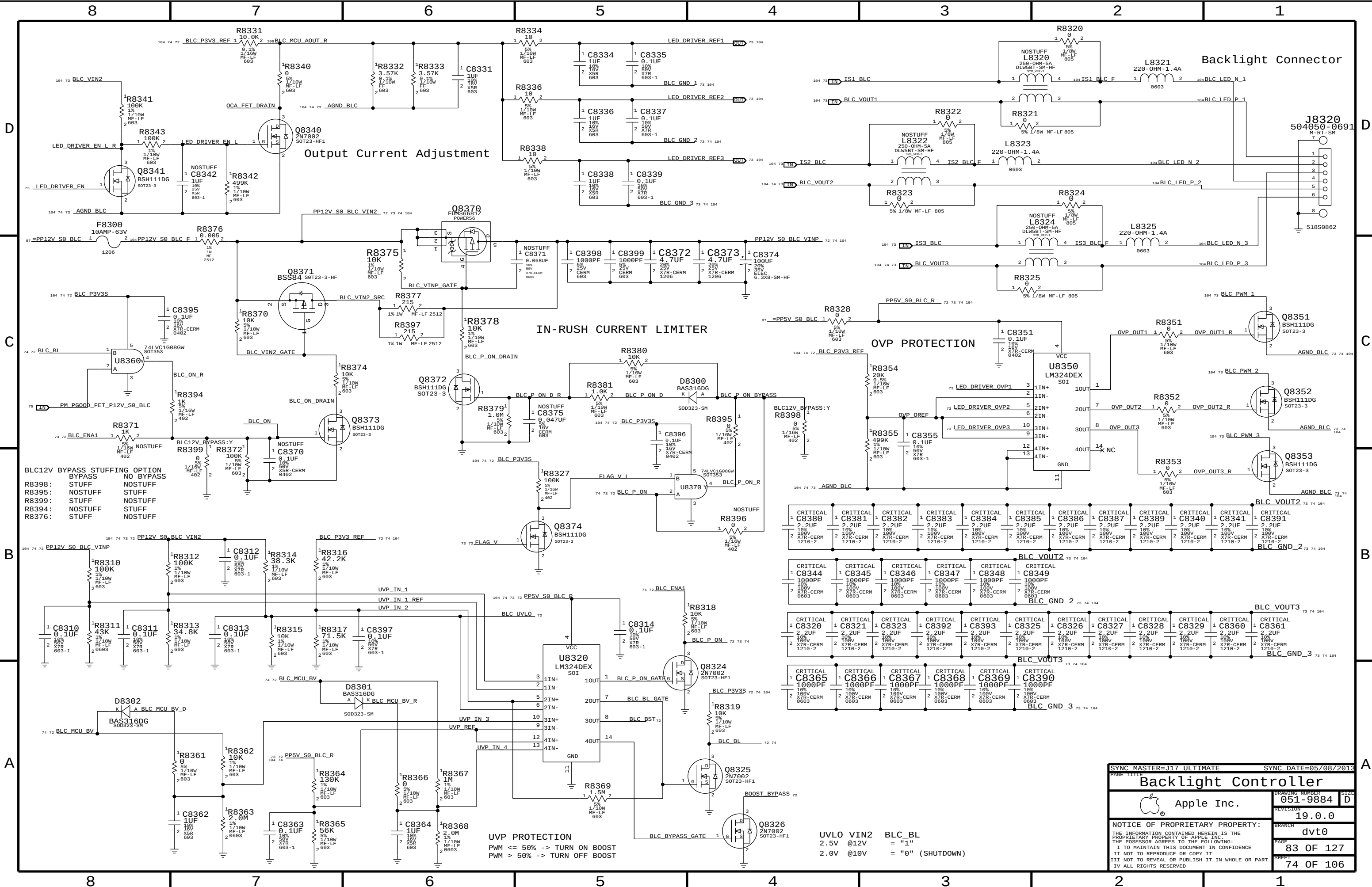
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Output Current Adjustment

IN-RUSH CURRENT LIMITER

OVP PROTECTION

Backlight Connector

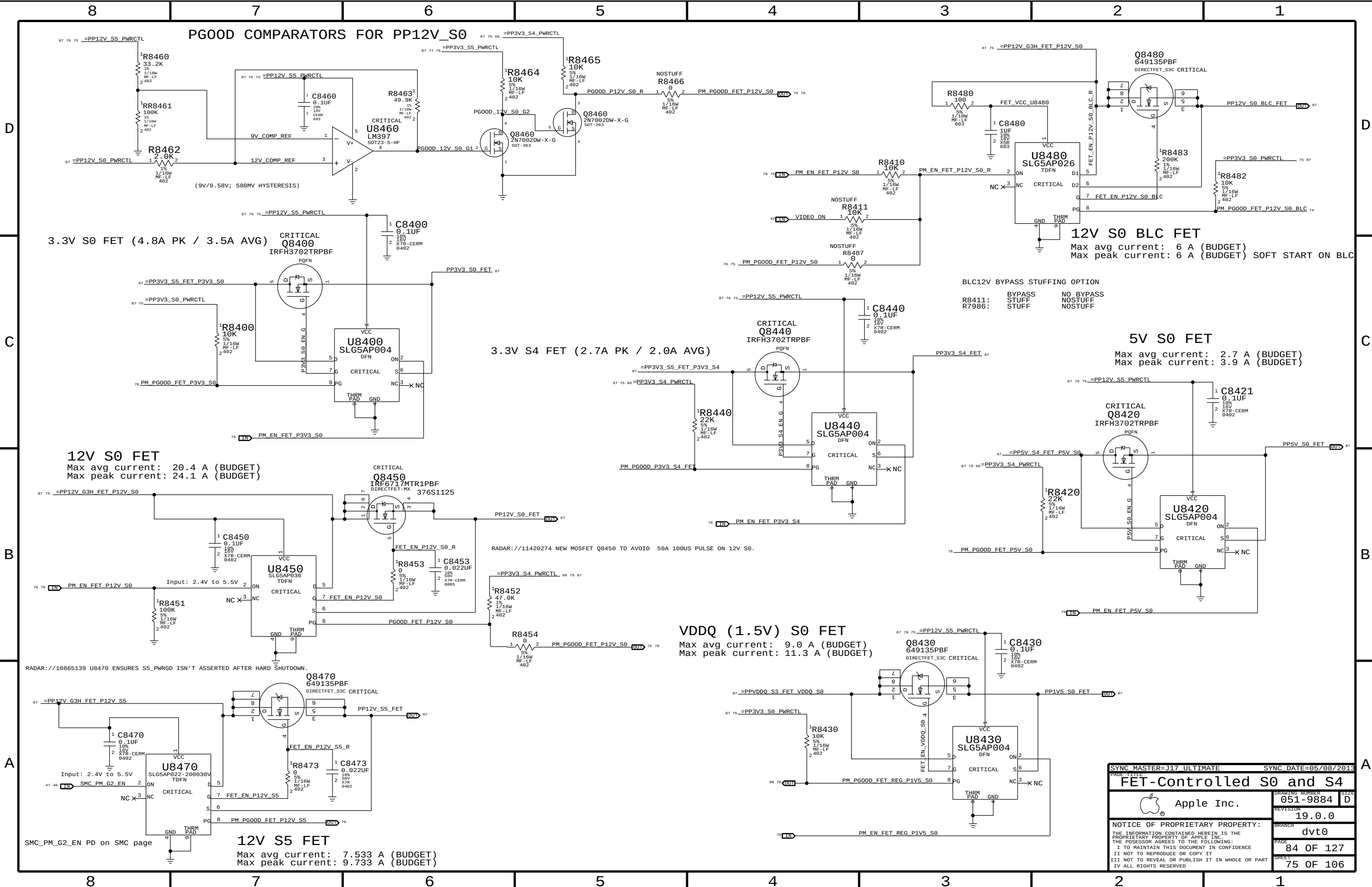
- BLC12V BYPASS STUFFING OPTION
BYPASS NO BYPASS
R8398: STUFF NOSTUFF
R8395: NOSTUFF STUFF
R8399: STUFF NOSTUFF
R8394: NOSTUFF STUFF
R8376: STUFF NOSTUFF

UVP PROTECTION
PWM <= 50% -> TURN ON BOOST
PWM > 50% -> TURN OFF BOOST

UVLO VIN2
2.5V @12V
2.0V @10V

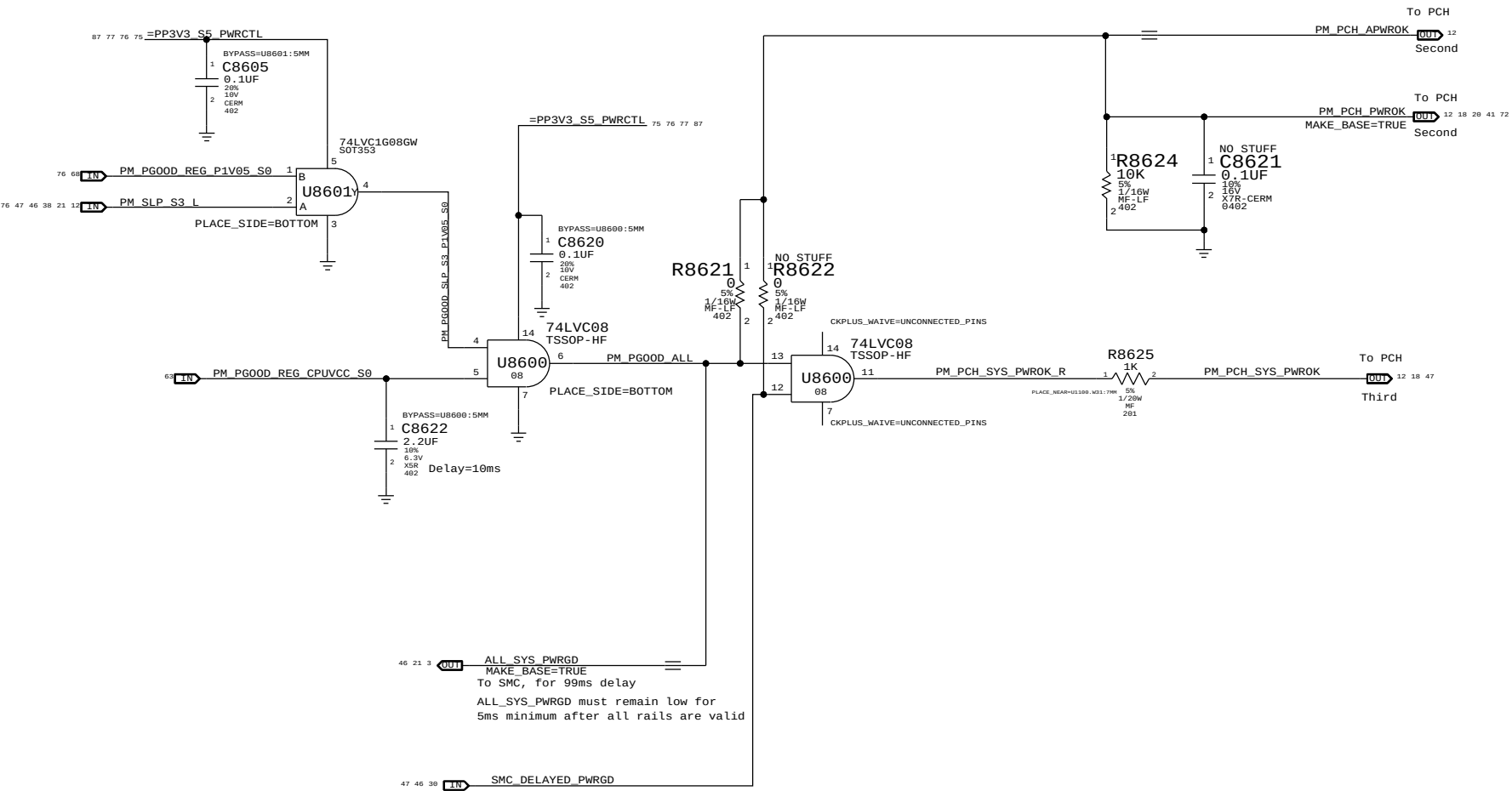
BLC_BL
= "1"
= "0" (SHUTDOWN)

SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/08/2013	
PAGE TITLE			
Backlight Controller			
Apple Inc.		DRAWING NUMBER	051-9884
		REVISION	19.0.0
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		PAGE	83 OF 127
		SHEET	74 OF 106



ALL_SYS_PWRGD, PCH_PWROK & SYS_PWROK Generation

PCH Power Goods



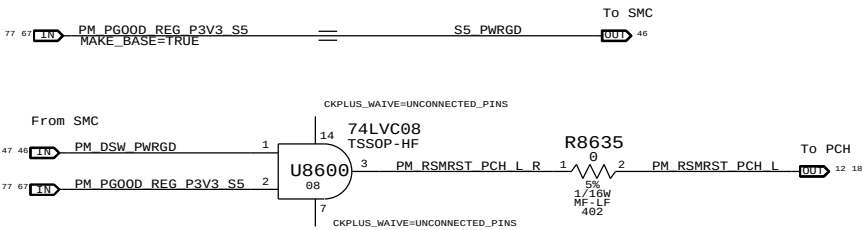
Resume Reset

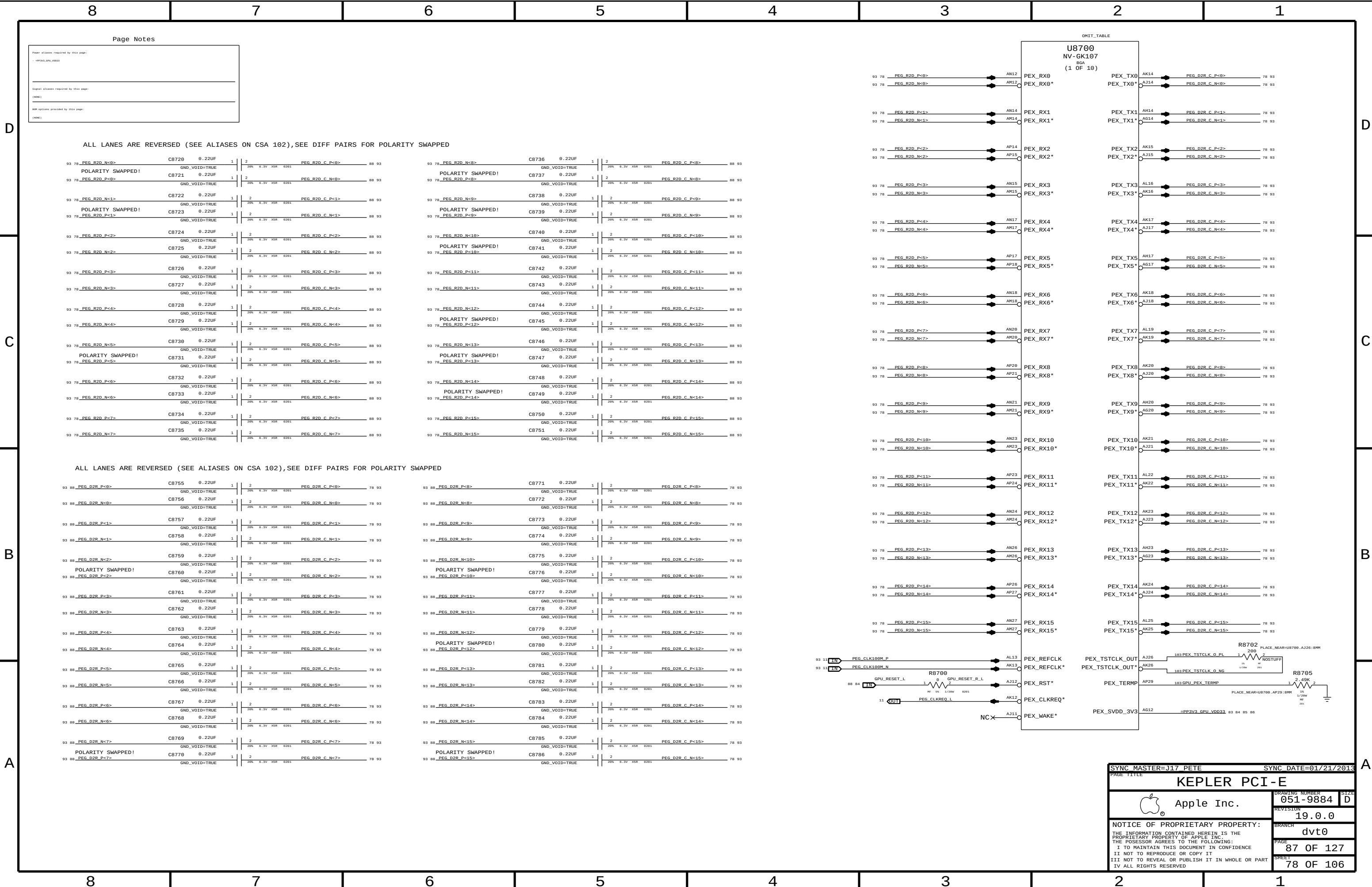
Intel Doc# 29517 Maho Bay PDG, Section 22.13
Intel Doc# 29562 Panther Point EDS, Section 8.7 and 8.8

Note:
The iMac J16/J17 designs does not support Deep Sx modes so both DPWROK and RSMRST# signals are shorted together

Requirements:
Power on:
Asserted at least 10 ms after all suspend well power is valid
Power off or loss of AC:
Transition to 0.8V or less before VccSUS3_3 drops to 2.90 V
to allow PCH to switch suspend well to battery without excessive loading

Method:
The SMC guarantees proper assertion and de-assertion of RSMRST# for normal operation via PM_DSW_PWRGD.
RSMRST# is asserted when power good from regulator is de-asserted in the event AC is lost. Power good de-assertion should happen quickly enough to meet Intel spec.





Page Notes

Power aliases required by this page:

...PP3V3_GPU_VDD33

Signal aliases required by this page:

(NONE)

ROM options provided by this page:

(NONE)

ALL LANES ARE REVERSED (SEE ALIASES ON CSA 102),SEE DIFF PAIRS FOR POLARITY SWAPPED

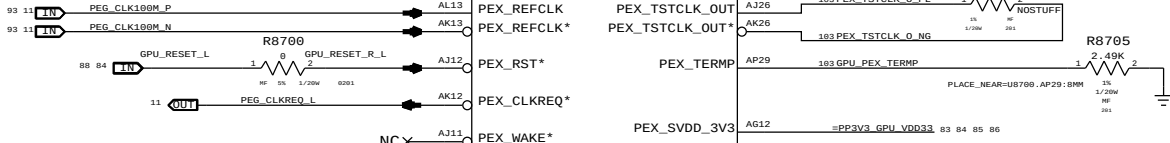
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78 OF 106		78 OF 106	

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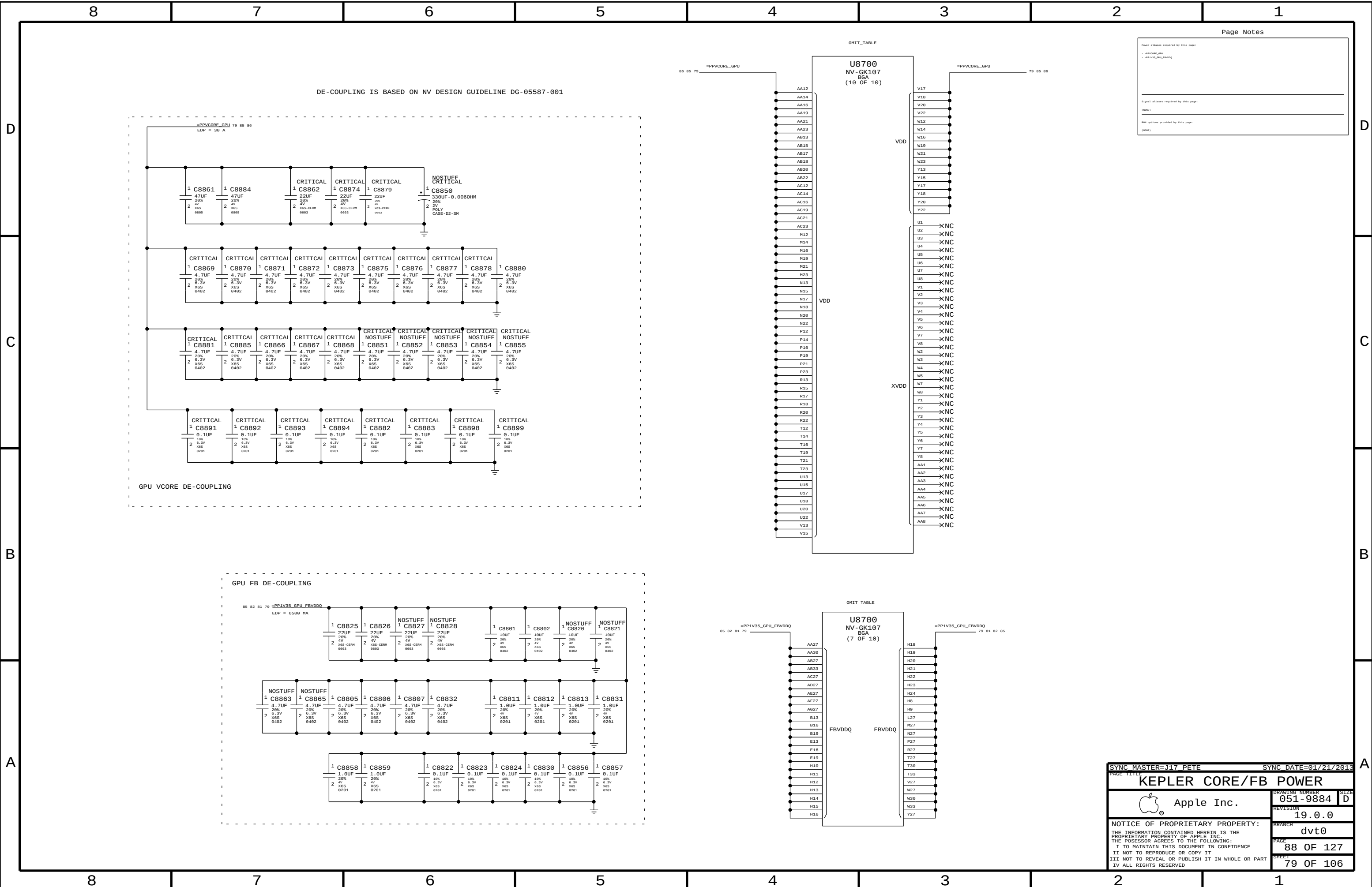
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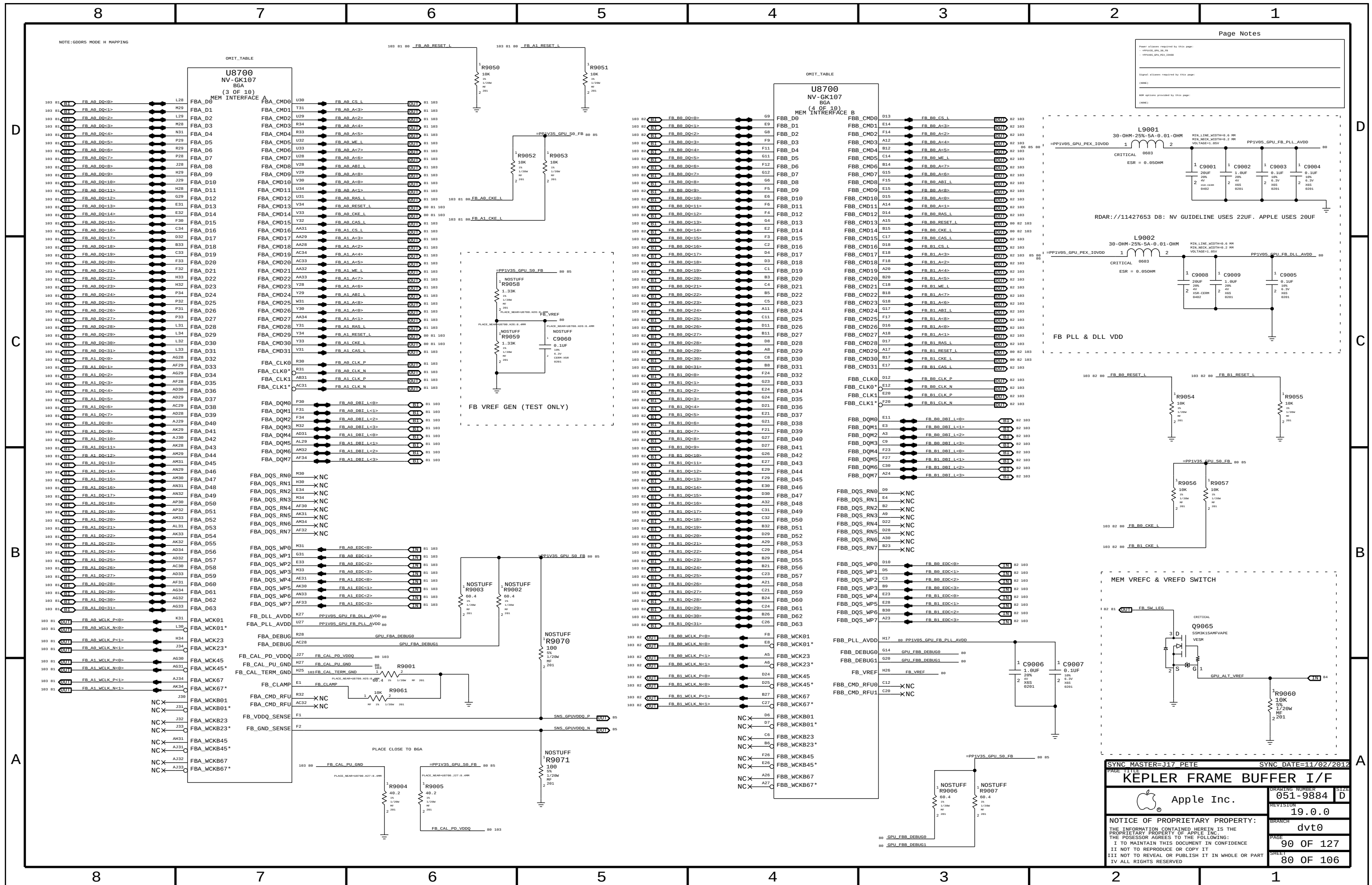
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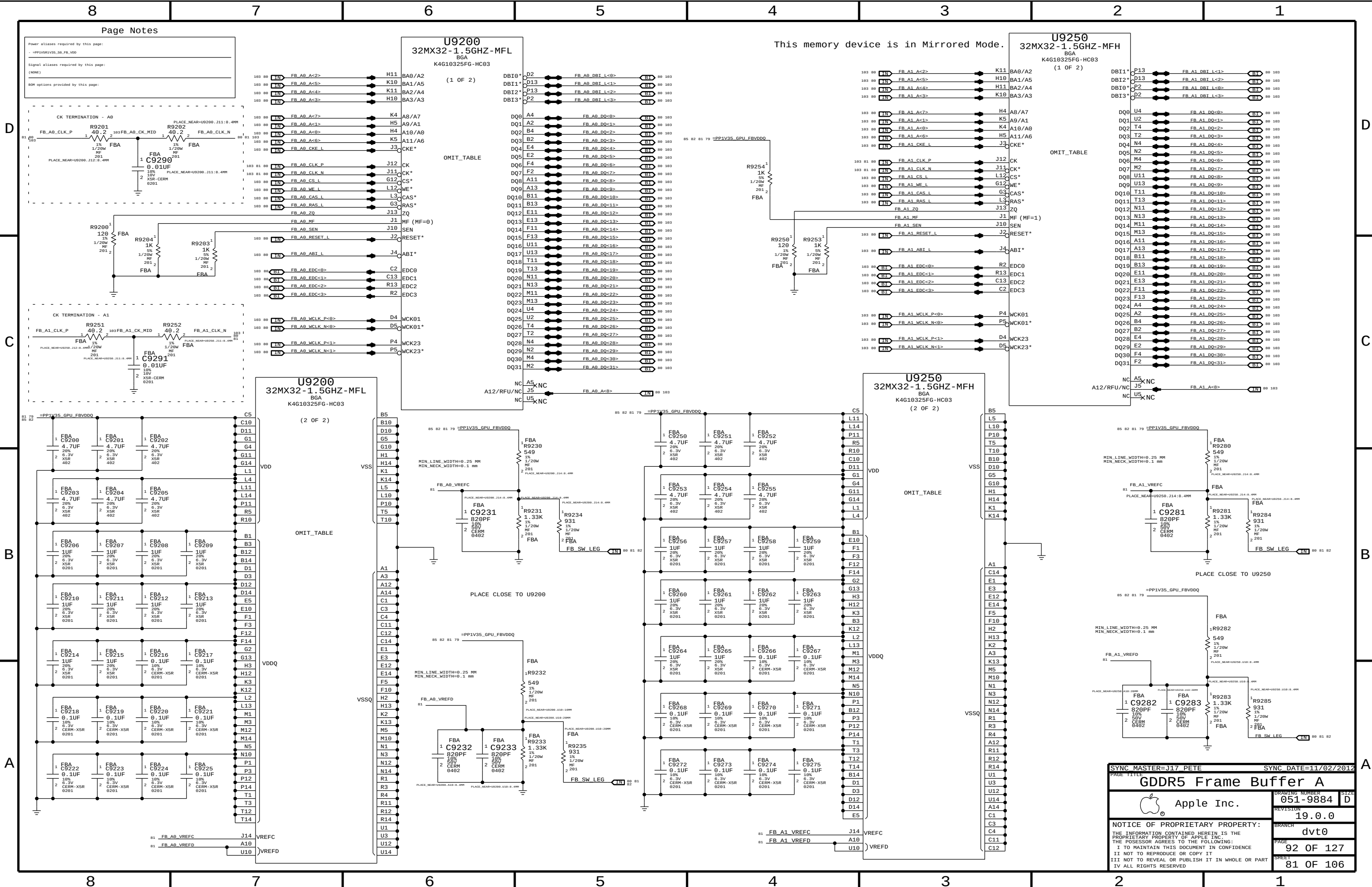
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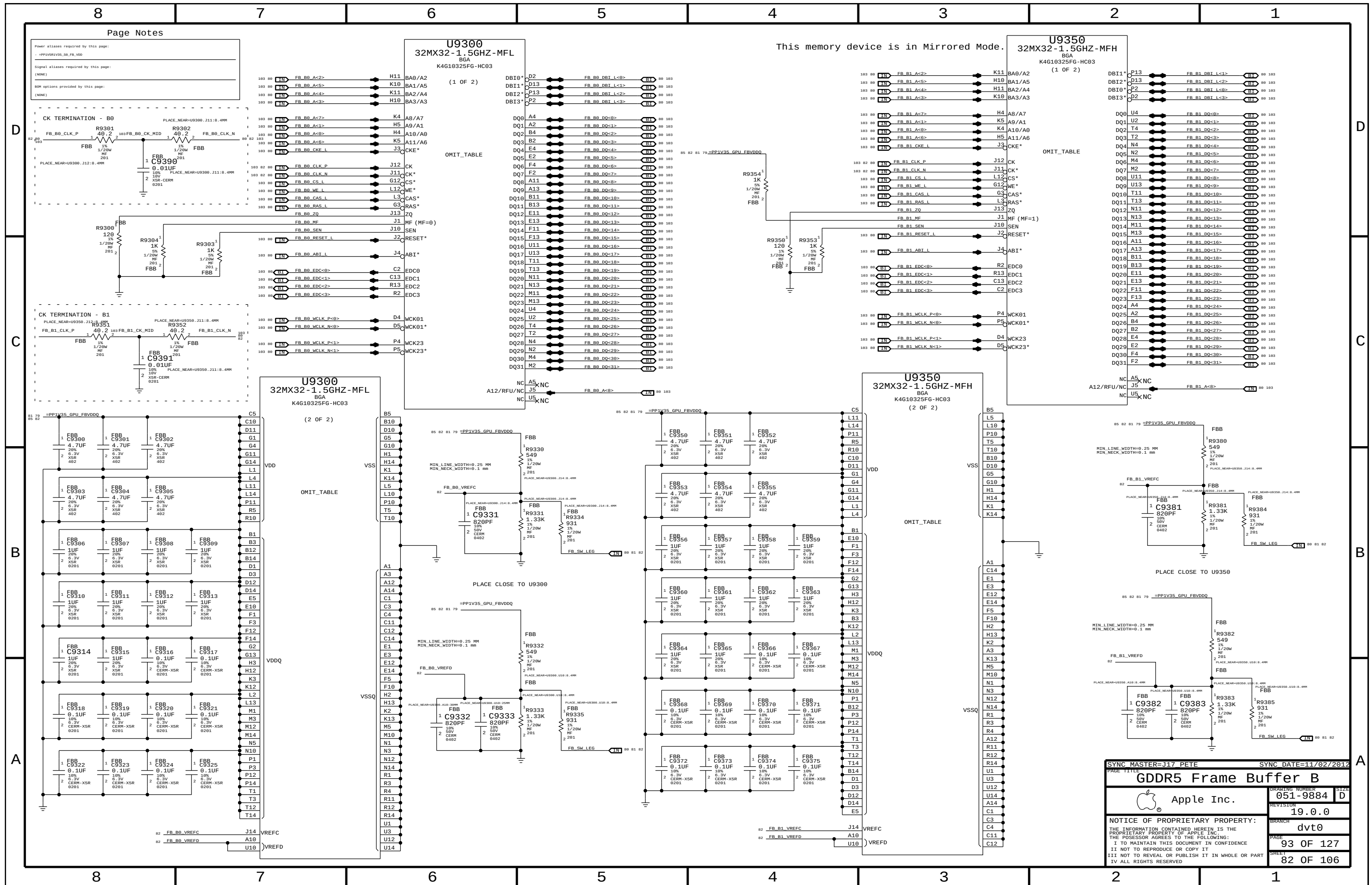
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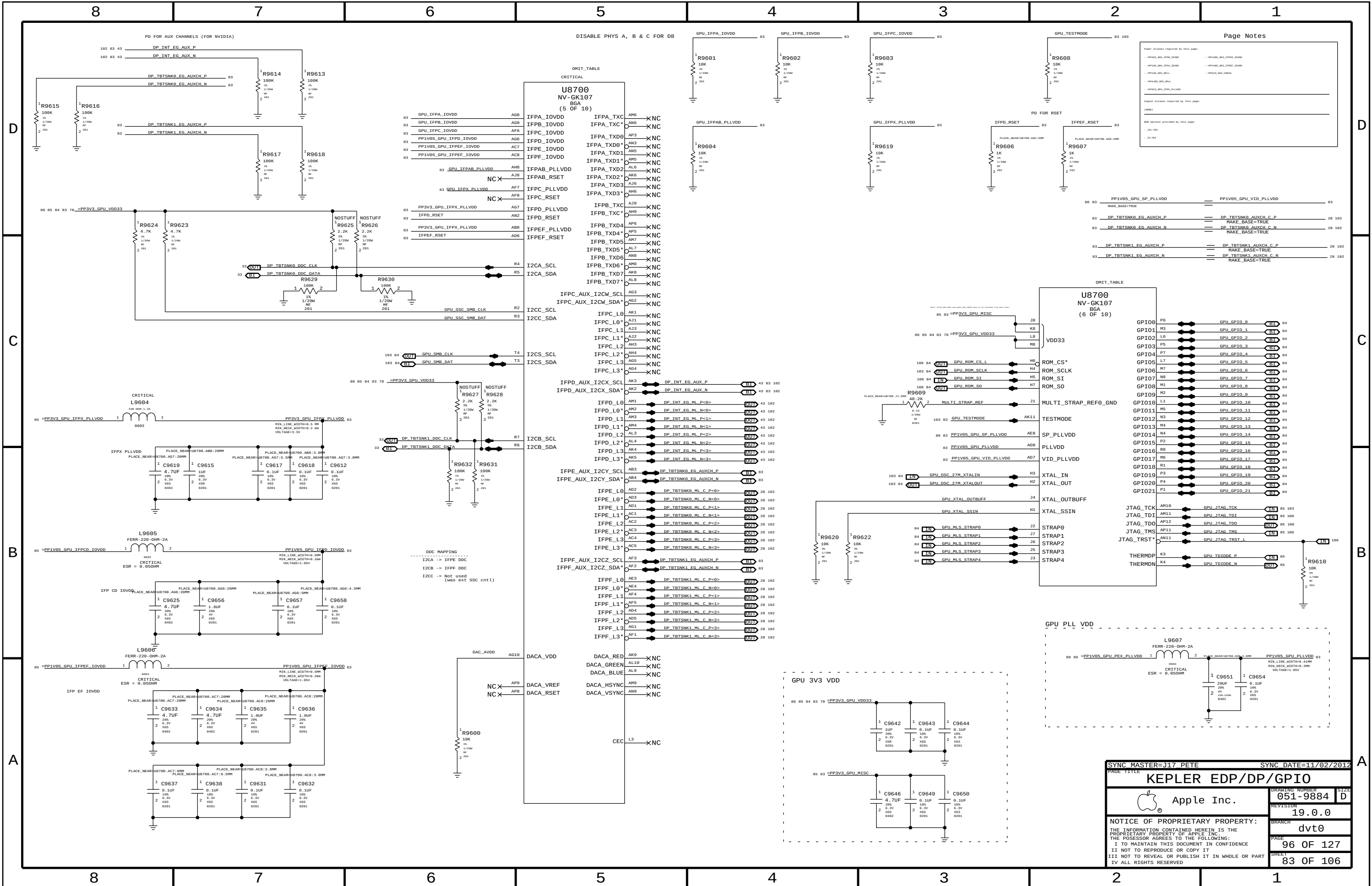
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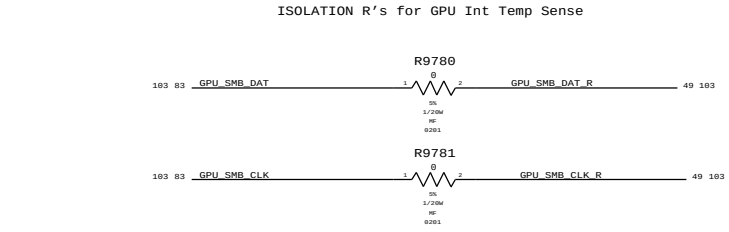
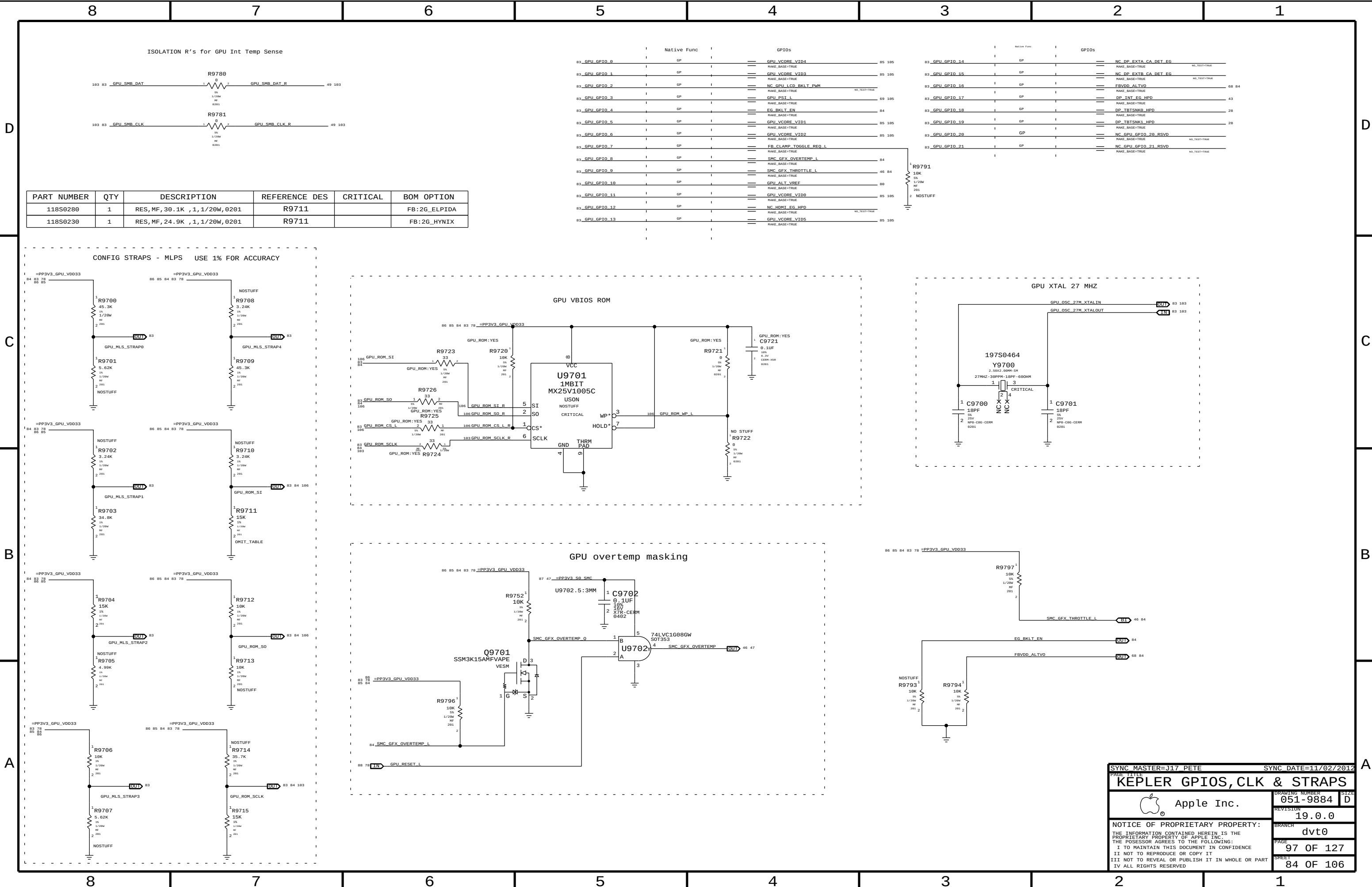
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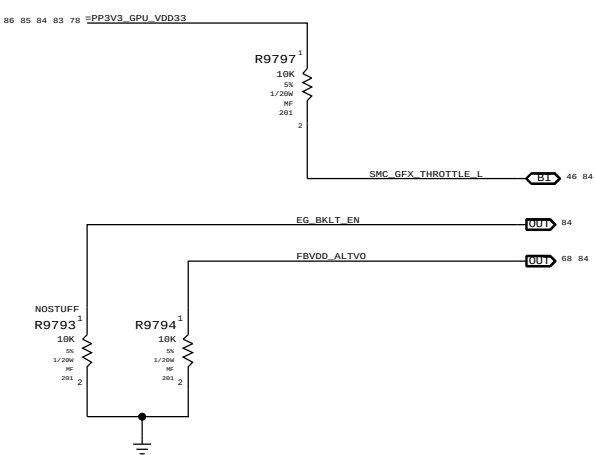
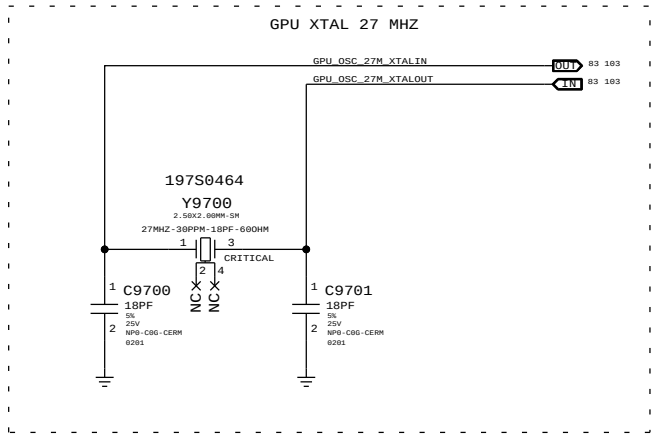
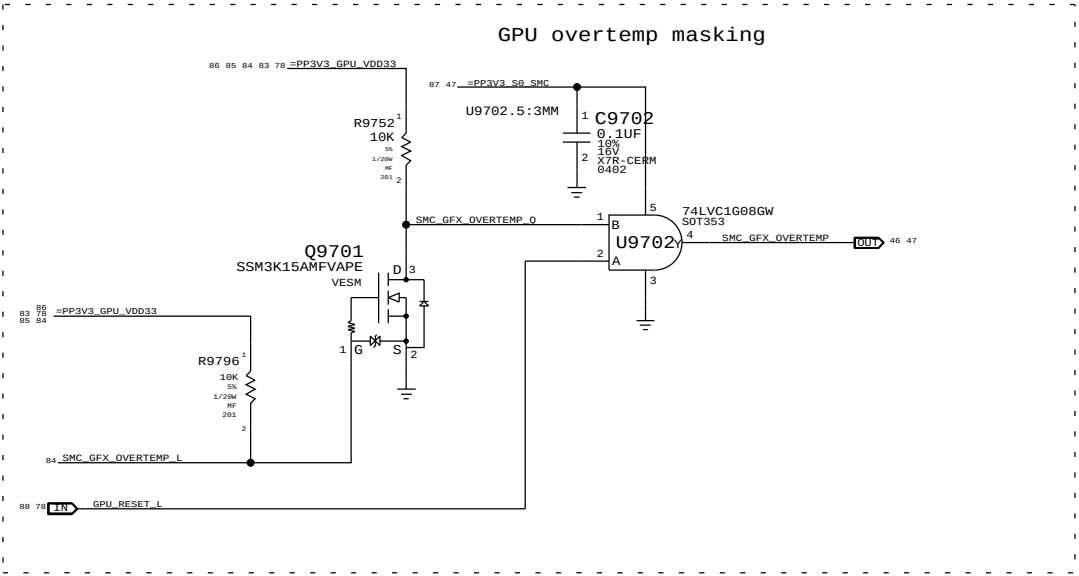
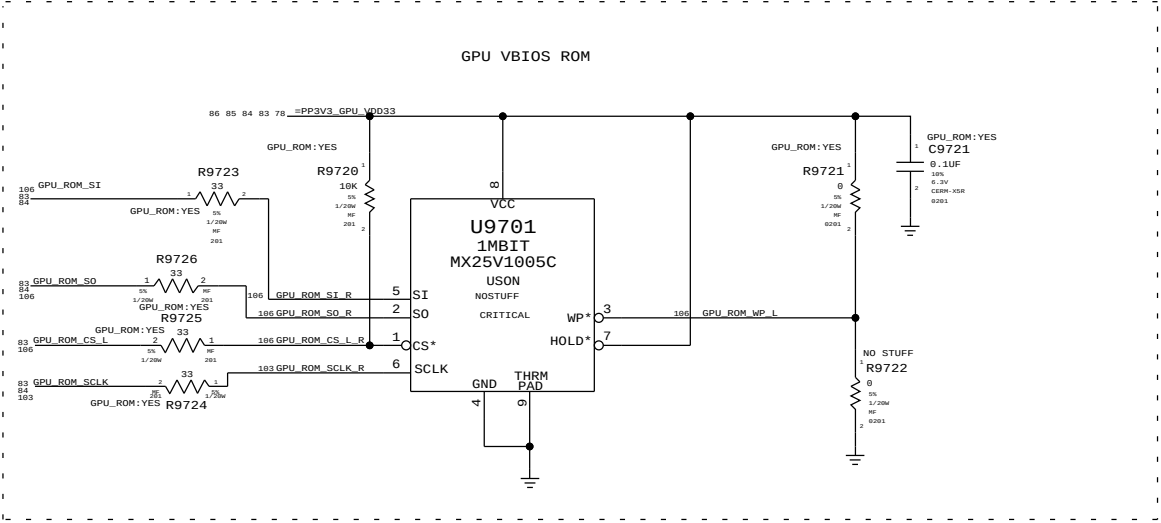
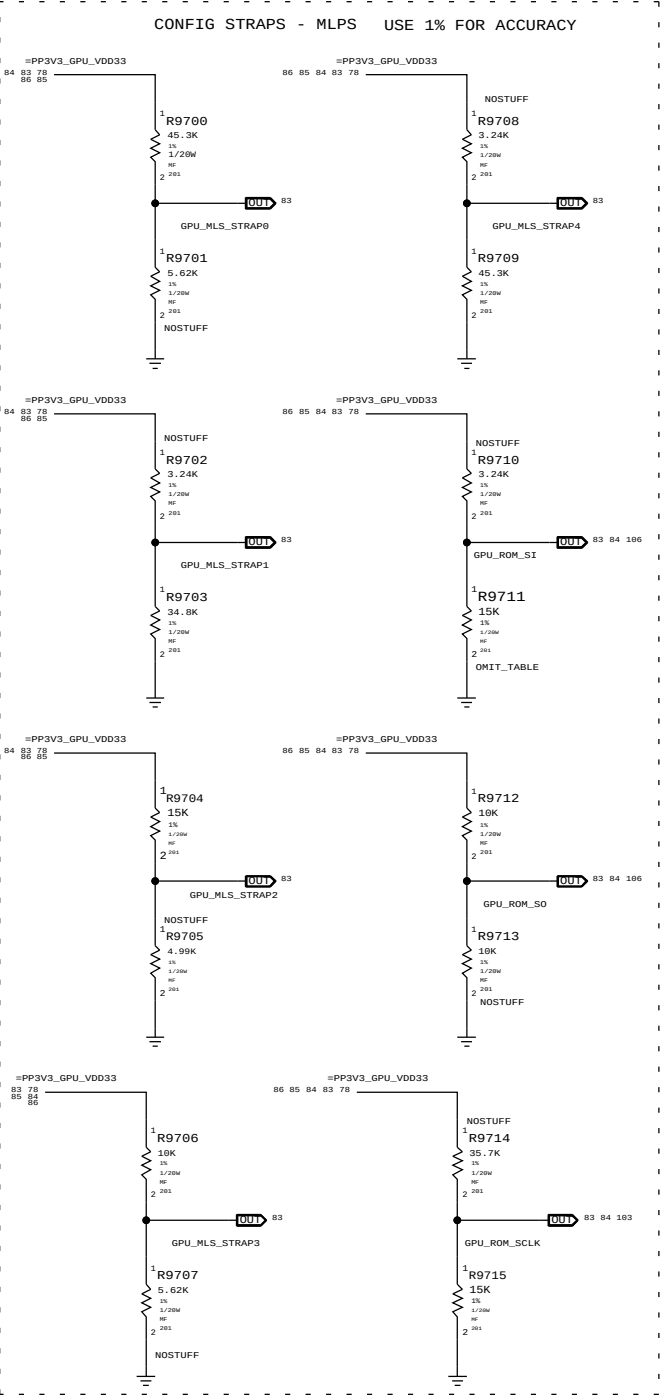
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Apple Inc.		DRAWING NUMBER	051-9884
		REVISION	19.0.0
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		PAGE	92 OF 127
		SHEET	81 OF 106







PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
118S0280	1	RES,MF,30.1K ,1,1/20W,0201	R9711		FB:2G_ELPIDA
118S0230	1	RES,MF,24.9K ,1,1/20W,0201	R9711		FB:2G_HYNIX



	Native Func	GPIOs
83_GPU_GPIO_0	GP	GPU_VCORE_VID4
83_GPU_GPIO_1	GP	GPU_VCORE_VID3
83_GPU_GPIO_2	GP	NC_GPU_LCD_BKLT_PWM
83_GPU_GPIO_3	GP	GPU_PSI_L
83_GPU_GPIO_4	GP	EG_BKLT_EN
83_GPU_GPIO_5	GP	GPU_VCORE_VID1
83_GPU_GPIO_6	GP	GPU_VCORE_VID2
83_GPU_GPIO_7	GP	FB_CLAMP_TOGGLE_REQ_L
83_GPU_GPIO_8	GP	SMC_GFX_OVERTEMP_L
83_GPU_GPIO_9	GP	SMC_GFX_THROTTLE_L
83_GPU_GPIO_10	GP	GPU_ALT_VREF
83_GPU_GPIO_11	GP	GPU_VCORE_VID0
83_GPU_GPIO_12	GP	NC_HDMI_EG_HPD
83_GPU_GPIO_13	GP	GPU_VCORE_VID5

	Native Func	GPIOs
83_GPU_GPIO_14	GP	NC_DP_EXTN_CA_DET_EG
83_GPU_GPIO_15	GP	NC_DP_EXTN_CA_DET_EG
83_GPU_GPIO_16	GP	FBVDD_ALTVDD
83_GPU_GPIO_17	GP	DP_INT_EG_HPD
83_GPU_GPIO_18	GP	DP_TBTSNK0_HPD
83_GPU_GPIO_19	GP	DP_TBTSNK1_HPD
83_GPU_GPIO_20	GP	NC_GPU_GPIO_20_RSVD
83_GPU_GPIO_21	GP	NC_GPU_GPIO_21_RSVD

SYNC MASTER=J17 PETE

SYNC DATE=11/02/2012

KEPLER GPIOs,CLK & STRAPS

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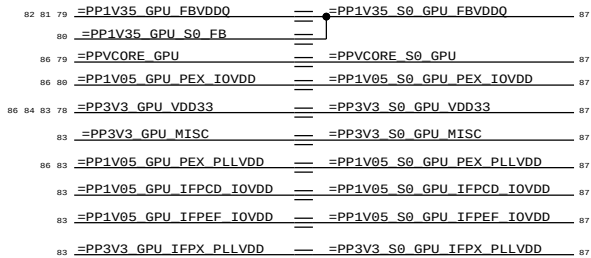
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PAGE
97 OF 127

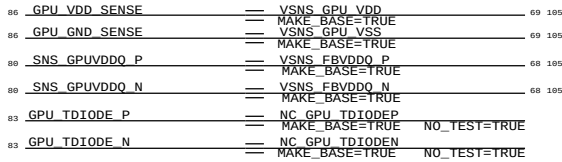
SHEET
84 OF 106

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GPU POWER ALIAS

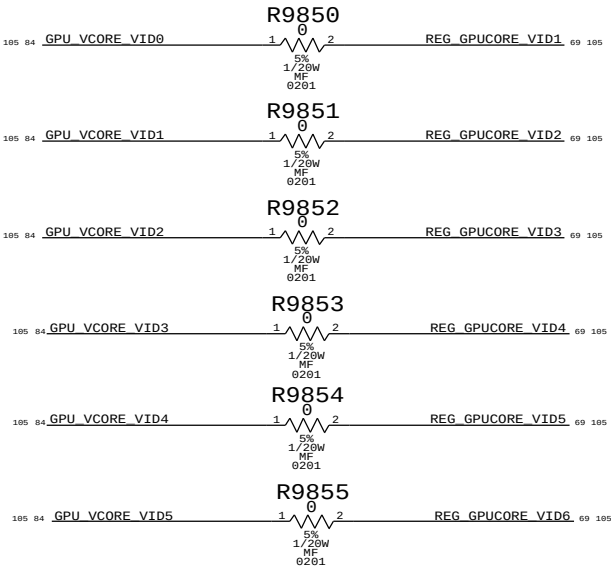


GPU SIGNAL & SENSE ALIAS



C

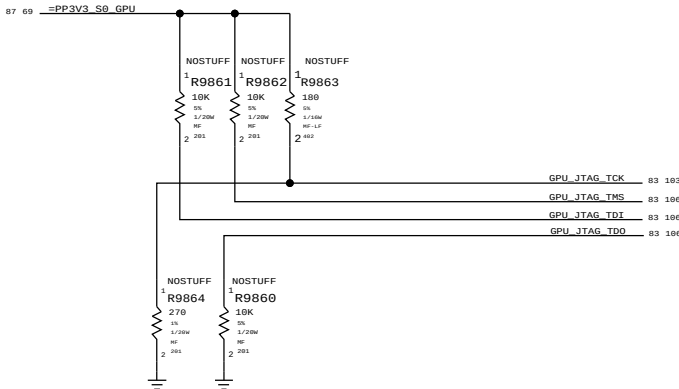
GPU VIDS ALIAS (VR VID0 IS TIED LOW)



D

B

PU/PD IS BASED ON RECOMMENDATION FROM NVIDIA FOR THEIR GPU JTAG DEBUGGER

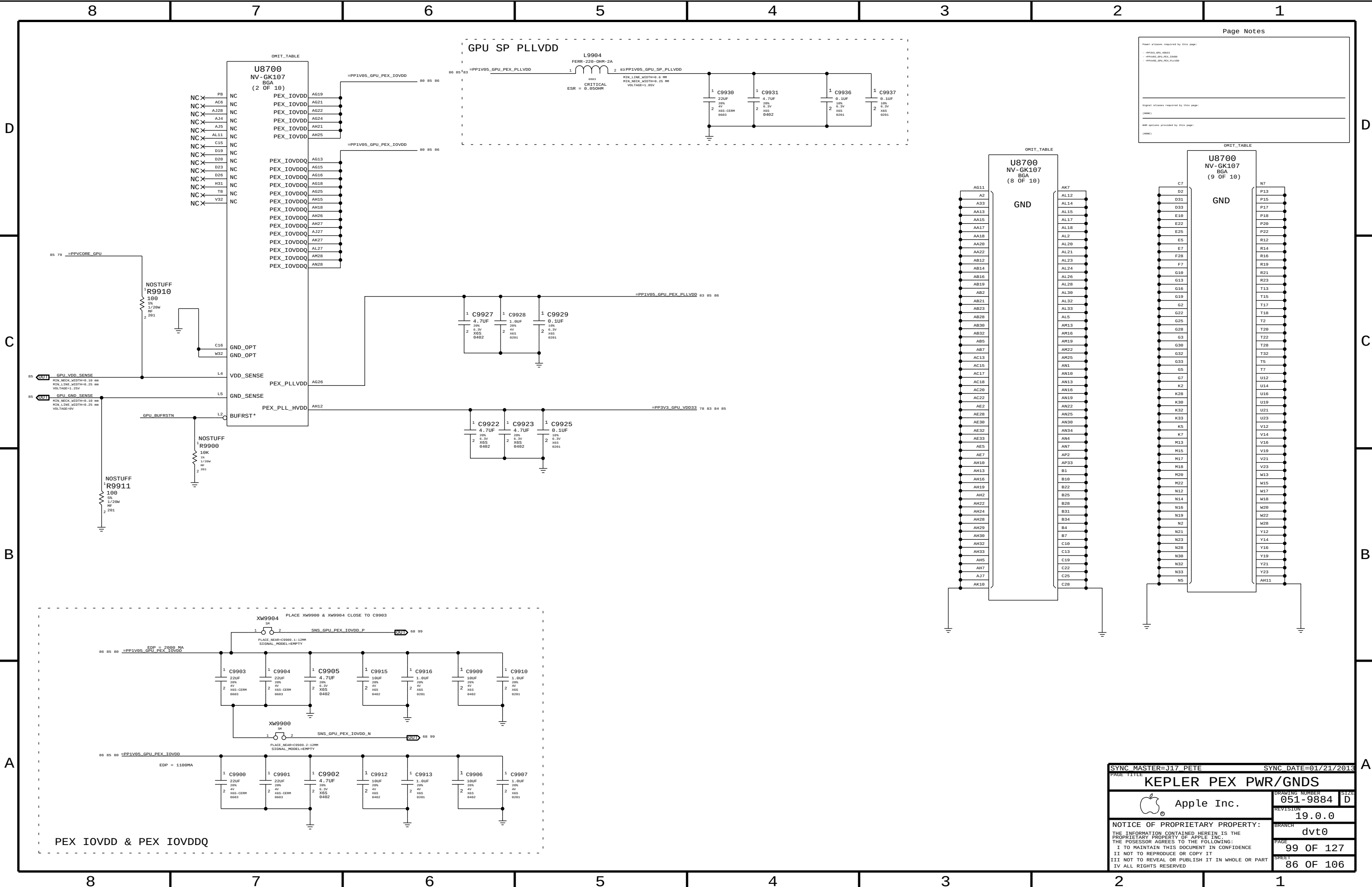


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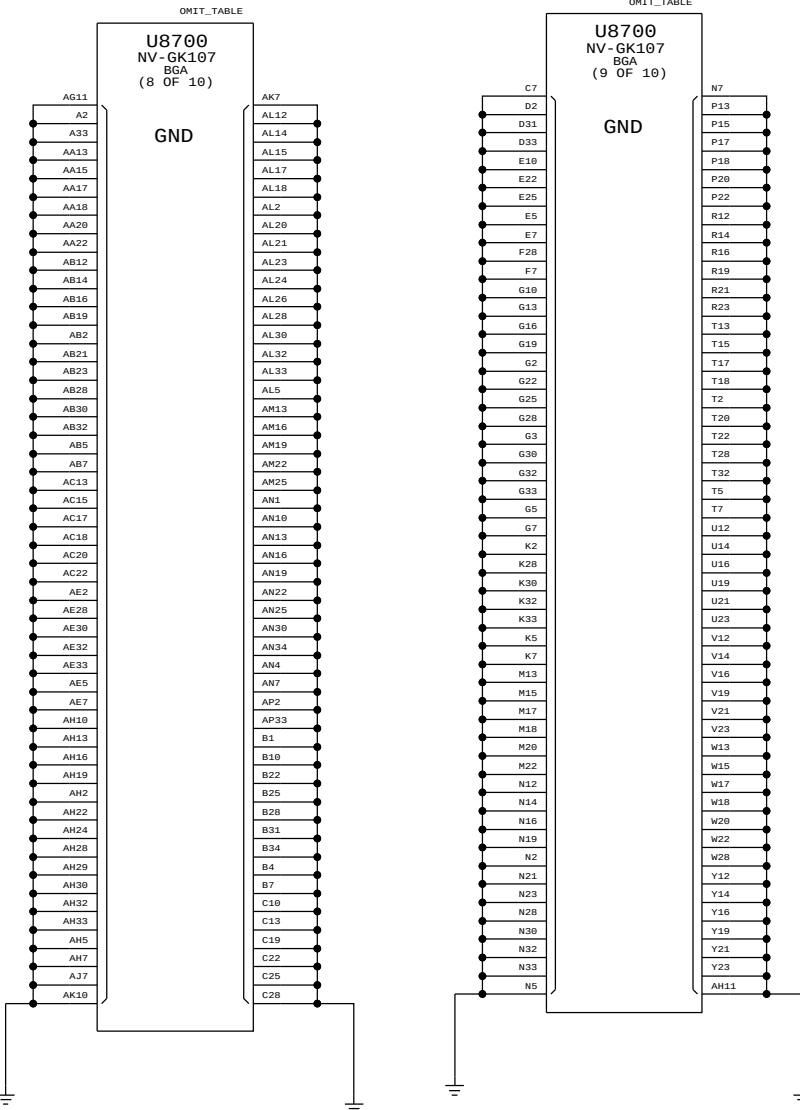
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		REVISION	19.0.0
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		PAGE	98 OF 127
		SHEET	85 OF 106

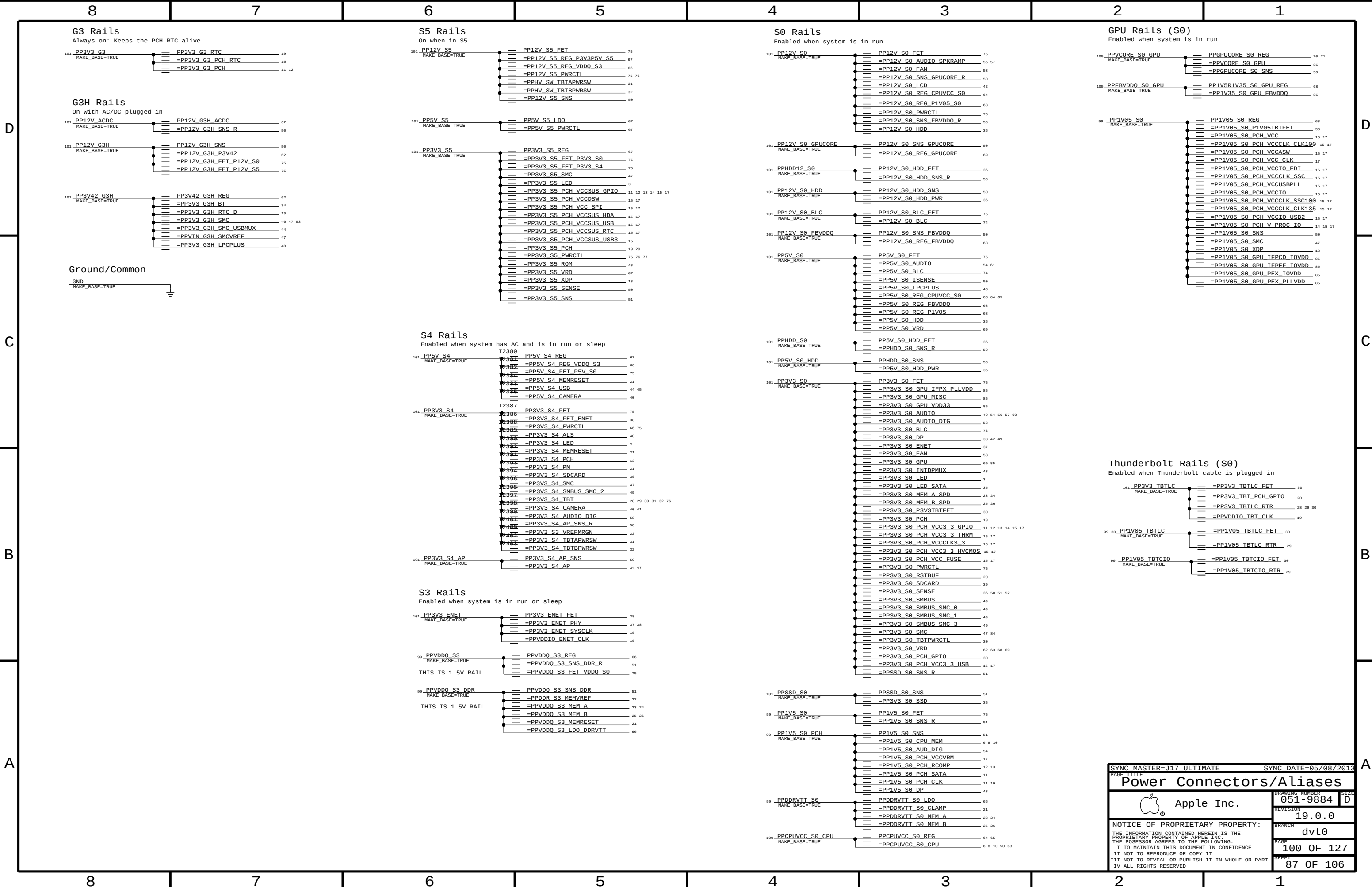
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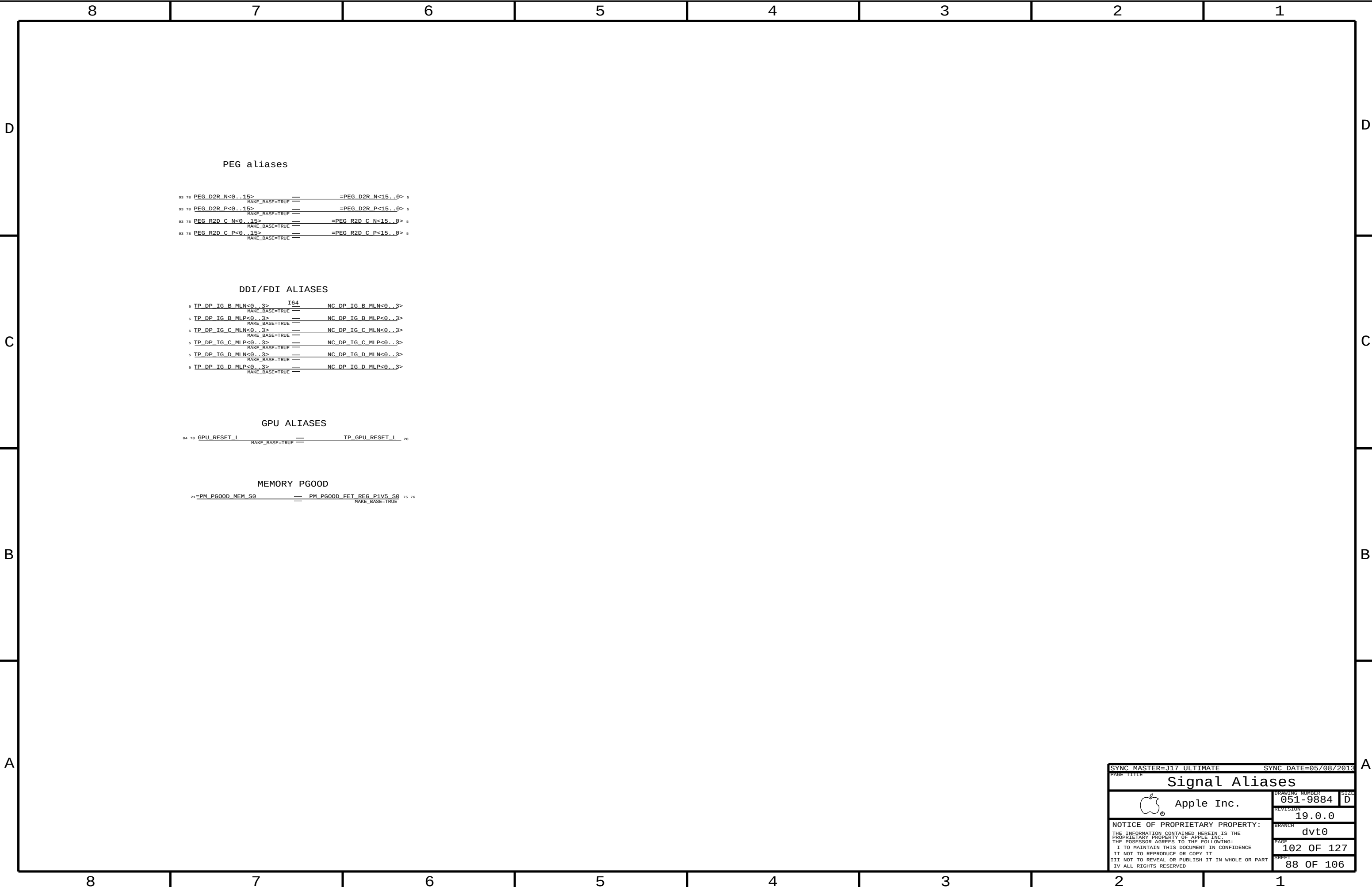


Page Notes	
Power aliases required by this page:	
- PP1V05_GPU_VDDQ1	
- PP1V05_GPU_PEX_PLLVDD	
- PP1V05_GPU_PEX_PLLVDD	
Signal aliases required by this page:	
(NONE)	
SOP options provided by this page:	
(NONE)	



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PEG aliases

93 78 PEG_D2R_N<0..15> == =PEG_D2R_N<15..0> 5
MAKE_BASE=TRUE ==
93 78 PEG_D2R_P<0..15> == =PEG_D2R_P<15..0> 5
MAKE_BASE=TRUE ==
93 78 PEG_R2D_C_N<0..15> == =PEG_R2D_C_N<15..0> 5
MAKE_BASE=TRUE ==
93 78 PEG_R2D_C_P<0..15> == =PEG_R2D_C_P<15..0> 5
MAKE_BASE=TRUE ==

DDI/FDI ALIASES

5 TP_DP_IG_B_MLN<0..3> I64 == NC_DP_IG_B_MLN<0..3>
MAKE_BASE=TRUE ==
5 TP_DP_IG_B_MLP<0..3> == NC_DP_IG_B_MLP<0..3>
MAKE_BASE=TRUE ==
5 TP_DP_IG_C_MLN<0..3> == NC_DP_IG_C_MLN<0..3>
MAKE_BASE=TRUE ==
5 TP_DP_IG_C_MLP<0..3> == NC_DP_IG_C_MLP<0..3>
MAKE_BASE=TRUE ==
5 TP_DP_IG_D_MLN<0..3> == NC_DP_IG_D_MLN<0..3>
MAKE_BASE=TRUE ==
5 TP_DP_IG_D_MLP<0..3> == NC_DP_IG_D_MLP<0..3>
MAKE_BASE=TRUE ==

GPU ALIASES

84 78 GPU_RESET_L == TP_GPU_RESET_L 20
MAKE_BASE=TRUE ==

MEMORY PG00D

21 ==PM_PG00D_MEM_S0 == PM_PG00D_FET_REG_P1V5_S0 75 76
MAKE_BASE=TRUE ==

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Signal Aliases

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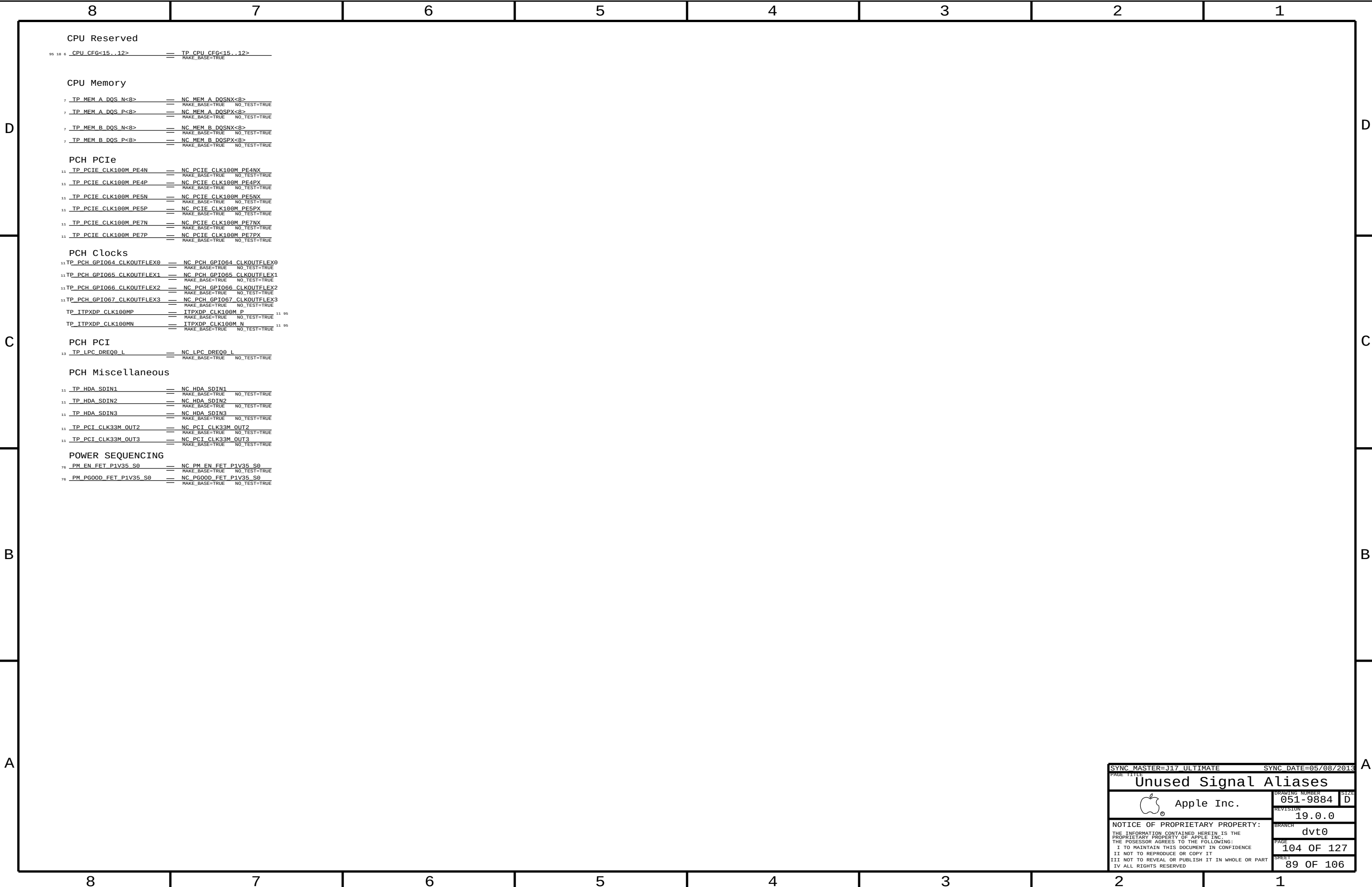
dvt0

PAGE

102 OF 127

SHEET

88 OF 106



8	7	6	5	4	3	2	1
D							D
C							C
B							B
A							A
8	7	6	5	4	3	2	1

SYNC MASTER=J17 ULTIMATE

SYNC DATE=05/08/2013

PAGE TITLE

Functional / ICT Test

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19.0.0

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PAGE

105 OF 127

SHEET

90 OF 106

DDR3

DDR3-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DDR_34S	*	=34_OHM_SE	=34_OHM_SE	=34_OHM_SE	=34_OHM_SE	=STANDARD	=STANDARD
DDR_39S	*	=39_OHM_SE	=39_OHM_SE	=39_OHM_SE	=39_OHM_SE	=STANDARD	=STANDARD
DDR_42S	*	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=STANDARD	=STANDARD
DDR_42S_D	*	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	0.1016 MM	0.1016 MM
DDR_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
DDR_68D	*	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF
DDR_COMP	*	Y	0.305 MM	0.105 MM	=STANDARD	=STANDARD	=STANDARD

Minimum diff spacing is 4 mil
Table 4-5, Intel Doc# 486712

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
POWER_ODR_P4MM	*	Y	0.400 MM	0.100 MM	3.0 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
POWER_DDR	*	POWER_DDR_P4MM
DDR_CLK_PHY	*	DDR_68D
DDR_CTRL_PHY	*	DDR_39S
DDR_CMD_PHY	*	DDR_34S
DDR_DQ_PHY	*	DDR_42S
DDR_DQS_PHY	*	DDR_42S_D
DDR_COMP_PHY	*	DDR_COMP

DDR3 Power-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
POWER_DDR	*	=2:1_SPACING	?

DDR3-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DDR_CLK_ISO	*	=5:1_SPACING	?
DDR_CTRL_ISO	*	=3.5:1_SPACING	?
DDR_CTRL2CTRL	*	=2.5:1_SPACING	?
DDR_CMD_ISO	*	=3.5:1_SPACING	?
DDR_CMD2CMD	*	=2:1_SPACING	?
DDR_DATA_ISO	*	=3:1_SPACING	?
DDR_DQ2DQ	*	=2:1_SPACING	900
DDR_DQ2DQS	*	=3:1_SPACING	?
DDR_BL2BL	*	=3:1_SPACING	?
DDR_CH2CH	*	=6:5:1_SPACING	?
DDR_COMP_ISO	*	0.381 MM	?

Main Segment Min Spacing Rules (mils) (Shark Bay PDG, Intel Doc# 486712)

Table	Trace	Design	Iso	Design	Comments
4-2	4	(diff)	15	19.69	CLK trace spacing controlled by =68_OHM_DIFF
4-3	8	9.84	12	13.78	
4-4	6	7.87	12	13.78	
4-5	8.5	7.87	12	11.81	DQ or DQS to other signals not in the same bytelane (but not ch)
					DQ to DQ in the same bytelane of the same channel
			10	11.81	DQ to DQS in the same bytelane of the same channel
			12	11.81	DQ or DQS in different bytelanes of the same channel
			25	25.59	DQ or DQS in different channels
			-	25.59	DDR3 to any other signal not DDR3

Constraints

Clocks: CK[3:0], CK#[3:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_CLK	*	*	DDR_CLK_ISO

Control: CS#[3:0], CKE[3:0], ODT[3:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_CTRL	*	*	DDR_CTRL_ISO
DDR_CTRL	DDR_CTRL	*	DDR_CTRL2CTRL

Command: MA[15:0], RAS#, CAS#, WE# BS[2:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_CMD	*	*	DDR_CMD_ISO
DDR_CMD	DDR_CMD	*	DDR_CMD2CMD

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_COMP	*	*	DDR_COMP_ISO

Data: DQS[7:0], DQS#[7:0], DQ[63:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_A_DQ_BYTE*	*	*	DDR_DATA_ISO
DDR_A_DQS*	*	*	DDR_DATA_ISO
DDR_B_DQ_BYTE*	*	*	DDR_DATA_ISO
DDR_B_DQS*	*	*	DDR_DATA_ISO
DDR_*_DQ_BYTE*	=SAME	*	DDR_DQ2DQ
DDR_A_DQ_BYTE*	DDR_A_DQS*	*	DDR_DQ2DQS
DDR_A_DQ_BYTE*	DDR_A_DQ_BYTE*	*	DDR_BL2BL
DDR_B_DQ_BYTE*	DDR_B_DQS*	*	DDR_DQ2DQS
DDR_B_DQ_BYTE*	DDR_B_DQ_BYTE*	*	DDR_BL2BL
DDR_A_*	DDR_B_*	*	DDR_CH2CH

See Note (3)

See Note (1)

See Note (3)

See Note (1)

See Note (2)

Note (1):

Deliberately set DQ to DQS spacing to 3:1 to avoid adding complexity to constraints, even though it can be less. Only one rule per channel is needed by trading off a little space.

Note (2):

Intel suggested 25 mil (0.65 mm) spacing for via to channel, and via to pad to two different channels. D0R3 draws about 20 mA per trace with edge rates in the 100s of ps. The main coupling mechanism is capacitive. A 0.65 mm spacing is used for power nets, which draw far more current (inductive coupling however). These rules are far too conservative. To meet these rules, the spacing must be applied to the net.

Note (3):

In order for the constraints $DDR_*_DQ_BYTE*$ to =SAME to win out over $DDR\{A,B\}_DQ_BYTE*$ to $DDR\{A,B\}_DQ_BYTE*$ so that the small intra-bytelane spacing is used, the spacing rule DDR_DQ2DQ must have a weight greater than DDR_BL2BL .

DDR3

Electrical Constraint Set		Physical	Spacing	
Channel A				
H177	DDR_A_CLK0	DDR_CLK_PHY	DDR_CLK	MEM_A_CLK_P<1..0> 7 23
H177	DDR_A_CLK0	DDR_CLK_PHY	DDR_CLK	MEM_A_CLK_N<1..0> 7 23
H255	DDR_A_CLK1	DDR_CLK_PHY	DDR_CLK	MEM_A_CLK_P<3..2> 7 24
H255	DDR_A_CLK1	DDR_CLK_PHY	DDR_CLK	MEM_A_CLK_N<3..2> 7 24
H189	DDR_A_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_A_CKE<1..0> 7 23
H189	DDR_A_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_A_CS_L<1..0> 7 23
H189	DDR_A_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_A_ODT<1..0> 7 23
H240	DDR_A_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_A_CKE<3..2> 7 24
H240	DDR_A_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_A_CS_L<3..2> 7 24
H240	DDR_A_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_A_ODT<3..2> 7 24
H189	DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM_A_A<15..0> 7 23 24
H216	DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM_A_BA<2..0> 7 23 24
H255	DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM_A_RAS_L 7 23 24
H255	DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM_A_CAS_L 7 23 24
H189	DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM_A_WE_L 7 23 24
H189	DDR_A_DQ_BYTE0	DDR_DQ_PHY	DDR_A_DQ_BYTE0	MEM_A_DQ<7..0> 7 27
H189	DDR_A_DQ_BYTE1	DDR_DQ_PHY	DDR_A_DQ_BYTE1	MEM_A_DQ<15..8> 7 27
H189	DDR_A_DQ_BYTE2	DDR_DQ_PHY	DDR_A_DQ_BYTE2	MEM_A_DQ<23..16> 7 27
H189	DDR_A_DQ_BYTE3	DDR_DQ_PHY	DDR_A_DQ_BYTE3	MEM_A_DQ<31..24> 7 27
H189	DDR_A_DQ_BYTE4	DDR_DQ_PHY	DDR_A_DQ_BYTE4	MEM_A_DQ<39..32> 7 27
H189	DDR_A_DQ_BYTE5	DDR_DQ_PHY	DDR_A_DQ_BYTE5	MEM_A_DQ<47..40> 7 27
H189	DDR_A_DQ_BYTE6	DDR_DQ_PHY	DDR_A_DQ_BYTE6	MEM_A_DQ<55..48> 7 27
H189	DDR_A_DQ_BYTE7	DDR_DQ_PHY	DDR_A_DQ_BYTE7	MEM_A_DQ<63..56> 7 27
H189	DDR_A_DQS0	DDR_DQS_PHY	DDR_A_DQS0	MEM_A_DQS_P<0> 7 27
H189	DDR_A_DQS0	DDR_DQS_PHY	DDR_A_DQS0	MEM_A_DQS_N<0> 7 27
H189	DDR_A_DQS1	DDR_DQS_PHY	DDR_A_DQS1	MEM_A_DQS_P<1> 7 27
H189	DDR_A_DQS1	DDR_DQS_PHY	DDR_A_DQS1	MEM_A_DQS_N<1> 7 27
H189	DDR_A_DQS2	DDR_DQS_PHY	DDR_A_DQS2	MEM_A_DQS_P<2> 7 27
H189	DDR_A_DQS2	DDR_DQS_PHY	DDR_A_DQS2	MEM_A_DQS_N<2> 7 27
H189	DDR_A_DQS3	DDR_DQS_PHY	DDR_A_DQS3	MEM_A_DQS_P<3> 7 27
H189	DDR_A_DQS3	DDR_DQS_PHY	DDR_A_DQS3	MEM_A_DQS_N<3> 7 27
H189	DDR_A_DQS4	DDR_DQS_PHY	DDR_A_DQS4	MEM_A_DQS_P<4> 7 27
H189	DDR_A_DQS4	DDR_DQS_PHY	DDR_A_DQS4	MEM_A_DQS_N<4> 7 27
H189	DDR_A_DQS5	DDR_DQS_PHY	DDR_A_DQS5	MEM_A_DQS_P<5> 7 27
H189	DDR_A_DQS5	DDR_DQS_PHY	DDR_A_DQS5	MEM_A_DQS_N<5> 7 27
H189	DDR_A_DQS6	DDR_DQS_PHY	DDR_A_DQS6	MEM_A_DQS_P<6> 7 27
H189	DDR_A_DQS6	DDR_DQS_PHY	DDR_A_DQS6	MEM_A_DQS_N<6> 7 27
H189	DDR_A_DQS7	DDR_DQS_PHY	DDR_A_DQS7	MEM_A_DQS_P<7> 7 27
H189	DDR_A_DQS7	DDR_DQS_PHY	DDR_A_DQS7	MEM_A_DQS_N<7> 7 27
Channel B				
H226	DDR_B_CLK0	DDR_CLK_PHY	DDR_CLK	MEM_B_CLK_P<1..0> 7 25
H226	DDR_B_CLK0	DDR_CLK_PHY	DDR_CLK	MEM_B_CLK_N<1..0> 7 25
H255	DDR_B_CLK1	DDR_CLK_PHY	DDR_CLK	MEM_B_CLK_P<3..2> 7 26
H255	DDR_B_CLK1	DDR_CLK_PHY	DDR_CLK	MEM_B_CLK_N<3..2> 7 26
H226	DDR_B_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_B_CKE<1..0> 7 25
H226	DDR_B_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_B_CS_L<1..0> 7 25
H226	DDR_B_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_B_ODT<1..0> 7 25
H226	DDR_B_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_B_CKE<3..2> 7 26
H226	DDR_B_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_B_CS_L<3..2> 7 26
H226	DDR_B_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_B_ODT<3..2> 7 26
H226	DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM_B_A<15..0> 7 25 26
H226	DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM_B_BA<2..0> 7 25 26
H226	DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM_B_RAS_L 7 25 26
H226	DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM_B_CAS_L 7 25 26
H226	DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM_B_WE_L 7 25 26
H226	DDR_B_DQ_BYTE0	DDR_DQ_PHY	DDR_B_DQ_BYTE0	MEM_B_DQ<7..0> 7 27
H226	DDR_B_DQ_BYTE1	DDR_DQ_PHY	DDR_B_DQ_BYTE1	MEM_B_DQ<15..8> 7 27
H226	DDR_B_DQ_BYTE2	DDR_DQ_PHY	DDR_B_DQ_BYTE2	MEM_B_DQ<23..16> 7 27
H226	DDR_B_DQ_BYTE3	DDR_DQ_PHY	DDR_B_DQ_BYTE3	MEM_B_DQ<31..24> 7 27
H226	DDR_B_DQ_BYTE4	DDR_DQ_PHY	DDR_B_DQ_BYTE4	MEM_B_DQ<39..32> 7 27
H226	DDR_B_DQ_BYTE5	DDR_DQ_PHY	DDR_B_DQ_BYTE5	MEM_B_DQ<47..40> 7 27
H226	DDR_B_DQ_BYTE6	DDR_DQ_PHY	DDR_B_DQ_BYTE6	MEM_B_DQ<55..48> 7 27
H226	DDR_B_DQ_BYTE7	DDR_DQ_PHY	DDR_B_DQ_BYTE7	MEM_B_DQ<63..56> 7 27
H226	DDR_B_DQS0	DDR_DQS_PHY	DDR_B_DQS0	MEM_B_DQS_P<0> 7 27
H226	DDR_B_DQS0	DDR_DQS_PHY	DDR_B_DQS0	MEM_B_DQS_N<0> 7 27
H226	DDR_B_DQS1	DDR_DQS_PHY	DDR_B_DQS1	MEM_B_DQS_P<1> 7 27
H226	DDR_B_DQS1	DDR_DQS_PHY	DDR_B_DQS1	MEM_B_DQS_N<1> 7 27
H226	DDR_B_DQS2	DDR_DQS_PHY	DDR_B_DQS2	MEM_B_DQS_P<2> 7 27
H226	DDR_B_DQS2	DDR_DQS_PHY	DDR_B_DQS2	MEM_B_DQS_N<2> 7 27
H226	DDR_B_DQS3	DDR_DQS_PHY	DDR_B_DQS3	MEM_B_DQS_P<3> 7 27
H226	DDR_B_DQS3	DDR_DQS_PHY	DDR_B_DQS3	MEM_B_DQS_N<3> 7 27
H226	DDR_B_DQS4	DDR_DQS_PHY	DDR_B_DQS4	MEM_B_DQS_P<4> 7 27
H226	DDR_B_DQS4	DDR_DQS_PHY	DDR_B_DQS4	MEM_B_DQS_N<4> 7 27
H226	DDR_B_DQS5	DDR_DQS_PHY	DDR_B_DQS5	MEM_B_DQS_P<5> 7 27
H226	DDR_B_DQS5	DDR_DQS_PHY	DDR_B_DQS5	MEM_B_DQS_N<5> 7 27
H226	DDR_B_DQS6	DDR_DQS_PHY	DDR_B_DQS6	MEM_B_DQS_P<6> 7 27
H226	DDR_B_DQS6	DDR_DQS_PHY	DDR_B_DQS6	MEM_B_DQS_N<6> 7 27
H226	DDR_B_DQS7	DDR_DQS_PHY	DDR_B_DQS7	MEM_B_DQS_P<7> 7 27
H226	DDR_B_DQS7	DDR_DQS_PHY	DDR_B_DQS7	MEM_B_DQS_N<7> 7 27
Reset				
H240		DDR_50S		MEM_RESET_L 21 23 24
SM COMP				
H240		DDR_COMP_PHY	DDR_COMP	CPU_SM_RCOMP<0..2> 6

SYNC MASTER=317 ULTIMATE		SYNC DATE=05/08/2013	
PAGE TITLE			
DDR3 Constraints			
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	051-9884	D	
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	SHEET	92 OF 106	

D

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11

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3

DCBA

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
PCIE_PHY	*	PCIE_85D
COMP_DMI_PHY	*	DMI_COMP

PCIE-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE_SAME_DIR	TOP,BOTTOM	=5X_DIELECTRIC	?
PCIE_SAME_DIR	*	=3.5X_DIELECTRIC	?
PCIE_ALT_DIR	*	=7X_DIELECTRIC	?
PCIE_ISO	*	=4:1_SPACING	?

TBT x4 PCIE Spacing Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE_TBT_R2D	PCIE_TBT_R2D	*	PCIE_SAME_DIR
PCIE_TBT_D2R	PCIE_TBT_D2R	*	PCIE_SAME_DIR
PCIE_TBT_D2R	PCIE_TBT_R2D	*	PCIE_ALT_DIR
PCIE_TBT_D2R	*	*	PCIE_ISO
PCIE_TBT_R2D	*	*	PCIE_ISO

PCH x1 PCIE Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE	*	*	PCIE_ISO

DMI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DMI_SAME_DIR	TOP,BOTTOM	=5X_DIELECTRIC	?
DMI_SAME_DIR	*	=4X_DIELECTRIC	?
DMI_ALT_DIR	*	=5X_DIELECTRIC	?
DMI_ISO	*	=4X_DIELECTRIC	?

DMI x4 PCIE Spacing Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DMI_N2S	DMI_N2S	*	DMI_SAME_DIR
DMI_S2N	DMI_S2N	*	DMI_SAME_DIR
DMI_N2S	DMI_S2N	*	DMI_ALT_DIR
DMI_N2S	*	*	DMI_ISO
DMI_S2N	*	*	DMI_ISO

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DMI_COMP	*	Y	0.2032 MM	0.2032 MM	3 MM	=STANDARD	=STANDARD

PCIE (PCH)

Electrical Constraint Set	Physical	Spacing
x4 Thunderbolt		
PCIE_GEN2_R2D	PCIE_PHY	PCIE_TBT_R2D
PCIE_GEN2_R2D	PCIE_PHY	PCIE_TBT_R2D
PCIE_GEN2_R2D	PCIE_PHY	PCIE_TBT_R2D
PCIE_GEN2_R2D	PCIE_PHY	PCIE_TBT_R2D
PCIE_GEN2_D2R	PCIE_PHY	PCIE_TBT_D2R
PCIE_GEN2_D2R	PCIE_PHY	PCIE_TBT_D2R
PCIE_GEN2_D2R	PCIE_PHY	PCIE_TBT_D2R
PCIE_GEN2_D2R	PCIE_PHY	PCIE_TBT_D2R
PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE
PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE
x1 AirPort		
PCIE_GEN2_R2D_CONN_AP	PCIE_PHY	PCIE
PCIE_GEN2_R2D_CONN_AP	PCIE_PHY	PCIE
PCIE_GEN2_R2D_CONN_AP	PCIE_PHY	PCIE
PCIE_GEN2_R2D_CONN_AP	PCIE_PHY	PCIE
PCIE_GEN2_D2R_CONN_AP	PCIE_PHY	PCIE
PCIE_GEN2_D2R_CONN_AP	PCIE_PHY	PCIE
PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE
PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE
x1 Caesar IV		
PCIE_GEN2_R2D	PCIE_PHY	PCIE
PCIE_GEN2_R2D	PCIE_PHY	PCIE
PCIE_GEN2_R2D	PCIE_PHY	PCIE
PCIE_GEN2_R2D	PCIE_PHY	PCIE
PCIE_GEN2_D2R	PCIE_PHY	PCIE
PCIE_GEN2_D2R	PCIE_PHY	PCIE
PCIE_GEN2_D2R	PCIE_PHY	PCIE
PCIE_GEN2_D2R	PCIE_PHY	PCIE
PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE
PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE
x2 SSD		
PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE
PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE
PCH PCIE Compensation		
COMP_DMI_PHY	COMP_PCIE	PCH_PCIE_RCOMP

CPU DP REF CLK

Electrical Constraint Set	Physical	Spacing
CPU DP REF CLK		
CPU_CLK135_PL1	PCIE_PHY	CLK_PCIE
CPU_CLK135_PL1	PCIE_PHY	CLK_PCIE
CPU_CLK135_PL1	PCIE_PHY	CLK_PCIE
CPU_CLK135_PL1	PCIE_PHY	CLK_PCIE

DMI

Electrical Constraint Set	Physical	Spacing
DMI		
DMI_N2S	PCIE_PHY	DMI_N2S
DMI_N2S	PCIE_PHY	DMI_N2S
DMI_S2N	PCIE_PHY	DMI_S2N
DMI_S2N	PCIE_PHY	DMI_S2N
PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE
PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE
DMI Compensation		
COMP_DMI_PHY	COMP_PCIE	PCH_DMI_RCOMP

PCH

PCH-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_PCH_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

PCH-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCH	*	=4:1_SPACING	?
COMP_PCH	*	=2:1_SPACING	?

PCI

PCI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_PCI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

PCI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCI	*	=2:1_SPACING	?

LPC

LPC-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

LPC-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC	*	=1.5:1_SPACING	?
CLK_LPC	*	=2:1_SPACING	?

HDA

HDA-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

HDA-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA	*	=2x_DIELECTRIC	?

Crystal

Crystal-specific Physical Rules

[illegible]

Crystal-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
XTAL	*	=4X_DIELECTRIC	?

SPI

SPI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI	*	=2:1_SPACING	?

PCI

Electrical Constraint Set	Physical	Spacing		
PCI Clock				
IEEE	CLK_PCT_55S	CLK_PCT	PCH_CLK33M_PCIIN	11 19
IEEE	CLK_PCT_55S	CLK_PCT	PCH_CLK33M_PCIOUT	11 19

LPC

Electrical Constraint Set	Physical	Spacing	
LPC			
1633	LPC_55S	LPC	LPC_AD<3..0> 13 46 48
1636	LPC_55S	LPC	LPC_AD_R<3..0> 13
1638	LPC_55S	LPC	LPC_FRAME_L 13 46 48
1637	LPC_55S	LPC	LPC_FRAME_R_L 13
LPC Clocks			
1639	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_LPCPLUS 19 48
1640	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_LPCPLUS_R 11 19
1641	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_SMC 19 46
1642	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_SMC_R 11 19

PCH Clocks

Electrical Constraint Set	Physical	Spacing		
PCH Reference Clock				
	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_SB	11 19
	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_SB_R	11
PCH RTC 32K				
	CLK_XTAL	XTAL	PCH_CLK32K_RTCX1	11 19
	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2	11 19
	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2_R	19
SMC 32K				
	CLK_PCH_55S	CLK_PCH	PM_CLK32K_SUSCLK_R	12 47
	CLK_PCH_55S	CLK_PCH	SMC_CLK32K	46 47

25 MHz Reference Clocks

Electrical Constraint Set	Physical	Spacing	
25M Reference Crystal			
 CLK_XTAL	XTAL	SYSCLK CLK25M_X1	19
 CLK_XTAL	XTAL	SYSCLK CLK25M_X2	19
 CLK_XTAL	XTAL	SYSCLK CLK25M_X2_R	19
25M Reference Clocks			
 CLK_PCH_55S	CLK_PCH	SYSCLK CLK25M_ENET	19 37
 CLK_PCH_55S	CLK_PCH	SYSCLK CLK25M_ENET_R	19
 CLK_PCH_55S	CLK_PCH	SYSCLK CLK25M_TBT	19 28
 CLK_PCH_55S	CLK_PCH	SYSCLK CLK25M_TBT_R	28

HDA

Electrical Constraint Set	Physical	Spacing
HDA		
H098	HDA_55S	HDA HDA_BIT_CLK
H099	HDA_55S	HDA HDA_BIT_CLK_R
H09B	HDA_55S	HDA HDA_RST_L
H09C	HDA_55S	HDA HDA_RST_R_L
H09D	HDA_55S	HDA HDA_SDOUT
H09E	HDA_55S	HDA HDA_SDOUT_R
H09F	HDA_55S	HDA HDA_SYNC
H09G	HDA_55S	HDA HDA_SYNC_R
H09H	HDA_55S	HDA HDA_SDI0
H09J	HDA_55S	HDA AUD_SDI_R
SPDIF		
H37A		HDA AUD_SPDIF_CHIP
H37B		HDA AUD_SPDIF_OUT

SPI Bootrom

Electrical Constraint Set	Physical	Spacing	
SPI ROM			
H009	SPI_50S	SPT	SPI_CLK_R 13 48
H097	SPT_50S	SPT	SPI_CLK 48
H098	SPT_50S	SPT	SPI_ALT_CLK 48
H480	SPT_50S	SPT	SPI_SMC_CLK 48 48
H489	SPT_50S	SPT	SPI_MLB_CLK 48
H092	SPT_50S	SPT	SPI_CS0_R_L 13 48
H095	SPT_50S	SPT	SPI_CS0_L 48
H096	SPT_50S	SPT	SPI_ALT_CS_L 48
H099	SPT_50S	SPT	SPI_SMC_CS_L 48 48
H099	SPT_50S	SPT	SPI_MLB_CS_L 48
H090	SPT_50S	SPT	SPI_MOSI_R 13 48
H400	SPT_50S	SPT	SPI_MOSI 48
H093	SPT_50S	SPT	SPI_ALT_MOSI 48
H210	SPT_50S	SPT	SPI_SMC_MOSI 48 48
H415	SPT_50S	SPT	SPI_MLB_MOSI 48
H480	SPT_50S	SPT	SPI_MISO 13 48
H095	SPT_50S	SPT	SPI_ALT_MISO 48
H439	SPT_50S	SPT	SPI_SMC_MISO 48 48
H415	SPT_50S	SPT	SPI_MLB_MISO 48
H095	SPT_50S	SPT	SPIROM_USE_MLB 14 48

USB

USB-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
USB_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
USB_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
USB2_PHY	*	USB_90D
USB3_PHY	*	USB_85D

USB-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB2_ISO	*	=3:1_SPACING	?
USB2_ISO	TOP,BOTTOM	=3:1_SPACING	?
USB3_ISO	*	=5.5:1_SPACING	?
USB3_ISO	TOP,BOTTOM	=5.5:1_SPACING	?

USB Min Spacing Rules (mils) (Maho Bay PDG, Intel Doc# 473718)

Section	Imp	Design	Iso	Design	Comments
12.2.1	90	90	12	11.81	USB 2.0
13.3.1	85	85	20	21.65	USB 3.0

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
USB2	*	*	USB2_ISO
USB3	*	*	USB3_ISO

Caesar IV (Ethernet/SD)

CIV-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
ENET_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
ENET_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
SD_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
ENET_COMP_PHY	*	ENET_50S
ENET_DIFF_PHY	*	ENET_100D
SD_PHY	*	SD_50S
CIV_SPI	*	SPI_55S

CIV-specific Spacing Definitions

Ethernet

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENET_DIFF_ISO	*	=6:1_SPACING	?
ENET_DIFF2DIFF	*	=3:1_SPACING	?
ENET_TRANS_ISO	*	1.27 MM	?
COMP_ENET_ISO	*	=4:1_SPACING	?

Constraints Ethernet

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
ENET_DIFF	*	*	ENET_DIFF_ISO
ENET_DIFF	ENET_DIFF	*	ENET_DIFF2DIFF
ENET_TRANS	*	*	ENET_TRANS_ISO
COMP_ENET	*	*	COMP_ENET_ISO
ENET_TRANS	ENET_TRANS	*	ENET_DIFF2DIFF

2 kV isolation

SD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SD_ISO	*	=3:1_SPACING	?

SD

NET_spacing_type1	NET_spacing_type2	AREA_type	SPACING_RULE_SET
SD	*	*	SD_ISO

Camera Processor-to-Camera Sensor I/F (SMIA/MIPI)

Camera Processor's SMIA Interface Physical Rules

[illegible]

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SMIA_DIFF_PHY	*	SMIA_100D

Camera Processor's SMIA Interface Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMIA_DIFF_ISO	*	=6:1_SPACING	?
SMIA_DIFF2DIFF	*	=3:1_SPACING	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SMIA_DIFF	*	*	SMIA_DIFF_ISO
SMIA_DIFF	SMIA_DIFF	*	SMIA_DIFF2DIFF

USB 3.0 and USB 2.0 Trixies Muxing

Electrical Constraint Set		Physical	Spacing	
External Port A (J4600)				
H320	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXT_A_RX_P
H322	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXT_A_RX_N
H323		USB3_PHY	USB3	USB3_EXT_A_RX_F_P
H324		USB3_PHY	USB3	USB3_EXT_A_RX_F_N
H325	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_A_TX_P
H326	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_A_TX_N
H327		USB3_PHY	USB3	USB3_EXT_A_TX_F_P
H328		USB3_PHY	USB3	USB3_EXT_A_TX_F_N
H329		USB3_PHY	USB3	USB3_EXT_A_TX_C_P
H330		USB3_PHY	USB3	USB3_EXT_A_TX_C_N
H331	USB2_MUXED_M0J0_CONN	USB2_PHY	USB2	USB_EXT_A_0_P
H332	USB2_MUXED_M0J0_CONN	USB2_PHY	USB2	USB_EXT_A_0_N
H333		USB2_PHY	USB2	USB2_EXT_A_MUXED_P
H334		USB2_PHY	USB2	USB2_EXT_A_MUXED_N
H335		USB2_PHY	USB2	USB2_EXT_A_P
H336		USB2_PHY	USB2	USB2_EXT_A_N
External Port B (J4610)				
H337	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXT_B_RX_P
H338	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXT_B_RX_N
H339		USB3_PHY	USB3	USB3_EXT_B_RX_F_P
H340		USB3_PHY	USB3	USB3_EXT_B_RX_F_N
H341	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_B_TX_P
H342	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_B_TX_N
H343		USB3_PHY	USB3	USB3_EXT_B_TX_F_P
H344		USB3_PHY	USB3	USB3_EXT_B_TX_F_N
H345		USB3_PHY	USB3	USB3_EXT_B_TX_C_P
H346		USB3_PHY	USB3	USB3_EXT_B_TX_C_N
H347	USB2_CONN	USB2_PHY	USB2	USB_EXT_B_0_P
H348	USB2_CONN	USB2_PHY	USB2	USB_EXT_B_0_N
H349		USB2_PHY	USB2	USB2_EXT_B_P
H350		USB2_PHY	USB2	USB2_EXT_B_N
External Port C (J4700)				
H351	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXT_C_RX_P
H352	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXT_C_RX_N
H353		USB3_PHY	USB3	USB3_EXT_C_RX_F_P
H354		USB3_PHY	USB3	USB3_EXT_C_RX_F_N
H355	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_C_TX_P
H356	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_C_TX_N
H357		USB3_PHY	USB3	USB3_EXT_C_TX_F_P
H358		USB3_PHY	USB3	USB3_EXT_C_TX_F_N
H359		USB3_PHY	USB3	USB3_EXT_C_TX_C_P
H360		USB3_PHY	USB3	USB3_EXT_C_TX_C_N
H361	USB2_CONN	USB2_PHY	USB2	USB_EXT_C_1_P
H362	USB2_CONN	USB2_PHY	USB2	USB_EXT_C_1_N
H363		USB2_PHY	USB2	USB2_EXT_C_P
H364		USB2_PHY	USB2	USB2_EXT_C_N
External Port D (J4710)				
H365	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXT_D_RX_P
H366	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXT_D_RX_N
H367		USB3_PHY	USB3	USB3_EXT_D_RX_F_P
H368		USB3_PHY	USB3	USB3_EXT_D_RX_F_N
H369	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_D_TX_P
H370	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_D_TX_N
H371		USB3_PHY	USB3	USB3_EXT_D_TX_F_P
H372		USB3_PHY	USB3	USB3_EXT_D_TX_F_N
H373		USB3_PHY	USB3	USB3_EXT_D_TX_C_P
H374		USB3_PHY	USB3	USB3_EXT_D_TX_C_N
H375	USB2_CONN	USB2_PHY	USB2	USB_EXT_D_0_P
H376	USB2_CONN	USB2_PHY	USB2	USB_EXT_D_0_N
H377		USB2_PHY	USB2	USB2_EXT_D_P
H378		USB2_PHY	USB2	USB2_EXT_D_N
Camera (J3510)				
H379	USB2_CONN_TNT	USB2_PHY	USB2	USB_CAMERA_P
H380	USB2_CONN_TNT	USB2_PHY	USB2	USB_CAMERA_N
PCH USB Compensation				
H381		PCH_5SS	COMP_PCH	PCH_USB_RBIA5

RMH Love

Electrical Constraint Set		Physical	Spacing	
FE00	USB2_MUXED_BT	USB2_PHY	USB2	USB_BT_P 13 34
FE01	USB2_MUXED_BT	USB2_PHY	USB2	USB_BT_N 13 34
FE02	USB2_MUXED_BT	USB2_PHY	USB2	USB_BT_MUX_P 34
FE03	USB2_MUXED_BT	USB2_PHY	USB2	USB_BT_MUX_N 34

Et tu Brute?

Electrical Constraint Set		Physical	Spacing	
Ethernet				
ENET0	ENET_MDI	ENET_DTEF_PHY	ENET_DTEF	ENETCONN_MDI_P<3..0>
ENET0	ENET_MDI	ENET_DTEF_PHY	ENET_DTEF	ENETCONN_MDI_N<3..0>
ENET0		ENET_DTEF_PHY	ENET_TRANS	ENETCONN_MDI_T_P<3..0>
ENET0		ENET_DTEF_PHY	ENET_TRANS	ENETCONN_MDI_T_N<3..0>
ENET0			ENET_TRANS	ENETCONN_MCT0
ENET0			ENET_TRANS	ENETCONN_MCT1
ENET0			ENET_TRANS	ENETCONN_MCT2
ENET0			ENET_TRANS	ENETCONN_MCT3
ENET0			ENET_TRANS	ENETCONN_MCT_BS
ENET0		ENET_COMP_PHY	COMP_ENET	ENET_RDAC
SD				
SD0	SD_DATA	SD_PHY	SD	ENET_CR_DATA<7..0>
SD0		SD_PHY	SD	SDCONN_DATA<7..0>
SD0		SD_PHY	SD	SDCONN_DATA_R<7..0>
SD0	SD_CMD	SD_PHY	SD	ENET_SD_CMD
SD0		SD_PHY	SD	SDCONN_CMD
SD0		SD_PHY	SD	SDCONN_CMD_R
SD0	SD_CLK	SD_PHY	SD	ENET_SD_CLK
SD0		SD_PHY	SD	SDCONN_CLK
SD0		SD_PHY	SD	SDCONN_CLK_R
SD0		SD_PHY	SD	ENET_MEDIA_SENSE
SD0		SD_PHY	SD	ENET_SD_DETECT_L
CIV SPI				
CS0		CIV_SPI	SPT	ENET_SCLK
MISO0		CIV_SPI	SPT	ENET_MISO
MOSI0		CIV_SPI	SPT	ENET_MOSI
CS1		CIV_SPI	SPT	ENET_CS_L

Camera Processor-Camera Sensor I/F

Electrical Constraint Set		Physical	Spacing	
1E9D0	SMIA_DP	SMIA_DIEF_PHY	SMIA_DIEF	SMIA_DATA_P 40
1E9D0	SMIA_DP	SMIA_DIEF_PHY	SMIA_DIEF	SMIA_DATA_N 40
1E9D0	SMIA_DP	SMIA_DIEF_PHY	SMIA_DIEF	SMIA_CLK_P 40
1E9D0	SMIA_DP	SMIA_DIEF_PHY	SMIA_DIEF	SMIA_CLK_N 40
1E9D0		SPT_5G5	SPT	CAM_SF_CLK 40
1E9D0		SPT_5G5	SPT	CAM_SF_CLK_R 40
1E9D0		SPT_5G5	SPT	CAM_SF_DIN 40
1E9D0		SPT_5G5	SPT	CAM_SF_DIN_R 40
1E9D0		SPT_5G5	SPT	CAM_SF_CS_L 40
1E9D0		SPT_5G5	SPT	CAM_SF_WP_L 40
1E9D0		SPT_5G5	SPT	CAM_SF_DOUT 40
1E9D0		SPT_5G5	SPT	CAM_SF_DOUT_R 40
1E9D0		SMB_PHY	SMB	I2C_CAMSENSOR_SDA 40
1E9D0		SMB_PHY	SMB	I2C_CAMSENSOR_SCL 40

SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/08/2013	
PAGE TITLE			
USB/Ethernet/SD Constraints			
 Apple Inc.		DRAWING NUMBER 051-9884	SIZE D
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		SHEET 97 OF 106	

SMBus

SMBus-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SMB_PHY	*	SMB_55S

SMBus-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB_ISO	*	=2x_DIELECTRIC	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SMB	*	*	SMB_ISO

Sensor

Sensor-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1:1_DIFFPAIR	*	Y	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.085 MM

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SNS_DIFF_PHY	*	1:1_DIFFPAIR

Sensor-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SENSE_ISO	*	=1.5:1_SPACING	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SENSE	*	*	SENSE_ISO
SENSE	POWER	*	PWR_P2MM
SENSE	GND	*	GND_P2MM

SMC Generic Control Line Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMC_ISO	*	=1:1_SPACING	?
SMC_3xISO	*	=3:1_SPACING	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SMC_CTRL	*	*	SMC_ISO
SMC_3XCTRL	*	*	SMC_3xISO

SMC Generic Control Line Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMC_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SMC_GEN	*	SMC_50S

Current/Voltage Sense

Electrical Constraint Set		Physical	Spacing	
Common				
IE1		SENSE	GND_SMC_AVSS	46 47 50 51
12V S5 (System Total)				
IE5	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	SNS_P12VG3H_P
IE6	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	SNS_P12VG3H_N
IE8			SENSE	ISNS_P12VG3H_R
IE9			SENSE	ISNS_P12VG3H
IE10			SENSE	VSNS_P12VG3H
12V S0 (GPU Core)				
IE20	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	SNS_P12VS0_GPUCORE_P
IE21	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	SNS_P12VS0_GPUCORE_N
IE22			SENSE	ISNS_P12VS0_GPUCORE_R
IE23			SENSE	ISNS_P12VS0_GPUCORE
IE24			SENSE	VSNS_P12VS0_GPUCORE
HDD				
IE30	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	SNS_HDD_P
IE31	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	SNS_HDD_N
IE33			SENSE	ISNS_HDD50_R
IE34			SENSE	ISNS_HDD50
IE35			SENSE	VSNS_HDD50
IE36	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	SNS_P12VS0_HDD_P
IE37	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	SNS_P12VS0_HDD_N
IE38			SENSE	ISNS_P12VS0_HDD_R
IE39			SENSE	ISNS_P12VS0_HDD
SSD				
IE05	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	SNS_SSD_P
IE07	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	SNS_SSD_N
IE08			SENSE	ISNS_SSD50_R
IE09			SENSE	ISNS_SSD50
IE09			SENSE	VSNS_P3V3S5
GPU Core (alternate low side sense)				
IE15	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	ISNS_GPUCORE_P
IE16	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	ISNS_GPUCORE_N
IE18			SENSE	ISNS_GPUCORE_ALT
IE19			SENSE	ISNS_GPUCORE_FB
IE20			SENSE	VSNS_GPUCORE_ALT
VDDQ S3 (DDR)				
IE33	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	SNS_VDDQ53_DDR_P
IE37	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	SNS_VDDQ53_DDR_N
IE38			SENSE	ISNS_VDDQ53_DDR_R
IE39			SENSE	ISNS_VDDQ53_DDR
IE40			SENSE	VSNS_VDDQ53_DDR
VDDQ S0 (GPU FBVDDQ)				
IE37	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	SNS_P12VS0_FBVDDQ_P
IE38	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	SNS_P12VS0_FBVDDQ_N
IE39			SENSE	ISNS_P12VS0_FBVDDQ_R
IE40			SENSE	ISNS_P12VS0_FBVDDQ
CPU Core				
IE19	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	ISNS_CPUVCC_P
IE20	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	ISNS_CPUVCC_N
IE43			SENSE	ISNS_CPUVCC_FB
IE45			SENSE	ISNS_CPUVCC
IE46			SENSE	VSNS_CPUVCC
PP1V05_S0_PCH				
IE55		SENSE	VSNS_P1V05S0_PCH	47 50
PP1V5_S0				
IE11	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	SNS_P1V5S0_P
IE12	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	SNS_P1V5S0_N
IE13			SENSE	ISNS_P1V5S0_R
IE14			SENSE	ISNS_P1V5S0
IE15			SENSE	VSNS_P1V5S0
Airport				
IE02	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	SNS_P3V354_AP_P
IE03	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	SNS_P3V354_AP_N
IE05			SENSE	ISNS_P3V354_AP_R
IE06			SENSE	ISNS_P3V354_AP

SMC

Electrical Constraint Set	Physical	Spacing	
SMC			
E1	CLK_XTAL	XTAL	SMC XTAL 46 47
E2	CLK_XTAL	XTAL	SMC XTAL 46 47
E91	SMC_GEN	SMC_CTL1	SMC LRESET L 20 46
E90	SMC_GEN	SMC_CTL1	SMC RUNTIME SCI L 14 46
E92	SMC_GEN	SMC_CTL1	SMC WAKE SCI L 14 46
E93	SMC_GEN	SMC_CTL1	SMC FAN_0 CTL 46 63
E94	SMC_GEN	SMC_CTL1	SMC FAN_0 TACH 46 63
E95	CPU1_ASYNC_PHY	CPU1_ASYNC	SMC PECT L 46 47 506
E96	SMC_GEN	SMC_3VCTL1	CPU PECT 0 14 46 47

SMBus

Electrical Constraint Set	Physical	Spacing	
SMC			
	SMB_PHY	SMB	SMBUS SMC 0 S0_SCL 46 49
	SMB_PHY	SMB	SMBUS SMC 0 S0_SDA 46 49
	SMB_PHY	SMB	SMBUS SMC 1 S0_SCL 46 49
	SMB_PHY	SMB	SMBUS SMC 1 S0_SDA 46 49
	SMB_PHY	SMB	SMBUS SMC 2 S4_SCL 46 49
	SMB_PHY	SMB	SMBUS SMC 2 S4_SDA 46 49
	SMB_PHY	SMB	SMBUS SMC 3_SCL 46 49
	SMB_PHY	SMB	SMBUS SMC 3_SDA 46 49
	SMB_PHY	SMB	SMBUS SMC 5 G3H_SCL 46 47
	SMB_PHY	SMB	SMBUS SMC 5 G3H_SDA 46 47
PCH			
	SMB_PHY	SMB	SMBUS PCH_CLK 13 49
	SMB_PHY	SMB	SMBUS PCH_DATA 13 49
	SMB_PHY	SMB	SML PCH 0_CLK 13 49
	SMB_PHY	SMB	SML PCH 0_DATA 13 49
Display Tcon			
	SMB_PHY	SMB	SMB_DP_TCON_SCL 42 49
	SMB_PHY	SMB	SMB_DP_TCON_SDA 42 49

Temperature Sense

Electrical Constraint Set	Physical	Spacing		
EMC1414-1 (Production)				
E57 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS T1 1 P	
E58 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS T1 1 N	
E59 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS T1 2 P	
E60 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS T1 2 N	
E65 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS ACDC P	52 62
E66 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS ACDC N	52 62
E920	SNS_DIFF_PHY	SENSE	SNS T1 3 P	
E921	SNS_DIFF_PHY	SENSE	SNS T1 3 N	
TMP423 (Development)				
E49 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS T2 1 P	
E61 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS T2 1 N	
E65 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS T2 2 P	
E70 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS T2 2 N	
E61 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS SKIN P	
E62 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS SKIN N	
E64 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS T2 3 P	
E65 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS T2 3 N	
HDD Out-of-Band				
E880		SENSE	SMC 00B1 D2R L	35 46
E881		SENSE	SMC 00B1 R2D L	46 47
E882		SENSE	SMC 00B1 D2R R	36
SSD Out-of-Band				
E883		SENSE	SMC 00B2 R2D L	35 47
E884		SENSE	SMC 00B2 D2R L	35 47

SYNCH MASTER=J17 ULTIMATE		SYNCH DATE=05/08/2013	
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SMBus/Sensor Constraints			
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		BRANCH	
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		PAGE	
			117 OF 127
		SHEET	
			98 OF 106

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<table><tr><th>Electrical Constraint Set</th><th>Physical</th><th>Spacing</th><th>Voltage</th><th>DIDT</th><th>NO_TEST</th></tr><tr><td colspan="6">Input Bus</td></tr><tr><td>PP12V_S0_CPUVCC_FLT</td><td>POWER</td><td>POWER</td><td>12V</td><td></td><td></td></tr><tr><td>REG_VCC_U7000</td><td>POWER</td><td>POWER</td><td>5V</td><td></td><td></td></tr><tr><td colspan="6">Local Ground</td></tr><tr><td>AGND_CPU</td><td>GND</td><td>GND</td><td>0V</td><td></td><td></td></tr><tr><td colspan="6">Phase 1</td></tr><tr><td>REG_TD_1</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_PWM_CPUVCC_1</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_PWM_CPUVCC_1_R</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_PHASE_CPUVCC_1</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td></tr><tr><td>REG_BOOT_CPUVCC_1</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td></tr><tr><td>REG_BOOT_CPUVCC_1_RC</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td></tr><tr><td>REG_UGATE_CPUVCC_1</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td></tr><tr><td>REG_LGATE_CPUVCC_1</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td></tr><tr><td>REG_SNUBBER_CPUVCC_1</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td></tr><tr><td>PPCPUVCC_S0_SENSE_1</td><td>POWER</td><td>POWER</td><td>1.8V</td><td></td><td></td></tr><tr><td>REG_ISENVCC_1_P</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td></td></tr><tr><td>REG_ISENVCC_1_N</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td></td></tr><tr><td>REG_ISENVCC_1_NR</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td></td></tr><tr><td colspan="6">Phase 2</td></tr><tr><td>REG_TD_2</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_PWM_CPUVCC_2</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_PWM_CPUVCC_2_R</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_PHASE_CPUVCC_2</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td></tr><tr><td>REG_BOOT_CPUVCC_2</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td></tr><tr><td>REG_BOOT_CPUVCC_2_RC</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td></tr><tr><td>REG_UGATE_CPUVCC_2</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td></tr><tr><td>REG_LGATE_CPUVCC_2</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td></tr><tr><td>REG_SNUBBER_CPUVCC_2</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td></tr><tr><td>PPCPUVCC_S0_SENSE_2</td><td>POWER</td><td>POWER</td><td>1.8V</td><td></td><td></td></tr><tr><td>REG_ISENVCC_2_P</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td></td></tr><tr><td>REG_ISENVCC_2_N</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td></td></tr><tr><td>REG_ISENVCC_2_NR</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td></td></tr><tr><td colspan="6">Phase 3</td></tr><tr><td>REG_TD_3</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_PWM_CPUVCC_3</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_PWM_CPUVCC_3_R</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_PHASE_CPUVCC_3</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td></tr><tr><td>REG_BOOT_CPUVCC_3</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td></tr><tr><td>REG_BOOT_CPUVCC_3_RC</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td></tr><tr><td>REG_UGATE_CPUVCC_3</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td></tr><tr><td>REG_LGATE_CPUVCC_3</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td></tr><tr><td>REG_SNUBBER_CPUVCC_3</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td></tr><tr><td>PPCPUVCC_S0_SENSE_3</td><td>POWER</td><td>POWER</td><td>1.8V</td><td></td><td></td></tr><tr><td>REG_ISENVCC_3_P</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td></td></tr><tr><td>REG_ISENVCC_3_N</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td></td></tr><tr><td>REG_ISENVCC_3_NR</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td></td></tr><tr><td colspan="6">Phase 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Constraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST	Input Bus						PP12V_S0_CPUVCC_FLT	POWER	POWER	12V			REG_VCC_U7000	POWER	POWER	5V			Local Ground						AGND_CPU	GND	GND	0V			Phase 1						REG_TD_1	VR_CTL_PHY	VR_CTL				REG_PWM_CPUVCC_1	VR_CTL_PHY	VR_CTL				REG_PWM_CPUVCC_1_R	VR_CTL_PHY	VR_CTL				REG_PHASE_CPUVCC_1	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUVCC_1	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUVCC_1_RC	VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_UGATE_CPUVCC_1	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_CPUVCC_1	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_CPUVCC_1	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		PPCPUVCC_S0_SENSE_1	POWER	POWER	1.8V			REG_ISENVCC_1_P	SNS_DIFF_PHY	SENSE				REG_ISENVCC_1_N	SNS_DIFF_PHY	SENSE				REG_ISENVCC_1_NR	SNS_DIFF_PHY	SENSE				Phase 2						REG_TD_2	VR_CTL_PHY	VR_CTL				REG_PWM_CPUVCC_2	VR_CTL_PHY	VR_CTL				REG_PWM_CPUVCC_2_R	VR_CTL_PHY	VR_CTL				REG_PHASE_CPUVCC_2	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUVCC_2	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUVCC_2_RC	VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_UGATE_CPUVCC_2	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_CPUVCC_2	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_CPUVCC_2	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		PPCPUVCC_S0_SENSE_2	POWER	POWER	1.8V			REG_ISENVCC_2_P	SNS_DIFF_PHY	SENSE				REG_ISENVCC_2_N	SNS_DIFF_PHY	SENSE				REG_ISENVCC_2_NR	SNS_DIFF_PHY	SENSE				Phase 3						REG_TD_3	VR_CTL_PHY	VR_CTL				REG_PWM_CPUVCC_3	VR_CTL_PHY	VR_CTL				REG_PWM_CPUVCC_3_R	VR_CTL_PHY	VR_CTL				REG_PHASE_CPUVCC_3	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUVCC_3	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUVCC_3_RC	VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_UGATE_CPUVCC_3	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_CPUVCC_3	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_CPUVCC_3	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		PPCPUVCC_S0_SENSE_3	POWER	POWER	1.8V			REG_ISENVCC_3_P	SNS_DIFF_PHY	SENSE				REG_ISENVCC_3_N	SNS_DIFF_PHY	SENSE				REG_ISENVCC_3_NR	SNS_DIFF_PHY	SENSE				Phase 4						REG_TD_4	VR_CTL_PHY	VR_CTL				REG_PWM_CPUVCC_4	VR_CTL_PHY	VR_CTL				REG_PWM_CPUVCC_4_R	VR_CTL_PHY	VR_CTL				REG_PHASE_CPUVCC_4	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUVCC_4	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUVCC_4_RC	VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_UGATE_CPUVCC_4	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_CPUVCC_4	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_CPUVCC_4	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		PPCPUVCC_S0_SENSE_4	POWER	POWER	1.8V			REG_ISENVCC_4_P	SNS_DIFF_PHY	SENSE				REG_ISENVCC_4_N	SNS_DIFF_PHY	SENSE				REG_ISENVCC_4_NR	SNS_DIFF_PHY	SENSE				<table><tr><th>Electrical Constraint Set</th><th>Physical</th><th>Spacing</th><th>Voltage</th><th>DIDT</th><th>NO_TEST</th></tr><tr><td colspan="6">ISL6372</td></tr><tr><td>REG_CPUVCC_DVC</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>CPUVCC_DVC_RC</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>CPUVCC_FB_RC_2</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_CPUVCC_COMP</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>CPUVCC_COMP_RC</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_CPUVCC_FB</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>CPUVCC_FB_RC</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>CPUVCC_FB_R_1</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>CPUVCC_FB_R_2</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>CPUVCC_PSICOMP_RC</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_CPUVCC_PSICOMP</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_CPUVCC_HFCOMP</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>CPU_VCCSENSE_P</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td></td></tr><tr><td>CPU_VCCSENSE_N</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td></td></tr><tr><td>CPU_VCCSENSE_R_P</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td></td></tr><tr><td>CPU_VCCSENSE_R_N</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td></td></tr><tr><td>SNS_VCC_XW_P</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td>1.8V</td><td></td><td></td></tr><tr><td>SNS_VCC_XW_N</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td>0V</td><td></td><td></td></tr><tr><td>REG_CPUVCC_VSEN</td><td></td><td>SENSE</td><td></td><td></td><td></td></tr><tr><td>REG_CPUVCC_RGND</td><td></td><td>SENSE</td><td></td><td></td><td></td></tr><tr><td>REG_CPUVCC_VIN</td><td></td><td>SENSE</td><td></td><td></td><td></td></tr><tr><td>REG_CPUVCC_IMON</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>CPUVCC_IMON_R</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_CPUVCC_TM</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_CPUVCC_IMX</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_CPUVCC_NPSI</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_CPUVCC_FDVID</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_CPUVCC_TMX</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_CPUVCC_MEMVRSEL</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_CPUVCC_RSET</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>CPU_VIDSCLK</td><td>VR_VID_PHY</td><td>VR_VID</td><td></td><td></td><td></td></tr><tr><td>CPU_VIDCLK_R</td><td>VR_VID_PHY</td><td>VR_VID</td><td></td><td></td><td></td></tr><tr><td>CPU_VIDALERT_L</td><td>VR_VID_PHY</td><td>VR_VID</td><td></td><td></td><td></td></tr><tr><td>CPU_VIDALERT_R_L</td><td>VR_VID_PHY</td><td>VR_VID</td><td></td><td></td><td></td></tr><tr><td>CPU_VIDSOUT</td><td>VR_VID_PHY</td><td>VR_VID</td><td></td><td></td><td></td></tr><tr><td>CPU_VIDSOUT_R</td><td>VR_VID_PHY</td><td>VR_VID</td><td></td><td></td><td></td></tr><tr><td colspan="6">Output Bus</td></tr><tr><td>PPCPUVCC_S0_CPU</td><td>POWER</td><td>POWER</td><td>1.8V</td><td></td><td></td></tr></table>				Electrical Constraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST	ISL6372						REG_CPUVCC_DVC	VR_CTL_PHY	VR_CTL				CPUVCC_DVC_RC	VR_CTL_PHY	VR_CTL				CPUVCC_FB_RC_2	VR_CTL_PHY	VR_CTL				REG_CPUVCC_COMP	VR_CTL_PHY	VR_CTL				CPUVCC_COMP_RC	VR_CTL_PHY	VR_CTL				REG_CPUVCC_FB	VR_CTL_PHY	VR_CTL				CPUVCC_FB_RC	VR_CTL_PHY	VR_CTL				CPUVCC_FB_R_1	VR_CTL_PHY	VR_CTL				CPUVCC_FB_R_2	VR_CTL_PHY	VR_CTL				CPUVCC_PSICOMP_RC	VR_CTL_PHY	VR_CTL				REG_CPUVCC_PSICOMP	VR_CTL_PHY	VR_CTL				REG_CPUVCC_HFCOMP	VR_CTL_PHY	VR_CTL				CPU_VCCSENSE_P	SNS_DIFF_PHY	SENSE				CPU_VCCSENSE_N	SNS_DIFF_PHY	SENSE				CPU_VCCSENSE_R_P	SNS_DIFF_PHY	SENSE				CPU_VCCSENSE_R_N	SNS_DIFF_PHY	SENSE				SNS_VCC_XW_P	SNS_DIFF_PHY	SENSE	1.8V			SNS_VCC_XW_N	SNS_DIFF_PHY	SENSE	0V			REG_CPUVCC_VSEN		SENSE				REG_CPUVCC_RGND		SENSE				REG_CPUVCC_VIN		SENSE				REG_CPUVCC_IMON	VR_CTL_PHY	VR_CTL				CPUVCC_IMON_R	VR_CTL_PHY	VR_CTL				REG_CPUVCC_TM	VR_CTL_PHY	VR_CTL				REG_CPUVCC_IMX	VR_CTL_PHY	VR_CTL				REG_CPUVCC_NPSI	VR_CTL_PHY	VR_CTL				REG_CPUVCC_FDVID	VR_CTL_PHY	VR_CTL				REG_CPUVCC_TMX	VR_CTL_PHY	VR_CTL				REG_CPUVCC_MEMVRSEL	VR_CTL_PHY	VR_CTL				REG_CPUVCC_RSET	VR_CTL_PHY	VR_CTL				CPU_VIDSCLK	VR_VID_PHY	VR_VID				CPU_VIDCLK_R	VR_VID_PHY	VR_VID				CPU_VIDALERT_L	VR_VID_PHY	VR_VID				CPU_VIDALERT_R_L	VR_VID_PHY	VR_VID				CPU_VIDSOUT	VR_VID_PHY	VR_VID				CPU_VIDSOUT_R	VR_VID_PHY	VR_VID				Output 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051-9884

19.0.0

dvt0

119 OF 127

100 OF 106

DRIVING NUMBER

REVISION

BRANCH

PAGE

SHEET

051-9884

19.0.0

dvt0

119 OF 127

100 OF 106

DRIVING NUMBER

REVISION

BRANCH

PAGE

SHEET

051-9884

19.0.0

dvt0

119 OF 127

100 OF 106

DRIVING NUMBER

REVISION

BRANCH

PAGE

SHEET

D

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B

A

D

C

B

A

3.3V S5/5V S4

Physical	Spacing	Voltage	DIDT	NO_TEST	
Input Bus					
U624 POWER	POWER	12V			REG VIN U7600 67
U626 POWER	POWER	5V			REG VCC1 U7600 67
U625 POWER	POWER	5V			REG VCC2 U7600 67
3.3V S5					
U646 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG PHASE P3V3S5 67
U647 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG B00T P3V3S5 67
U653 VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG B00T P3V3S5_RC 67
U648 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG UGATE P3V3S5 67
U650 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG LGATE P3V3S5 67
U652 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG SNUBBER P3V3S5 67
U609 VR_CTL_PHY	VR_CTL				REG P3V3S5_ISEN 67
U607 VR_CTL_PHY	VR_CTL				REG P3V3S5_OCSET 67
U608 VR_CTL_PHY	VR_CTL				REG P3V3S5_FSET 67
U608 VR_CTL_PHY	VR_CTL				REG P3V3S5_VOUT 67
U609 VR_CTL_PHY	VR_CTL				REG P3V3S5_VOUT_R 67
U609 VR_CTL_PHY	VR_CTL				REG P3V3S5_FB 67
5V S3					
U613 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG PHASE P5VS4 67
U615 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG B00T P5VS4 67
U616 VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG B00T P5VS4_RC 67
U615 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG UGATE P5VS4 67
U616 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG LGATE P5VS4 67
U614 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG SNUBBER P5VS4 67
U616 VR_CTL_PHY	VR_CTL				REG P5VS4_ISEN 67
U616 VR_CTL_PHY	VR_CTL				REG P5VS4_OCSET 67
U616 VR_CTL_PHY	VR_CTL				REG P5VS4_FSET 67
U616 VR_CTL_PHY	VR_CTL				REG P5VS4_VOUT 67
U616 VR_CTL_PHY	VR_CTL				REG P5VS4_VOUT_R 67
U616 VR_CTL_PHY	VR_CTL				REG P5VS4_FB 67
Output Bus					
U615 POWER	POWER	5V			PP5V_S5 67
U616 POWER	POWER	5V			PP5V_S4 67
U616 POWER	POWER	3.3V			PP3V3_S5 67
FET Switched					
U626 POWER	POWER	5V			PP5V_S0 67
U624 POWER	POWER	3.3V			PP3V3_S4 67
U625 POWER	POWER	3.3V			PP3V3_S0 67
U625 POWER	POWER	3.3V			PP3V3_S0_SSD 67
U626 POWER	POWER	3.3V			PP3V3_ENET 67
U626 POWER	POWER	3.3V			PP3V3_TBTLIC 67
Sensed					
U625 POWER	POWER	3.3V			PPSSD_S0 67
U627 POWER	POWER	3.3V			PP3V3_S4_AP 67

3.42V G3H

Physical	Spacing	Voltage	DIDT	NO_TEST	
3.42V G3H					
U636 POWER	VR_SWITCH	12V	TRUE		P3V42G3H_BOOST 62
U636 POWER	VR_SWITCH	12V	TRUE		P3V42G3H_SW 62
U636 VR_CTL_PHY	VR_CTL				P3V42G3H_FB 62
U636 VR_CTL_PHY	VR_CTL				P3V42G3H_SHDN_L 62
Output Bus					
U636 POWER	POWER	3.425V			PP3V42_G3H 67

3.3V G3

Physical	Spacing	Voltage	DIDT	NO_TEST	
U637 POWER	POWER	3.3V			PP3V3_G3 67

HDD S0

Physical	Spacing	Voltage	DIDT	NO_TEST	
FET Switched					
U639 POWER	POWER	5V			PPHDD_S0 67
Sensed					
U640 POWER	POWER	5V			PP5V_S0_HDD 67

12V

Physical	Spacing	Voltage	DIDT	NO_TEST	
Input Bus					
U646 POWER	POWER	12V			PP12V_ACDC 67
FET Switched					
U646 POWER	POWER	12V			PP12V_S5 67
U646 POWER	POWER	12V			PP12V_S0 67
U647 POWER	POWER	12V			PP12V_S0_BLC 67
U648 POWER	POWER	12V			PPHDD12_S0 67
Sensed					
U648 POWER	POWER	12V			PP12V_G3H 67
U649 POWER	POWER	12V			PP12V_G3H_P3V42 62
U648 POWER	POWER	12V			PP12V_S0_GPUCORE 67
U648 POWER	POWER	12V			PP12V_S0_FBVDDQ 67
U649 POWER	POWER	12V			PP12V_S0_HDD 67

Ground/Common

Physical	Spacing	Voltage	DIDT	NO_TEST	
Common					
U650 GND	GND	0V			GND

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Platform VReg Constraints

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19.0.0

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PAGE
120 OF 127

SHEET
101 OF 106

Thunderbolt

Thunderbolt-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
TBT_I2C_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
TBT_SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
TBTD_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

Thunderbolt-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBT_2C	*	=2x_DIELECTRIC	?
TBT_SPI	*	=2x_DIELECTRIC	?
TBTDP	*	=5x_DIELECTRIC	?
TBTDP	TOP, BOTTOM	=7x_DIELECTRIC	?
BGA_TBT_AREA	*	0. 075MM	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA_TBT	BGA_TBT_AREA

SOURCE: Bill Cornelius's T29 Routing Notes

DisplayPort

DP-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	0.08MM	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

DP-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DISPLAYPORT	*	=3:1_SPACING	?

Pairs should be within 100 mils of clock length.

Max length of DisplayPort traces: 12 inches

DisplayPort intra-pair matching should be 5 ps. Inter-pair matching should be within 150 ps.

DisplayPort AUX channel intra-pair matching should be 5 ps. No relationship to other signals.

TBT IC Net Properties

Electrical Constraint Set	Physical	Spacing			
EE000	DP 850	DISPLAYPORT	DP TBTSNK0 ML C P<3...0>	28 83	
EE001	DP 850	DISPLAYPORT	DP TBTSNK0 ML C N<3...0>	28 83	
EE002	DP TBTSNK0_ML	DP 850	DISPLAYPORT	DP TBTSNK0 ML P<3...0>	28
EE003	DP TBTSNK0_ML	DP 850	DISPLAYPORT	DP TBTSNK0 ML N<3...0>	28
EE004	DP 850	DISPLAYPORT	DP TBTSNK0 AUXCH C P	28 83	
EE005	DP 850	DISPLAYPORT	DP TBTSNK0 AUXCH C N	28	
EE006	DP TBTSNK0_AUX	DP 850	DISPLAYPORT	DP TBTSNK0 AUXCH P	28
EE007	DP TBTSNK0_AUX	DP 850	DISPLAYPORT	DP TBTSNK0 AUXCH N	28
EE008	DP 850	DISPLAYPORT	DP TBTSNK1 ML C P<3...0>	28 83	
EE009	DP 850	DISPLAYPORT	DP TBTSNK1 ML C N<3...0>	28 83	
EE010	DP TBTSNK1_ML	DP 850	DISPLAYPORT	DP TBTSNK1 ML P<3...0>	28
EE011	DP TBTSNK1_ML	DP 850	DISPLAYPORT	DP TBTSNK1 ML N<3...0>	28
EE012	DP 850	DISPLAYPORT	DP TBTSNK1 AUXCH C P	28 83	
EE013	DP 850	DISPLAYPORT	DP TBTSNK1 AUXCH C N	28 83	
EE014	DP TBTSNK1_AUX	DP 850	DISPLAYPORT	DP TBTSNK1 AUXCH P	28
EE015	DP TBTSNK1_AUX	DP 850	DISPLAYPORT	DP TBTSNK1 AUXCH N	28
EE016	DP INTPMI_TBT_ML_MUX	DP 850	DISPLAYPORT	DP TBTSRC ML P<3...0>	43
EE017	DP INTPMI_TBT_ML_MUX	DP 850	DISPLAYPORT	DP TBTSRC ML N<3...0>	43
EE018	DP INTPMI_TBT_ML_MUX	DP 850	DISPLAYPORT	DP TBTSRC ML C P<3...0>	43
EE019	DP INTPMI_TBT_ML_MUX	DP 850	DISPLAYPORT	DP TBTSRC ML C N<3...0>	43
EE020	DP INTPMI_TBT_AUX_MUX	DP 850	DISPLAYPORT	DP TBTSRC AUXCH P	43
EE021	DP INTPMI_TBT_AUX_MUX	DP 850	DISPLAYPORT	DP TBTSRC AUXCH N	43
EE022	DP 850	DISPLAYPORT	DP TBTSRC AUX C P	43	
EE023	DP 850	DISPLAYPORT	DP TBTSRC AUX C N	43	
EE024					
EE025	TBT T2C 55S	TBT T2C	I2C TBTTRT_SCL	28	
EE026	TBT T2C 55S	TBT T2C	I2C TBTTRT_SDA	28	
EE027	TBT_SPT_CLK	TBT_SPT 55S	TBT_SPT	TBT SPI_CLK	28
EE028	TBT_SPT_MOSI	TBT_SPT 55S	TBT_SPT	TBT SPI_MOSI	28
EE029	TBT_SPT_MISO	TBT_SPT 55S	TBT_SPT	TBT SPI_MISO	28
EE030	TBT_SPT_CS_L	TBT_SPT 55S	TBT_SPT	TBT SPI_CS_L	28

*: Only used on hosts supporting T29 video-in

DisplayPort

Electrical Constraint Set	Physical	Spacing	
Graphics Source			
FE050 DP INTPNL EG ML_MUX	DP 850	DISPLAYPORT	DP INT EG ML P<3..0> 43 83
FE050 DP INTPNL EG ML_MUX	DP 850	DISPLAYPORT	DP INT EG ML N<3..0> 43 83
FE050 DP INTPNL EG AUX_MUX	DP 850	DISPLAYPORT	DP INT EG AUX P 43 83
FE050 DP INTPNL EG AUX_MUX	DP 850	DISPLAYPORT	DP INT EG AUX N 43 83
FE050 DP INTPNL EG AUX_MUX	DP 850	DISPLAYPORT	DP INT EG AUX C P 43
FE050 DP INTPNL EG AUX_MUX	DP 850	DISPLAYPORT	DP INT EG AUX C N 43
Internal Panel			
FE050	DP 850	DISPLAYPORT	DP INTPNL ML C P<3..0> 43
FE050	DP 850	DISPLAYPORT	DP INTPNL ML C N<3..0> 43
FE050 DP INTPNL MI_CONN	DP 850	DISPLAYPORT	DP INTPNL ML P<3..0> 42 43
FE050 DP INTPNL MI_CONN	DP 850	DISPLAYPORT	DP INTPNL ML N<3..0> 42 43
FE050 DP INTPNL AUX_CONN	DP 850	DISPLAYPORT	DP INTPNL AUX P 42 43
FE050 DP INTPNL AUX_CONN	DP 850	DISPLAYPORT	DP INTPNL AUX N 42 43
Internal DP SPDIF			
FE050		HDA	DP_INT SPDIF AUDIO 42 54

TBT/DP Net Properties

Electrical Constraint Set	Physical	Spacing	
Port A			
E621 TBT_A_R2D1	TBTDP_960	TBTDP	TBT_A_R2D_C_P<1>
E622 TBT_A_R2D1	TBTDP_960	TBTDP	TBT_A_R2D_C_N<1>
E623 TBT_A_R2D8	TBTDP_960	TBTDP	TBT_A_R2D_C_P<0>
E624 TBT_A_R2D8	TBTDP_960	TBTDP	TBT_A_R2D_C_N<0>
E625	TBTDP_960	TBTDP	TBT_A_R2D_P<1...0>
E626	TBTDP_960	TBTDP	TBT_A_R2D_N<1...0>
E627 DP_TBTPA_ML1	DP_850	DISPLAYPORT	DP_TBTPA_ML_C_P<1>
E628 DP_TBTPA_ML1	DP_850	DISPLAYPORT	DP_TBTPA_ML_C_N<1>
E629 DP_TBTPA_ML3	DP_850	DISPLAYPORT	DP_TBTPA_ML_C_P<3>
E630 DP_TBTPA_ML3	DP_850	DISPLAYPORT	DP_TBTPA_ML_C_N<3>
E631	DP_850	DISPLAYPORT	DP_TBTPA_ML_P<1>
E632	DP_850	DISPLAYPORT	DP_TBTPA_ML_N<1>
E633	DP_850	DISPLAYPORT	DP_TBTPA_ML_P<3>
E634	DP_850	DISPLAYPORT	DP_TBTPA_ML_N<3>
E635	DP_850	DISPLAYPORT	DP_A_LSX_ML_P<1>
E636 DP_A_LSX	DP_850	DISPLAYPORT	DP_A_LSX_ML_N<1>
E637	TBTDP_960	TBTDP	TBT_A_D2R_C_P<1...0>
E638	TBTDP_960	TBTDP	TBT_A_D2R_C_N<1...0>
E639 TBT_A_D2R1	TBTDP_960	TBTDP	TBT_A_D2R_P<1>
E640 TBT_A_D2R1	TBTDP_960	TBTDP	TBT_A_D2R_N<1>
E641 TBT_A_D2R8	TBTDP_960	TBTDP	TBT_A_D2R_P<0>
E642 TBT_A_D2R8	TBTDP_960	TBTDP	TBT_A_D2R_N<0>
E643 TBT_A_AUXCH	DP_850	DISPLAYPORT	DP_TBTPA_AUXCH_C_P
E644 TBT_A_AUXCH	DP_850	DISPLAYPORT	DP_TBTPA_AUXCH_C_N
E645	DP_850	DISPLAYPORT	DP_TBTPA_AUXCH_P
E646	DP_850	DISPLAYPORT	DP_TBTPA_AUXCH_N
E647 DP_A_AUXCH_DDC	DP_850	DISPLAYPORT	DP_A_AUXCH_DDC_P
E648 DP_A_AUXCH_DDC	DP_850	DISPLAYPORT	DP_A_AUXCH_DDC_N
E649	TBTDP_960	TBTDP	TBT_A_D2R1_AUXDDC_P
E650	TBTDP_960	TBTDP	TBT_A_D2R1_AUXDDC_N
Port B			
E651 TBT_B_R2D1	TBTDP_960	TBTDP	TBT_B_R2D_C_P<1>
E652 TBT_B_R2D1	TBTDP_960	TBTDP	TBT_B_R2D_C_N<1>
E653 TBT_B_R2D8	TBTDP_960	TBTDP	TBT_B_R2D_C_P<0>
E654 TBT_B_R2D8	TBTDP_960	TBTDP	TBT_B_R2D_C_N<0>
E655	TBTDP_960	TBTDP	TBT_B_R2D_P<1...0>
E656	TBTDP_960	TBTDP	TBT_B_R2D_N<1...0>
E657 DP_TBTPB_ML1	DP_850	DISPLAYPORT	DP_TBTPB_ML_C_P<1>
E658 DP_TBTPB_ML1	DP_850	DISPLAYPORT	DP_TBTPB_ML_C_N<1>
E659 DP_TBTPB_ML3	DP_850	DISPLAYPORT	DP_TBTPB_ML_C_P<3>
E660 DP_TBTPB_ML3	DP_850	DISPLAYPORT	DP_TBTPB_ML_C_N<3>
E661	DP_850	DISPLAYPORT	DP_TBTPB_ML_P<1>
E662	DP_850	DISPLAYPORT	DP_TBTPB_ML_N<1>
E663	DP_850	DISPLAYPORT	DP_TBTPB_ML_P<3>
E664	DP_850	DISPLAYPORT	DP_TBTPB_ML_N<3>
E665 DP_B_LSX	DP_850	DISPLAYPORT	DP_B_LSX_ML_P<1>
E666 DP_B_LSX	DP_850	DISPLAYPORT	DP_B_LSX_ML_N<1>
E667	TBTDP_960	TBTDP	TBT_B_D2R_C_P<1...0>
E668	TBTDP_960	TBTDP	TBT_B_D2R_C_N<1...0>
E669 TBT_B_D2R1	TBTDP_960	TBTDP	TBT_B_D2R_P<1>
E670 TBT_B_D2R1	TBTDP_960	TBTDP	TBT_B_D2R_N<1>
E671 TBT_B_D2R8	TBTDP_960	TBTDP	TBT_B_D2R_P<0>
E672 TBT_B_D2R8	TBTDP_960	TBTDP	TBT_B_D2R_N<0>
E673 TBT_B_AUXCH	DP_850	DISPLAYPORT	DP_TBTPB_AUXCH_C_P
E674 TBT_B_AUXCH	DP_850	DISPLAYPORT	DP_TBTPB_AUXCH_C_N
E675	DP_850	DISPLAYPORT	DP_TBTPB_AUXCH_P
E676	DP_850	DISPLAYPORT	DP_TBTPB_AUXCH_N
E677 DP_B_AUXCH_DDC	DP_850	DISPLAYPORT	DP_B_AUXCH_DDC_P
E678 DP_B_AUXCH_DDC	DP_850	DISPLAYPORT	DP_B_AUXCH_DDC_N
E679	TBTDP_960	TBTDP	TBT_B_D2R1_AUXDDC_P
E680	TBTDP_960	TBTDP	TBT_B_D2R1_AUXDDC_N

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GDDR5

GDDR5-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
GDDR_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
GDDR_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	12 MM	=STANDARD	=STANDARD
GDDR_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
GDDR_MA_PHY	*	GDDR_45S
GDDR_ADBI_PHY	*	GDDR_45S
GDDR_CTRL_PHY	*	GDDR_45S
GDDR_CLK_PHY	*	GDDR_80D
GDDR_DQ_PHY	*	GDDR_45S
GDDR_EDC_PHY	*	GDDR_45S
GDDR_DBI_PHY	*	GDDR_45S
GDDR_WCK_PHY	*	GDDR_80D

Main Segment Min Spacing Rules for 4.5 Gbps or Less (AMD Doc# 49919)

Trace-to-Trace				Isolation				
Table	Micro	Design	Strip Design	Micro	Design	Strip Design	Comments	
5-6/5-7	2:1	2:1	2:1 2:1	5:1 5:1	5:1 5:1	5:1 5:1	Memory address (MA). Implemented 4.5 Gbps or less rules for K70.	
5-6/5-7	2:1	2:1	2:1 2:1	5:1 5:1	5:1 5:1	5:1 5:1	Address dynamic bus inversion (ADBI)	
5-6/5-7	2:1	2:1	2:1 2:1	5:1 5:1	5:1 5:1	5:1 5:1	Control (CTRL)	
5-6/5-7	5:1	5:1	5:1 5:1	5:1 5:1	5:1 5:1	5:1 5:1	Clock (CLK)	
5-6/5-7	3:1	3:1	3:1 3:1	5:1 5:1	5:1 5:1	5:1 5:1	Data (DQ)	
5-6/5-7	7:1	7:1	7:1 7:1	7:1 7:1	7:1 7:1	7:1 7:1	Error detection pins (EDC). Using larger isolation rules,	
5-6/5-7	3:1	3:1	3:1 3:1	5:1 5:1	5:1 5:1	5:1 5:1	Data dynamic bus inversion (DBI)	
5-6/5-7	5:1	5:1	5:1 5:1	5:1 5:1	5:1 5:1	5:1 5:1	Forwarded clock (WCK)	

GDDR5-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR_ISO	*	=3x_DIELECTRIC	?
GDDR_ISO	TOP, BOTTOM	=5x_DIELECTRIC	?
GDDR_MA2MA	*	=2x_DIELECTRIC	?
GDDR_MA2MA	TOP, BOTTOM	=3x_DIELECTRIC	?
GDDR_ADBI2ADBI	*	=2x_DIELECTRIC	?
GDDR_ADBI2ADBI	TOP, BOTTOM	=2x_DIELECTRIC	?
GDDR_CTRL2CTRL	*	=2x_DIELECTRIC	?
GDDR_CTRL2CTRL	TOP, BOTTOM	=2x_DIELECTRIC	?
GDDR_CLK2CLK	*	=3x_DIELECTRIC	?
GDDR_CLK2CLK	TOP, BOTTOM	=5x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR_DQ2DQ	*	=3x_DIELECTRIC	?
GDDR_DQ2DQ	TOP, BOTTOM	=3x_DIELECTRIC	?
GDDR_EDC_ISO	*	=3X_DIELECTRIC	?
GDDR_EDC_ISO	TOP, BOTTOM	=5X_DIELECTRIC	?
GDDR_EDC2EDC	*	=3X_DIELECTRIC	?
GDDR_EDC2EDC	TOP, BOTTOM	=5X_DIELECTRIC	?
GDDR_DBI2DBI	*	=3x_DIELECTRIC	?
GDDR_DBI2DBI	TOP, BOTTOM	=3x_DIELECTRIC	?
GDDR_WCK2WCK	*	=3X_DIELECTRIC	?
GDDR_WCK2WCK	TOP, BOTTOM	=5x_DIELECTRIC	?

Constraints ($x \in \{A, B\}, y \in \{0, 1\}$)

Memory Address: M_{Axy}[8:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_MA	*	*	GDDR_ISO
GDDR_*_*_MA	=SAME	*	GDDR_MA2MA

Address Dynamic Bus Inversion: ADBIx

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_ADBI	*	*	GDDR_ISO
GDDR_*_*_ADBI	=SAME	*	GDDR_ADBI2ADBI

Control: Reset, CKExy, CSxy, WExy, RASxy, CASxy

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_CTRL	*	*	GDDR_ISO
GDDR_*_*_CTRL	*	*	GDDR_ISO
GDDR_*_*_CTRL	=SAME	*	GDDR_CTRL2CTRL

Clock: CKxy

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_CLK	*	*	GDDR_ISO
GDDR_*_*_CLK	=SAME	*	GDDR_CLK2CLK

GPU

GPU-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_GPU_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

GPU-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_GPU_ISO	*	=4:1_SPACING	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_GPU	*	*	CLK_GPU_ISO

GDDR5 Frame Buffer A

Electrical Constraint Set		Physical	Spacing		
Memory Address					
MEM0	GDDR A0 MA	GDDR NA PHY	GDDR A 0 MA	FB A0 A<8..0>	NO TEST:TRUE
MEM1	GDDR A1 MA	GDDR NA PHY	GDDR A 1 MA	FB A1 A<8..0>	NO TEST:TRUE
Address Dynamic Bus Inv					
FE00	GDDR A0 ADBT	GDDR ADBT PHY	GDDR A 0 ADBT	FB A0 ABT L	NO TEST:TRUE
FE01	GDDR A1 ADBT	GDDR ADBT PHY	GDDR A 1 ADBT	FB A1 ABT L	NO TEST:TRUE
Control					
FE04	GDDR A0 CKE	GDDR CTRL PHY	GDDR A 0 CTRL	FB A0 CKE L	NO TEST:TRUE
FE05	GDDR A0 CTRL	GDDR CTRL PHY	GDDR A 0 CTRL	FB A0 CS L	NO TEST:TRUE
FE06	GDDR A0 CTRL	GDDR CTRL PHY	GDDR A 0 CTRL	FB A0 WE L	NO TEST:TRUE
FE07	GDDR A0 CTRL	GDDR CTRL PHY	GDDR A 0 CTRL	FB A0 CAS L	NO TEST:TRUE
FE08	GDDR A0 CTRL	GDDR CTRL PHY	GDDR A 0 CTRL	FB A0 RAS L	NO TEST:TRUE
FE09	GDDR A1 CKE	GDDR CTRL PHY	GDDR A 1 CTRL	FB A1 CKE L	NO TEST:TRUE
FE10	GDDR A1 CTRL	GDDR CTRL PHY	GDDR A 1 CTRL	FB A1 CS L	NO TEST:TRUE
FE11	GDDR A1 CTRL	GDDR CTRL PHY	GDDR A 1 CTRL	FB A1 WE L	NO TEST:TRUE
FE12	GDDR A1 CTRL	GDDR CTRL PHY	GDDR A 1 CTRL	FB A1 CAS L	NO TEST:TRUE
FE13	GDDR A1 CTRL	GDDR CTRL PHY	GDDR A 1 CTRL	FB A1 RAS L	NO TEST:TRUE
Clock					
FE16	GDDR A0 CLK	GDDR CLK PHY	GDDR A 0 CLK	FB A0 CLK P	NO TEST:TRUE
FE17	GDDR A0 CLK	GDDR CLK PHY	GDDR A 0 CLK	FB A0 CLK N	NO TEST:TRUE
FE18	GDDR A1 CLK	GDDR CLK PHY	GDDR A 1 CLK	FB A1 CLK P	NO TEST:TRUE
FE19	GDDR A1 CLK	GDDR CLK PHY	GDDR A 1 CLK	FB A1 CLK N	NO TEST:TRUE
Data					
FE20	GDDR A0 DQ BYTE0	GDDR DQ PHY	GDDR A 0 DQ	FB A0 DQ<7..0>	NO TEST:TRUE
FE21	GDDR A0 DQ BYTE1	GDDR DQ PHY	GDDR A 0 DQ	FB A0 DQ<15..8>	NO TEST:TRUE
FE22	GDDR A0 DQ BYTE2	GDDR DQ PHY	GDDR A 0 DQ	FB A0 DQ<23..16>	NO TEST:TRUE
FE23	GDDR A0 DQ BYTE3	GDDR DQ PHY	GDDR A 0 DQ	FB A0 DQ<31..24>	NO TEST:TRUE
FE24	GDDR A1 DQ BYTE0	GDDR DQ PHY	GDDR A 1 DQ	FB A1 DQ<7..0>	NO TEST:TRUE
FE25	GDDR A1 DQ BYTE1	GDDR DQ PHY	GDDR A 1 DQ	FB A1 DQ<15..8>	NO TEST:TRUE
FE26	GDDR A1 DQ BYTE2	GDDR DQ PHY	GDDR A 1 DQ	FB A1 DQ<23..16>	NO TEST:TRUE
FE27	GDDR A1 DQ BYTE3	GDDR DQ PHY	GDDR A 1 DQ	FB A1 DQ<31..24>	NO TEST:TRUE
Error Detection					
FE30	GDDR A0 EDC0	GDDR EDC PHY	GDDR A 0 EDC	FB A0 EDC<0>	NO TEST:TRUE
FE31	GDDR A0 EDC1	GDDR EDC PHY	GDDR A 0 EDC	FB A0 EDC<1>	NO TEST:TRUE
FE32	GDDR A0 EDC2	GDDR EDC PHY	GDDR A 0 EDC	FB A0 EDC<2>	NO TEST:TRUE
FE33	GDDR A0 EDC3	GDDR EDC PHY	GDDR A 0 EDC	FB A0 EDC<3>	NO TEST:TRUE
FE34	GDDR A1 EDC0	GDDR EDC PHY	GDDR A 1 EDC	FB A1 EDC<0>	NO TEST:TRUE
FE35	GDDR A1 EDC1	GDDR EDC PHY	GDDR A 1 EDC	FB A1 EDC<1>	NO TEST:TRUE
FE36	GDDR A1 EDC2	GDDR EDC PHY	GDDR A 1 EDC	FB A1 EDC<2>	NO TEST:TRUE
FE37	GDDR A1 EDC3	GDDR EDC PHY	GDDR A 1 EDC	FB A1 EDC<3>	NO TEST:TRUE
Data Dynamic Bus Inv					
FE39	GDDR A0 DBT0	GDDR DBT PHY	GDDR A 0 DBT	FB A0 DBT L<0>	NO TEST:TRUE
FE40	GDDR A0 DBT1	GDDR DBT PHY	GDDR A 0 DBT	FB A0 DBI L<1>	NO TEST:TRUE
FE41	GDDR A0 DBT2	GDDR DBT PHY	GDDR A 0 DBT	FB A0 DBI L<2>	NO TEST:TRUE
FE42	GDDR A0 DBT3	GDDR DBT PHY	GDDR A 0 DBT	FB A0 DBI L<3>	NO TEST:TRUE
FE43	GDDR A1 DBT0	GDDR DBT PHY	GDDR A 1 DBT	FB A1 DBI L<0>	NO TEST:TRUE
FE44	GDDR A1 DBT1	GDDR DBT PHY	GDDR A 1 DBT	FB A1 DBI L<1>	NO TEST:TRUE
FE45	GDDR A1 DBT2	GDDR DBT PHY	GDDR A 1 DBT	FB A1 DBI L<2>	NO TEST:TRUE
FE46	GDDR A1 DBT3	GDDR DBT PHY	GDDR A 1 DBT	FB A1 DBI L<3>	NO TEST:TRUE
Forwarded Clock					
FE48	GDDR A0 WCK0	GDDR WCK PHY	GDDR A 0 WCK	FB A0 WCLK P<0>	NO TEST:TRUE
FE49	GDDR A0 WCK0	GDDR WCK PHY	GDDR A 0 WCK	FB A0 WCLK N<0>	NO TEST:TRUE
FE50	GDDR A0 WCK1	GDDR WCK PHY	GDDR A 0 WCK	FB A0 WCLK P<1>	NO TEST:TRUE
FE51	GDDR A0 WCK1	GDDR WCK PHY	GDDR A 0 WCK	FB A0 WCLK N<1>	NO TEST:TRUE
FE52	GDDR A1 WCK0	GDDR WCK PHY	GDDR A 1 WCK	FB A1 WCLK P<0>	NO TEST:TRUE
FE53	GDDR A1 WCK0	GDDR WCK PHY	GDDR A 1 WCK	FB A1 WCLK N<0>	NO TEST:TRUE
FE54	GDDR A1 WCK1	GDDR WCK PHY	GDDR A 1 WCK	FB A1 WCLK P<1>	NO TEST:TRUE
FE55	GDDR A1 WCK1	GDDR WCK PHY	GDDR A 1 WCK	FB A1 WCLK N<1>	NO TEST:TRUE

GPU

Electrical Constraint Set	Physical	Spacing	
Clocks			
PEX0	CLK_GPU_55S	CLK_GPU	PEX_TSTCLK_0_PL 78
PEX0	CLK_GPU_55S	CLK_GPU	PEX_TSTCLK_0_NG 78
PEX0	CLK_PCTE_PHY	CLK_PCTE	GPU_TESTMODE 83
PEX0	CLK_PCTE_PHY	CLK_PCTE	GPU_PEX_TERM 78
PEX0	CLK_GPU_55S	CLK_GPU	FB_A0_CK_MID NO_TEST=TRUE 81
PEX0	CLK_GPU_55S	CLK_GPU	FB_A1_CK_MID NO_TEST=TRUE 81
PEX0	CLK_GPU_55S	CLK_GPU	FB_B0_CK_MID NO_TEST=TRUE 82
PEX0	CLK_GPU_55S	CLK_GPU	FB_B1_CK_MID NO_TEST=TRUE 82
PEX0	CLK_GPU_55S	CLK_GPU	FB_C0_CK_MID NO_TEST=TRUE 82
PEX0	CLK_GPU_55S	CLK_GPU	FB_C1_CK_MID NO_TEST=TRUE 82
PEX0	CLK_GPU_55S	CLK_GPU	FB_D0_CK_MID NO_TEST=TRUE 82
PEX0	CLK_GPU_55S	CLK_GPU	FB_D1_CK_MID NO_TEST=TRUE 82
PEX0	CLK_GPU_55S	CLK_GPU	GPU_JTAG_TCK 83 85
PEX0	CLK_GPU_55S	CLK_GPU	GPU_ROM_SCLK 83 84
PEX0	CLK_GPU_55S	CLK_GPU	GPU_ROM_SCLK_R 84
PEX0	CLK_GPU_55S	CLK_GPU	GPU_XTALOUT 83 84
PEX0	CLK_GPU_55S	CLK_GPU	GPU_OSC_27M_XTALOUT 83 84
PEX0	CLK_GPU_55S	CLK_GPU	GPU_OSC_27M_XTALIN 83 84
SMB			
ES00	SMB_PHY	SMB	GPU_SMB_CLK 83 84
ES00	SMB_PHY	SMB	GPU_SMB_DAT 83 84
ES00	SMB_PHY	SMB	GPU_SMB_CLK_R 49 84
ES00	SMB_PHY	SMB	GPU_SMB_DAT_R 49 84
PCIe Compensation			
ES00	PCIE_56S	COMP_PCIE	FB_CAL_PD_VDDQ 80
ES00	PCIE_56S	COMP_PCIE	FB_CAL_PU_GND 80
ES00	PCIE_56S	COMP_PCIE	FB_CAL_TERM_GND 80

GDDR5 Frame Buffer B

Electrical Constraint Set		Physical	Spacing		
Memory Address					
H009	GDDR_B0_MA	GDDR_MA_PHY	GDDR_B_0_MA	FB_B0_A<8..0>	NO TEST=TRUE
H009	GDDR_B1_MA	GDDR_MA_PHY	GDDR_B_1_MA	FB_B1_A<8..0>	NO TEST=TRUE
Address Dynamic Bus Inv					
H009	GDDR_B0_ADBT	GDDR_ADBT_PHY	GDDR_B_0_ADBT	FB_B0_ABT_L	NO TEST=TRUE
H009	GDDR_B1_ADBT	GDDR_ADBT_PHY	GDDR_B_1_ADBT	FB_B1_ABT_L	NO TEST=TRUE
Control					
H009	GDDR_B0_CKE	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_CKE_L	NO TEST=TRUE
H009	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_CS_L	NO TEST=TRUE
H009	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_WE_L	NO TEST=TRUE
H009	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_CAS_L	NO TEST=TRUE
H009	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_RAS_L	NO TEST=TRUE
H009	GDDR_B1_CKE	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_CKE_L	NO TEST=TRUE
H009	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_CS_L	NO TEST=TRUE
H009	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_WE_L	NO TEST=TRUE
H009	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_CAS_L	NO TEST=TRUE
H009	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_RAS_L	NO TEST=TRUE
Clock					
H009	GDDR_B0_CLK	GDDR_CLK_PHY	GDDR_B_0_CLK	FB_B0_CLK_P	NO TEST=TRUE
H009	GDDR_B0_CLK	GDDR_CLK_PHY	GDDR_B_0_CLK	FB_B0_CLK_N	NO TEST=TRUE
H009	GDDR_B1_CLK	GDDR_CLK_PHY	GDDR_B_1_CLK	FB_B1_CLK_P	NO TEST=TRUE
H009	GDDR_B1_CLK	GDDR_CLK_PHY	GDDR_B_1_CLK	FB_B1_CLK_N	NO TEST=TRUE
Data					
H009	GDDR_B0_DQ_BYTE0	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<7..0>	NO TEST=TRUE
H009	GDDR_B0_DQ_BYTE1	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<15..8>	NO TEST=TRUE
H009	GDDR_B0_DQ_BYTE2	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<23..16>	NO TEST=TRUE
H009	GDDR_B0_DQ_BYTE3	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<31..24>	NO TEST=TRUE
H009	GDDR_B1_DQ_BYTE0	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<7..0>	NO TEST=TRUE
H009	GDDR_B1_DQ_BYTE1	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<15..8>	NO TEST=TRUE
H009	GDDR_B1_DQ_BYTE2	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<23..16>	NO TEST=TRUE
H009	GDDR_B1_DQ_BYTE3	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<31..24>	NO TEST=TRUE
Error Detection					
H009	GDDR_B0_EDC0	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<0>	NO TEST=TRUE
H009	GDDR_B0_EDC1	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<1>	NO TEST=TRUE
H009	GDDR_B0_EDC2	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<2>	NO TEST=TRUE
H009	GDDR_B0_EDC3	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<3>	NO TEST=TRUE
H009	GDDR_B1_EDC0	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<0>	NO TEST=TRUE
H009	GDDR_B1_EDC1	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<1>	NO TEST=TRUE
H009	GDDR_B1_EDC2	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<2>	NO TEST=TRUE
H009	GDDR_B1_EDC3	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<3>	NO TEST=TRUE
Data Dynamic Bus Inv					
H009	GDDR_B0_DBT0	GDDR_DBT_PHY	GDDR_B_0_DBT	FB_B0_DBT_L<0>	NO TEST=TRUE
H009	GDDR_B0_DBT1	GDDR_DBT_PHY	GDDR_B_0_DBT	FB_B0_DBT_L<1>	NO TEST=TRUE
H009	GDDR_B0_DBT2	GDDR_DBT_PHY	GDDR_B_0_DBT	FB_B0_DBT_L<2>	NO TEST=TRUE
H009	GDDR_B0_DBT3	GDDR_DBT_PHY	GDDR_B_0_DBT	FB_B0_DBT_L<3>	NO TEST=TRUE
H009	GDDR_B1_DBT0	GDDR_DBT_PHY	GDDR_B_1_DBT	FB_B1_DBT_L<0>	NO TEST=TRUE
H009	GDDR_B1_DBT1	GDDR_DBT_PHY	GDDR_B_1_DBT	FB_B1_DBT_L<1>	NO TEST=TRUE
H009	GDDR_B1_DBT2	GDDR_DBT_PHY	GDDR_B_1_DBT	FB_B1_DBT_L<2>	NO TEST=TRUE
H009	GDDR_B1_DBT3	GDDR_DBT_PHY	GDDR_B_1_DBT	FB_B1_DBT_L<3>	NO TEST=TRUE
Forwarded Clock					
H009	GDDR_B0_WCLK	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_P<0>	NO TEST=TRUE
H009	GDDR_B0_WCK0	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_N<0>	NO TEST=TRUE
H009	GDDR_B0_WCK1	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_P<1>	NO TEST=TRUE
H009	GDDR_B0_WCK1	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_N<1>	NO TEST=TRUE
H009	GDDR_B1_WCK0	GDDR_WCK_PHY	GDDR_B_1_WCK	FB_B1_WCLK_P<0>	NO TEST=TRUE
H009	GDDR_B1_WCK0	GDDR_WCK_PHY	GDDR_B_1_WCK	FB_B1_WCLK_N<0>	NO TEST=TRUE
H009	GDDR_B1_WCK1	GDDR_WCK_PHY	GDDR_B_1_WCK	FB_B1_WCLK_P<1>	NO TEST=TRUE
H009	GDDR_B1_WCK1	GDDR_WCK_PHY	GDDR_B_1_WCK	FB_B1_WCLK_N<1>	NO TEST=TRUE

Frame Buffer Reset

Electrical Constraint Set		Physical	Spacing	
Reset				
H442	GD0R_A0_RESET	GD0R_56S	GD0R_CTR1	FB_A0_RESET_L
H445	GD0R_A1_RESET	GD0R_56S	GD0R_CTR1	FB_A1_RESET_L
H446	GD0R_B0_RESET	GD0R_56S	GD0R_CTR1	FB_B0_RESET_L
H447	GD0R_B1_RESET	GD0R_56S	GD0R_CTR1	FB_B1_RESET_L
H448	GD0R_C0_RESET	GD0R_56S	GD0R_CTR1	FB_C0_RESET_L
H449	GD0R_C1_RESET	GD0R_56S	GD0R_CTR1	FB_C1_RESET_L
H450	GD0R_D0_RESET	GD0R_56S	GD0R_CTR1	FB_D0_RESET_L
H451	GD0R_D1_RESET	GD0R_56S	GD0R_CTR1	FB_D1_RESET_L

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		BRANCH	dvt0
		PAGE	122 OF 127
		SHEET	103 OF 106

D

C

B

A

	Physical	Spacing	Voltage	DIDT	NO_TEST	
Input	Bus					
U755	POWER	POWER	12V			PP12V_S0_BLC_VIN2
U777	POWER	POWER	12V			PP12V_S0_BLC_VINP
U805	POWER	POWER	14V			PRE_REG_OUT
U809	POWER	POWER	3.3V			BLC_P3V3S
U80A	POWER	POWER	3.3V			BLC_P3V3_REF
U80P	POWER	POWER	3.3V			BLC_P3V3
U855	POWER	POWER	3.3V			PP3V3_S0_BLC_R
U84A	POWER	POWER	5V			SPTX_VIN
U84D	POWER	POWER	5V			PP8V_BLC
U84V	POWER	POWER	5V			BLC_VIN2
U80P	POWER	POWER	14V			BOOST_FET_DRAIN
U80P	POWER	POWER	12V			BOOST_VDD
U80S	POWER	POWER	5V			PP5V_S0_BLC_R
U80S	POWER	POWER	12V			PP12V_S0_BLC_F
Local Ground						
U755	BLC_CTL_PHY	BLC_PHASE	5V			BLC_GND_1
U80A	BLC_CTL_PHY	BLC_PHASE	5V			BLC_GND_2
U80D	BLC_CTL_PHY	BLC_PHASE	5V			BLC_GND_3
U81E	GND	GND	5V			AGND_BLC
Backlight						
U755	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE1
U755	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE1_R
U755	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE2
U755	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE2_R
U80P	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE3
U755	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE3_R
U755	BLC_CTL_PHY	BLC_CTL				LED_DRVRS_CS_RC_1
U755	BLC_CTL_PHY	BLC_CTL				LED_DRVRS_CS_RC_2
U755	BLC_CTL_PHY	BLC_CTL				LED_DRVRS_CS_RC_3
U80A	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_CS1
U80P	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_CS2
U80P	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_CS3
U80P	BLC_CTL_PHY	BLC_CTL				LED_DRVRS_CS_C1
U80A	BLC_CTL_PHY	BLC_CTL				LED_DRVRS_CS_C2
U80A	BLC_CTL_PHY	BLC_CTL				LED_DRVRS_CS_C3
U83A	BLC_CTL_PHY	BLC_HV	80V			LED_DRIVER_FDBK_R_1
U80A	BLC_CTL_PHY	BLC_HV	80V			LED_DRIVER_FDBK_R_2
U83A	BLC_CTL_PHY	BLC_HV	80V			LED_DRIVER_FDBK_R_3
U83D	BLC_CTL_PHY	BLC_CTL	80V			LED_DRIVER_FDBK1
U83D	BLC_CTL_PHY	BLC_CTL	80V			LED_DRIVER_FDBK2
U80A	BLC_CTL_PHY	BLC_CTL	80V			LED_DRIVER_FDBK3
U83A	BLC_CTL_PHY	BLC_CTL				BLC_PWM_1_R
U83A	BLC_CTL_PHY	BLC_CTL				BLC_PWM_2_R
U83A	BLC_CTL_PHY	BLC_CTL				BLC_PWM_3_R
U80P	BLC_CTL_PHY	BLC_CTL				BLC_PWM_1
U80P	BLC_CTL_PHY	BLC_CTL				BLC_PWM_2
U80P	BLC_CTL_PHY	BLC_CTL				BLC_PWM_3
U81E	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_REF1
U81E	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_REF2
U81E	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_REF3
U755	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_COMP1
U755	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_COMP2
U755	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_COMP3
U82P	BLC_CTL_PHY	BLC_CTL				BCOMP1
U82P	BLC_CTL_PHY	BLC_CTL				BCOMP2
U82P	BLC_CTL_PHY	BLC_CTL				BCOMP3
U80P	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_FLT1
U80P	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_FLT2
U80P	BLC_CTL_PHY	BLC_CTL				LED_FLT_R_1
U80P	BLC_CTL_PHY	BLC_CTL				LED_FLT_R_2
U80P	BLC_CTL_PHY	BLC_CTL				LED_FLT_R_3
U84D	BLC_CTL_PHY	BLC_CTL				PRE_REG_OUT_R
U855	BLC_CTL_PHY	BLC_CTL				BOOST_FB
U855	BLC_CTL_PHY	BLC_CTL				BOOST_COMP
U855	BLC_CTL_PHY	BLC_CTL				BOOST_COMP_C
U855	BLC_CTL_PHY	BLC_CTL		TRUE		BOOST_GDRV
U855	BLC_CTL_PHY	BLC_CTL		TRUE		BOOST_GDRV_R
U855	BLC_CTL_PHY	BLC_CTL				BOOST_ISNS
U855	BLC_CTL_PHY	BLC_CTL				BOOST_ISNS_R
U855	BLC_CTL_PHY	BLC_HV	80V			LED_DRVRS_DRAIN_1
U855	BLC_CTL_PHY	BLC_HV	80V			LED_DRVRS_DRAIN_2
U855	BLC_CTL_PHY	BLC_HV	80V			LED_DRVRS_DRAIN_3
OUTPUT BUS						
U80D	POWER_BLC_RET	BLC_HV	80V			IS1_BLC_F
U80D	POWER_BLC_RET	BLC_HV	80V			IS2_BLC_F
U80D	POWER_BLC_RET	BLC_HV	80V			IS3_BLC_F
U80D	POWER_BLC_RET	BLC_HV	80V			IS1_BLC
U80P	POWER_BLC_RET	BLC_HV	80V			IS2_BLC
U80D	POWER_BLC_RET	BLC_HV	80V			IS3_BLC
U80S	POWER_BLC_RET	BLC_HV	80V			BLC_LED_P_1
U80P	POWER_BLC_RET	BLC_HV	80V			BLC_LED_N_1
U80D	POWER_BLC_RET	BLC_HV	80V			BLC_LED_P_2
U80D	POWER_BLC_RET	BLC_HV	80V			BLC_LED_N_2
U80D	POWER_BLC_RET	BLC_HV	80V			BLC_LED_P_3
U80P	POWER_BLC_RET	BLC_HV	80V			BLC_LED_N_3
U82P	POWER_BLC	BLC_HV	80V			BLC_VOUT1
U82P	POWER_BLC	BLC_HV	80V			BLC_VOUT2
U82P	POWER_BLC	BLC_HV	80V			BLC_VOUT3

D

C

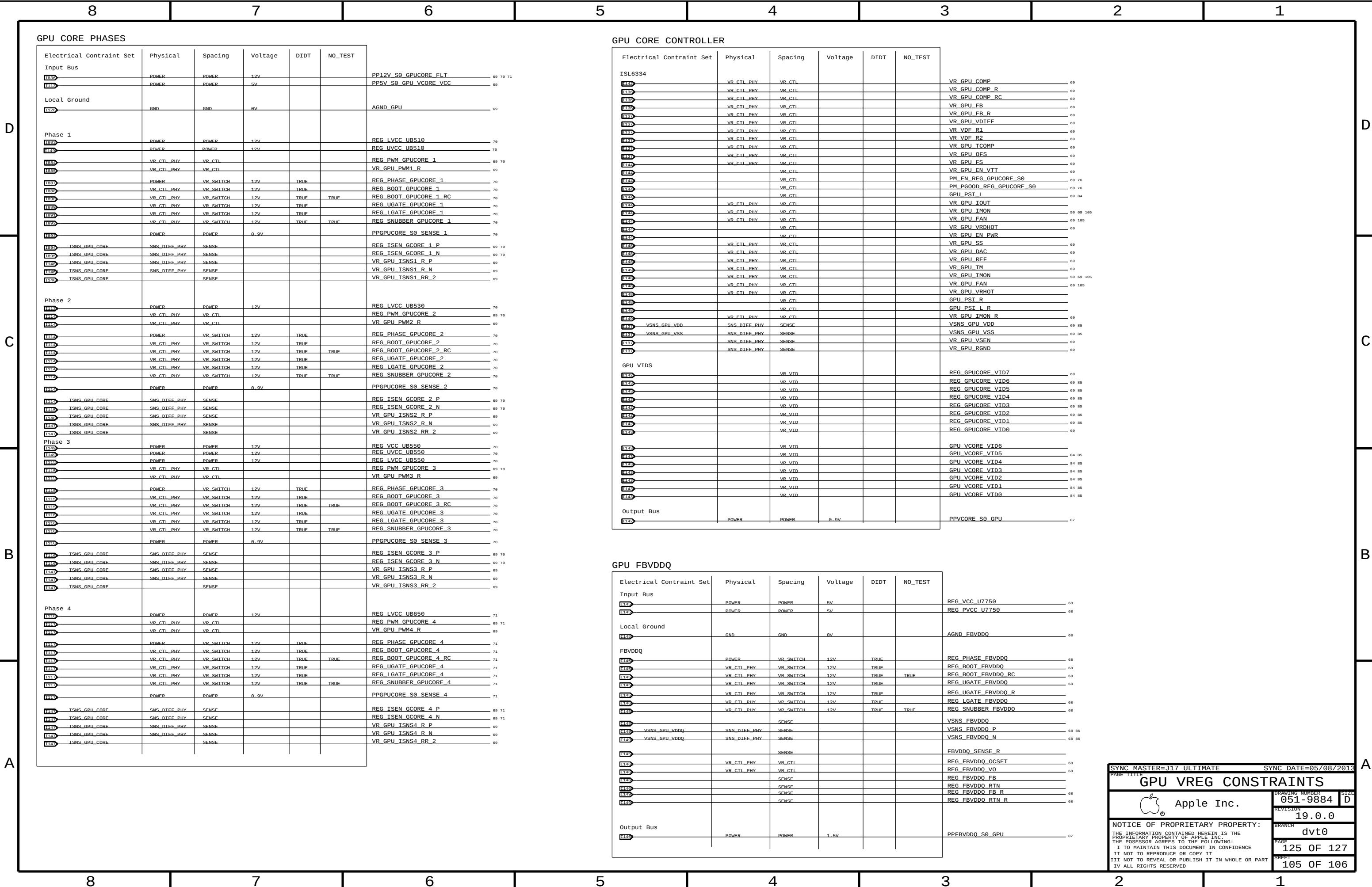
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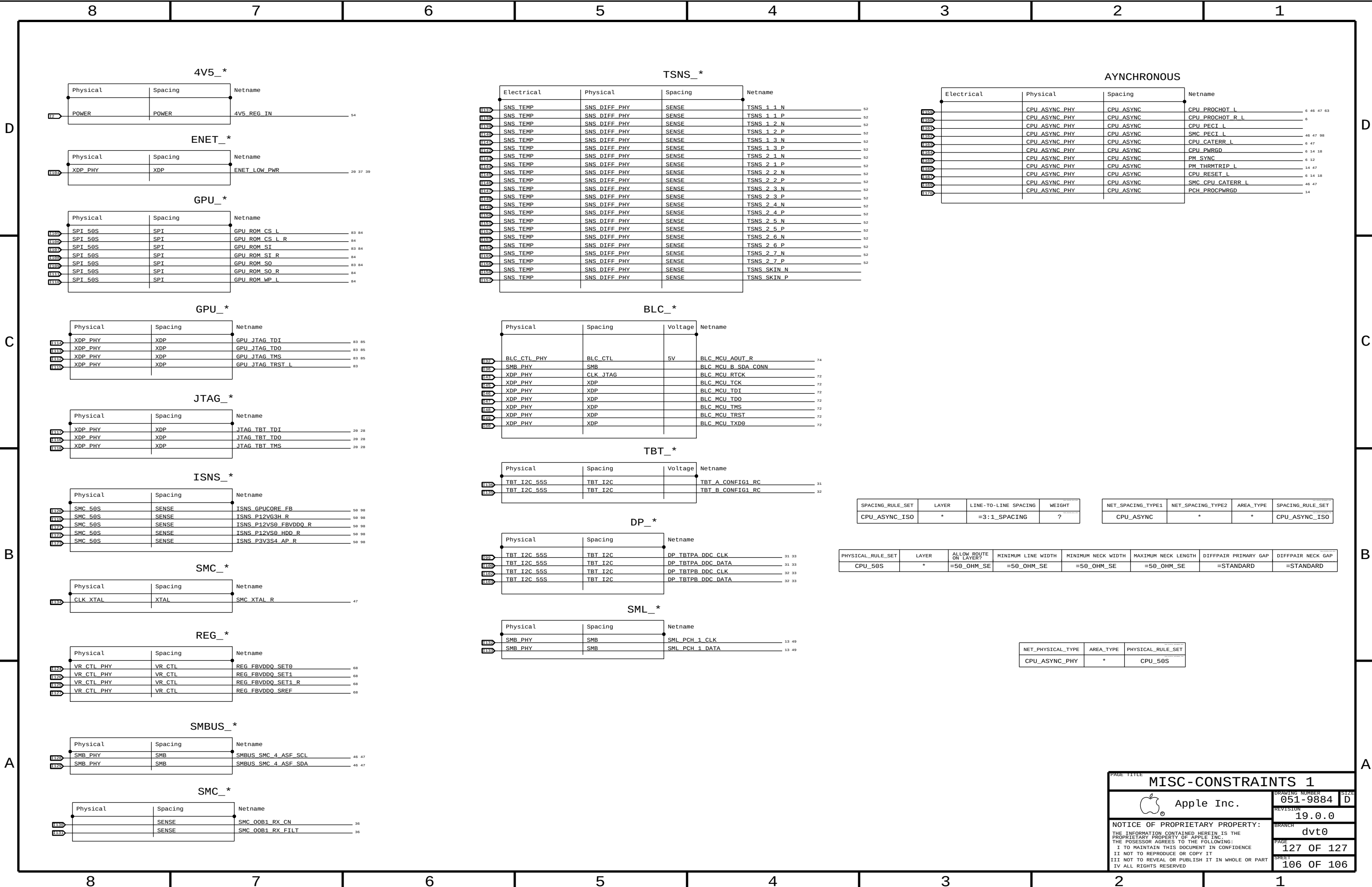
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Constraints

BLC High Voltage Output

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		19.0.0	
		BRANCH	
		dvt0	
		PAGE	
		124 OF 127	
		SHEET	
		104 OF 106	





MISC-CONSTRAINTS 1

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19.0.0

BRANCH
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PAGE
127 OF 127

SHEET
106 OF 106